ZOYASOFTS

Industry-Ready Internship Program

E-COMMERCE PRODUCT PRICE & REVIEW ANALYSIS

Project Report

Using Excel, SQL, Python and Power BI

Submitted By

Sayali Vishnu Patil

BCA Third Year

Submitted To

ZoyaSofts

Academic Year:2026-2027

College Name :- D.R.Mane College, Kagal

Project Overview

The project titled **"E-Commerce Product Price & Review Analysis"** focuses on analyzing product prices, customer ratings, reviews, and discounts using e-commerce product listing data collected from Flipkart. The dataset contains 103 product records. The main purpose of this project is to understand product pricing patterns, customer preferences, and differences in product ratings across categories and brands.

1. INTRODUCTION

E-commerce is the process of buying and selling products through online platforms. Customers can compare product prices, ratings, reviews, and discounts before purchasing products.

The main purpose of this project is to analysis e-commerce product data collected from Flipkart. The project focuses on product prices, brands, customer ratings, review counts, discounts, sellers, and delivery promises.

The collected data is analysed using Excel, PostgreSQL, Python, and Power BI to understand product information and present the analysis through visualizations and a dashboard.

2. PROBLEM STATEMENT

In e-commerce platforms, customers have access to a large number of products with different prices, brands, ratings, reviews, and discounts. Comparing this information manually can be time-consuming and difficult.

The main problem addressed by this project is to analyze the collected Flipkart product data systematically to understand price differences, customer ratings, review counts, discounts, and competition across product categories.

This project uses Excel, PostgreSQL, Python, and Power BI to organize, analyze, and visualize product data. The analysis helps identify useful patterns and generate business insights that support product comparison and better decision-making**.**

3. OBJECTIVES OF THE PROJECT

- To collect e-commerce product data from Flipkart.

- To organize product information in Excel.

- To store and analysis data using PostgreSQL and SQL queries.

- To perform Exploratory Data Analysis (EDA) using Python.

- To analysis product prices, brands, ratings, reviews, and discounts.

- To create a dashboard using Power BI.

- To present business insights and recommendations based on the collected data.

---

4. SCOPE OF THE PROJECT

This project focuses on analysing the collected e-commerce product dataset. It helps compare product categories, identify frequently appearing brands, examine ratings and review counts, compare discounts, and study competition across categories.

The analysis is limited to the collected dataset. Product prices, ratings, discounts, seller details, and delivery promises may change over time.

---

5. TOOLS AND TECHNOLOGIES USED

Microsoft Excel: Used to organize and prepare the collected product data.

PostgreSQL: Used to store product data in a database and perform SQL analysis.

Python: Used for data analysis and Exploratory Data Analysis (EDA).

Pandas and NumPy: Used for handling and analysing data.

Matplotlib: Used to create data visualizations where applicable.

Microsoft Power BI: Used to create charts and an interactive dashboard.

---

6. DATA COLLECTION

The dataset was manually collected from Flipkart product listings. It contains product-related information used for analysis, such as product names, categories, prices, ratings, reviews, brands, and discounts, where available.

- **Data Source:** Flipkart
- **Number of Records:** 103
- **Data Collection Method:** Manual data entry
- **Project Type:** Exploratory Data Analysis and Business Intelligence

Opened the Flipkart website using a web browser.

Searched for different products available on Flipkart.

Collected the product details displayed on the website.

Recorded product names, categories, brands, prices, discounts, ratings, and review counts.

Recorded seller names and delivery promises wherever publicly visible.

Entered the collected information into an Excel worksheet.

Prepared a dataset containing 103 product records for further analysis.

Saved the collected data for preprocessing and analysis using SQL, Python, and Power BI.

- Product Category
- Product Name
- Brand
- Price
- Discount
- Rating
- Review Count
- Seller
- Delivery Promise

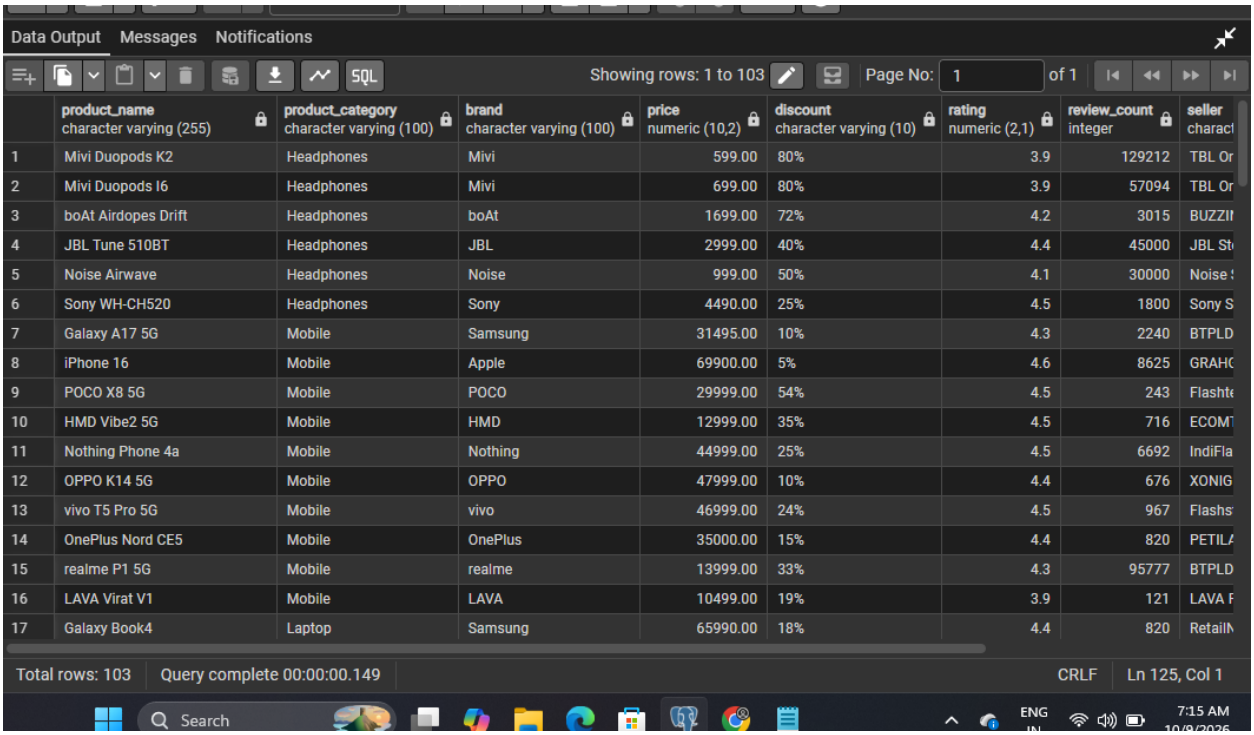
Seller and delivery information were included where publicly visible. The collected information was organized in Excel for further analysis.

---

7. DATA STORAGE

The collected product data was organized in Microsoft Excel and used for database analysis in PostgreSQL.

The dataset fields were used to store product details such as product name, category, brand, price, discount, rating, review count, seller, and delivery promise.



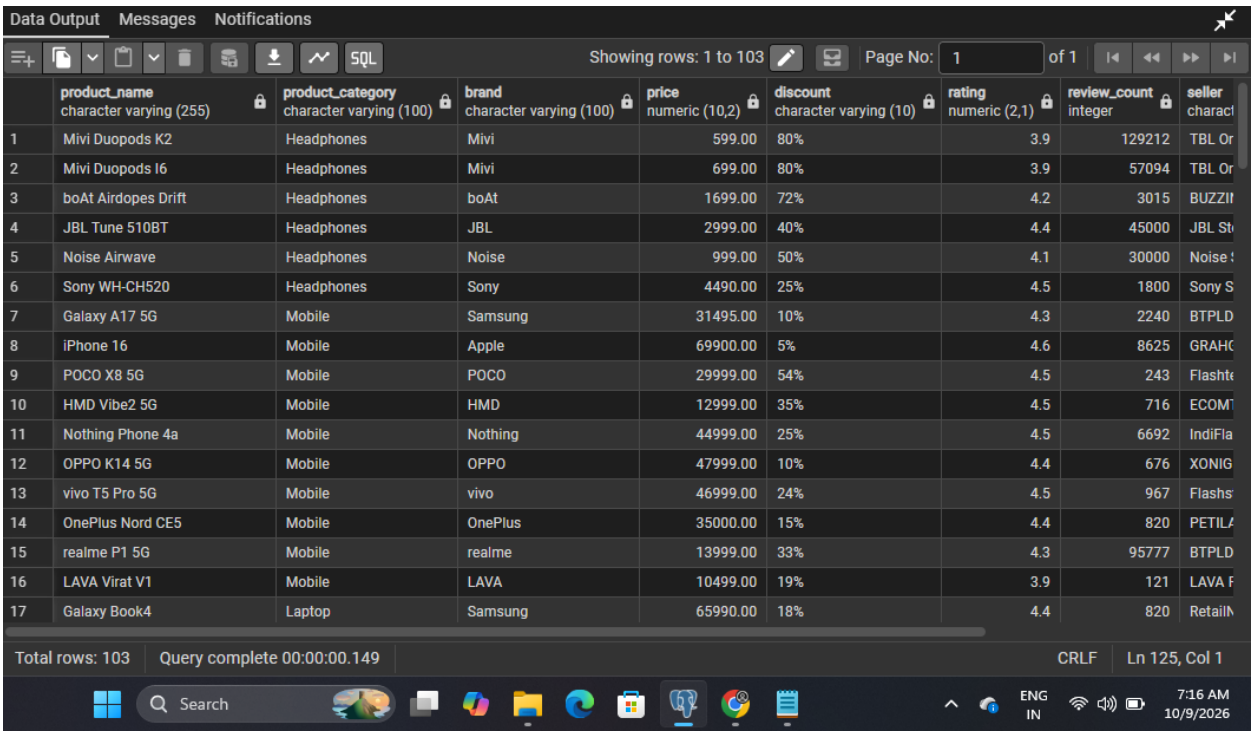
Data Output | Messages | Notifications

Showing rows: 1 to 103 | Page No: 1 of 1

	product_name character varying (255)	product_category character varying (100)	brand character varying (100)	price numeric (10,2)	discount character varying (10)	rating numeric (2,1)	review_count integer	seller charact
1	Mivi Duopods K2	Headphones	Mivi	599.00	80%	3.9	129212	TBL Or
2	Mivi Duopods I6	Headphones	Mivi	699.00	80%	3.9	57094	TBL Or
3	boAt Airdopes Drift	Headphones	boAt	1699.00	72%	4.2	3015	BUZZI
4	JBL Tune 510BT	Headphones	JBL	2999.00	40%	4.4	45000	JBL St
5	Noise Airwave	Headphones	Noise	999.00	50%	4.1	30000	Noise
6	Sony WH-CH520	Headphones	Sony	4490.00	25%	4.5	1800	Sony S
7	Galaxy A17 5G	Mobile	Samsung	31495.00	10%	4.3	2240	BTPLD
8	iPhone 16	Mobile	Apple	69900.00	5%	4.6	8625	GRAH
9	POCO X8 5G	Mobile	POCO	29999.00	54%	4.5	243	Flashte
10	HMD Vibe2 5G	Mobile	HMD	12999.00	35%	4.5	716	ECOM
11	Nothing Phone 4a	Mobile	Nothing	44999.00	25%	4.5	6692	IndiFla
12	OPPO K14 5G	Mobile	OPPO	47999.00	10%	4.4	676	XONIG
13	vivo T5 Pro 5G	Mobile	vivo	46999.00	24%	4.5	967	Flashs
14	OnePlus Nord CE5	Mobile	OnePlus	35000.00	15%	4.4	820	PETIL
15	realme P1 5G	Mobile	realme	13999.00	33%	4.3	95777	BTPLD
16	LAVA Virat V1	Mobile	LAVA	10499.00	19%	3.9	121	LAVA F
17	Galaxy Book4	Laptop	Samsung	65990.00	18%	4.4	820	RetailN

Total rows: 103 | Query complete 00:00:00.149 | CRLF | Ln 125, Col 1

Search | ENG IN | 7:15 AM 10/9/2026



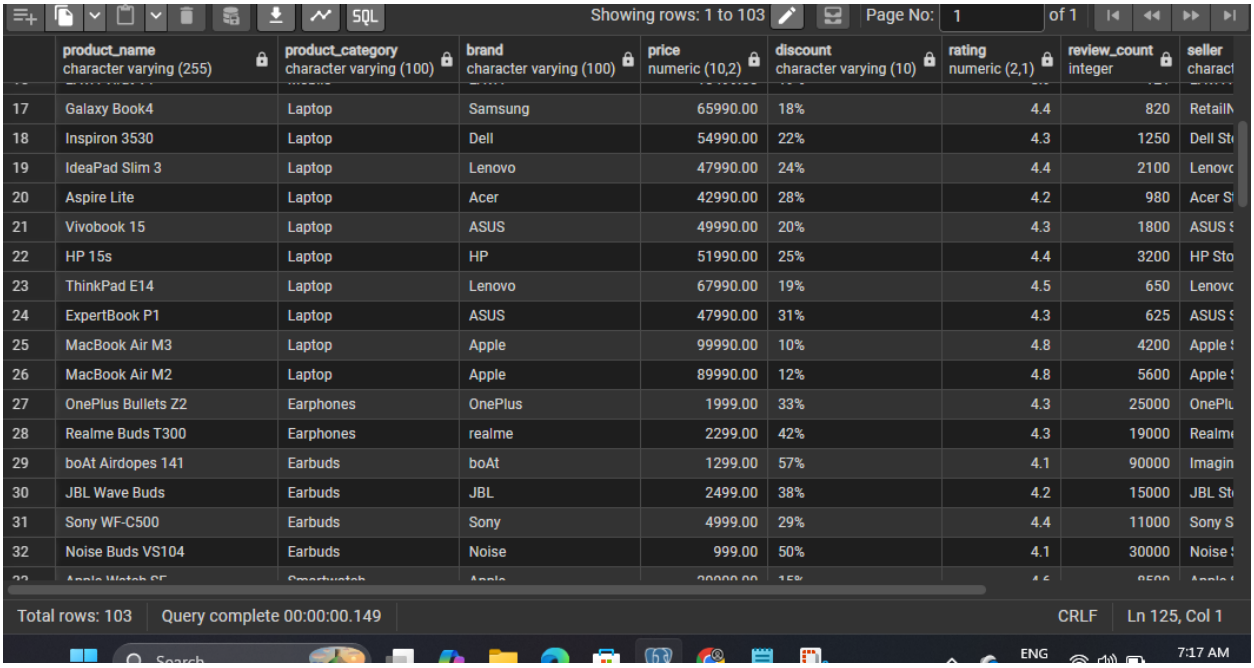
Data Output | Messages | Notifications

Showing rows: 1 to 103 | Page No: 1 of 1

	product_name character varying (255)	product_category character varying (100)	brand character varying (100)	price numeric (10,2)	discount character varying (10)	rating numeric (2,1)	review_count integer	seller charact
1	Mivi Duopods K2	Headphones	Mivi	599.00	80%	3.9	129212	TBL Or
2	Mivi Duopods I6	Headphones	Mivi	699.00	80%	3.9	57094	TBL Or
3	boAt Airdopes Drift	Headphones	boAt	1699.00	72%	4.2	3015	BUZZI
4	JBL Tune 510BT	Headphones	JBL	2999.00	40%	4.4	45000	JBL St
5	Noise Airwave	Headphones	Noise	999.00	50%	4.1	30000	Noise
6	Sony WH-CH520	Headphones	Sony	4490.00	25%	4.5	1800	Sony S
7	Galaxy A17 5G	Mobile	Samsung	31495.00	10%	4.3	2240	BTPLD
8	iPhone 16	Mobile	Apple	69900.00	5%	4.6	8625	GRAH
9	POCO X8 5G	Mobile	POCO	29999.00	54%	4.5	243	Flashte
10	HMD Vibe2 5G	Mobile	HMD	12999.00	35%	4.5	716	ECOM
11	Nothing Phone 4a	Mobile	Nothing	44999.00	25%	4.5	6692	IndiFla
12	OPPO K14 5G	Mobile	OPPO	47999.00	10%	4.4	676	XONIG
13	vivo T5 Pro 5G	Mobile	vivo	46999.00	24%	4.5	967	Flashs
14	OnePlus Nord CE5	Mobile	OnePlus	35000.00	15%	4.4	820	PETIL
15	realme P1 5G	Mobile	realme	13999.00	33%	4.3	95777	BTPLD
16	LAVA Virat V1	Mobile	LAVA	10499.00	19%	3.9	121	LAVA F
17	Galaxy Book4	Laptop	Samsung	65990.00	18%	4.4	820	RetailN

Total rows: 103 | Query complete 00:00:00.149 | CRLF | Ln 125, Col 1

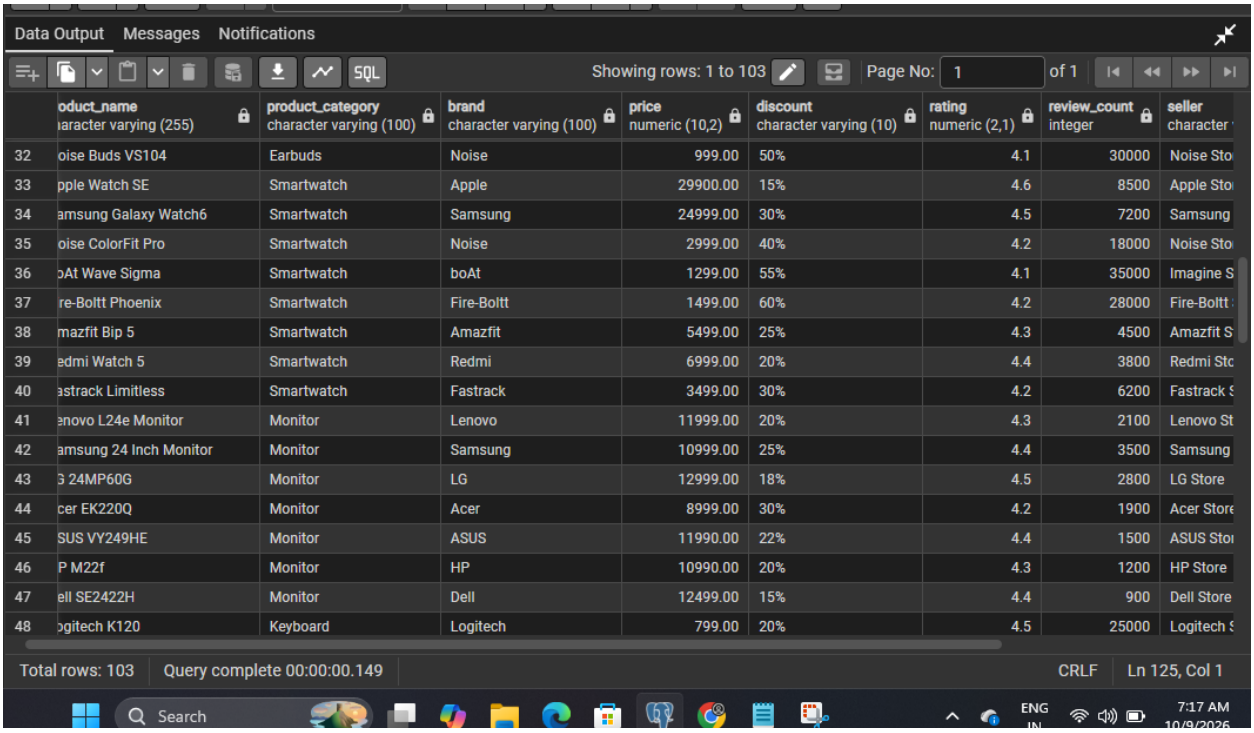
Search | ENG IN | 7:16 AM 10/9/2026



Showing rows: 1 to 103 | Page No: 1 of 1

	product_name character varying (255)	product_category character varying (100)	brand character varying (100)	price numeric (10,2)	discount character varying (10)	rating numeric (2,1)	review_count integer	seller charact
17	Galaxy Book4	Laptop	Samsung	65990.00	18%	4.4	820	RetailN
18	Inspiron 3530	Laptop	Dell	54990.00	22%	4.3	1250	Dell St
19	IdeaPad Slim 3	Laptop	Lenovo	47990.00	24%	4.4	2100	Lenovo
20	Aspire Lite	Laptop	Acer	42990.00	28%	4.2	980	Acer St
21	Vivobook 15	Laptop	ASUS	49990.00	20%	4.3	1800	ASUS S
22	HP 15s	Laptop	HP	51990.00	25%	4.4	3200	HP Sto
23	ThinkPad E14	Laptop	Lenovo	67990.00	19%	4.5	650	Lenovo
24	ExpertBook P1	Laptop	ASUS	47990.00	31%	4.3	625	ASUS S
25	MacBook Air M3	Laptop	Apple	99990.00	10%	4.8	4200	Apple S
26	MacBook Air M2	Laptop	Apple	89990.00	12%	4.8	5600	Apple S
27	OnePlus Bullets Z2	Earphones	OnePlus	1999.00	33%	4.3	25000	OnePlu
28	Realme Buds T300	Earphones	realme	2299.00	42%	4.3	19000	Realme
29	boAt Airdopes 141	Earbuds	boAt	1299.00	57%	4.1	90000	Imagin
30	JBL Wave Buds	Earbuds	JBL	2499.00	38%	4.2	15000	JBL St
31	Sony WF-C500	Earbuds	Sony	4999.00	29%	4.4	11000	Sony S
32	Noise Buds VS104	Earbuds	Noise	999.00	50%	4.1	30000	Noise S

Total rows: 103 | Query complete 00:00:00.149 | CRLF | Ln 125, Col 1



Data Output | Messages | Notifications

Showing rows: 1 to 103 | Page No: 1 of 1

	oduct_name aracter varying (255)	product_category character varying (100)	brand character varying (100)	price numeric (10,2)	discount character varying (10)	rating numeric (2,1)	review_count integer	seller character
32	oise Buds VS104	Earbuds	Noise	999.00	50%	4.1	30000	Noise Sto
33	pple Watch SE	Smartwatch	Apple	29900.00	15%	4.6	8500	Apple Sto
34	amsung Galaxy Watch6	Smartwatch	Samsung	24999.00	30%	4.5	7200	Samsung
35	oise ColorFit Pro	Smartwatch	Noise	2999.00	40%	4.2	18000	Noise Sto
36	oAt Wave Sigma	Smartwatch	boAt	1299.00	55%	4.1	35000	Imagine S
37	re-Boltt Phoenix	Smartwatch	Fire-Boltt	1499.00	60%	4.2	28000	Fire-Boltt
38	mazfit Bip 5	Smartwatch	Amazfit	5499.00	25%	4.3	4500	Amazfit S
39	edmi Watch 5	Smartwatch	Redmi	6999.00	20%	4.4	3800	Redmi Sto
40	astrack Limitless	Smartwatch	Fastrack	3499.00	30%	4.2	6200	Fastrack S
41	enovo L24e Monitor	Monitor	Lenovo	11999.00	20%	4.3	2100	Lenovo St
42	amsung 24 Inch Monitor	Monitor	Samsung	10999.00	25%	4.4	3500	Samsung
43	G 24MP60G	Monitor	LG	12999.00	18%	4.5	2800	LG Store
44	cer EK220Q	Monitor	Acer	8999.00	30%	4.2	1900	Acer Store
45	SUS VY249HE	Monitor	ASUS	11990.00	22%	4.4	1500	ASUS Sto
46	P M22f	Monitor	HP	10990.00	20%	4.3	1200	HP Store
47	ell SE2422H	Monitor	Dell	12499.00	15%	4.4	900	Dell Store
48	ogitech K120	Keyboard	Logitech	799.00	20%	4.5	25000	Logitech S

Total rows: 103 | Query complete 00:00:00.149 | CRLF | Ln 125, Col 1

Data Output | Messages | Notifications

Showing rows: 1 to 103 | Page No: 1 of 1

	product_name character varying (255)	product_category character varying (100)	brand character varying (100)	price numeric (10,2)	discount character varying (10)	rating numeric (2,1)	review_count integer	seller charact
47	Dell SE2422H	Monitor	Dell	12499.00	15%	4.4	900	Dell St
48	Logitech K120	Keyboard	Logitech	799.00	20%	4.5	25000	Logitec
49	HP K1500	Keyboard	HP	699.00	30%	4.2	9000	HP Sto
50	Dell KB216	Keyboard	Dell	899.00	18%	4.4	12000	Dell St
51	Redragon K552	Keyboard	Redragon	2999.00	25%	4.5	7500	Redrag
52	Logitech G213	Keyboard	Logitech	3999.00	20%	4.6	6800	Logitec
53	Logitech M185	Mouse	Logitech	799.00	20%	4.4	40000	Logitec
54	HP M100	Mouse	HP	499.00	30%	4.2	12000	HP Sto
55	Dell MS116	Mouse	Dell	599.00	25%	4.3	18000	Dell St
56	Razer DeathAdder Essential	Mouse	Razer	1999.00	35%	4.5	9500	Razer S
57	Logitech G102	Mouse	Logitech	1799.00	30%	4.6	15000	Logitec
58	Canon EOS 1500D	Camera	Canon	39999.00	20%	4.5	8500	Canon
59	Nikon D7500	Camera	Nikon	79999.00	15%	4.6	3200	Nikon S
60	Sony Alpha ZV-E10	Camera	Sony	59990.00	18%	4.5	5100	Sony S
61	Canon EOS R50	Camera	Canon	64999.00	12%	4.7	2700	Canon
62	GoPro HERO12	Camera	GoPro	39999.00	22%	4.6	6300	GoPro
63	Canon Pixma E3370	Printer	Canon	6999.00	25%	4.3	15000	Canon

Total rows: 103 | Query complete 00:00:00.149 | CRLF | Ln 125, Col 1

	product_name character varying (255)	product_category character varying (100)	brand character varying (100)	price numeric (10,2)	discount character varying (10)	rating numeric (2,1)	review_count integer	seller charact
63	Canon Pixma E3370	Printer	Canon	6999.00	25%	4.3	15000	Canon
64	HP DeskJet 2331	Printer	HP	4999.00	30%	4.2	22000	HP Sto
65	Epson L3250	Printer	Epson	13999.00	20%	4.5	18000	Epson
66	Brother DCP-T426W	Printer	Brother	13990.00	15%	4.4	6200	Brothe
67	HP LaserJet M1005	Printer	HP	18999.00	18%	4.3	8500	HP Sto
68	TP-Link Archer C6	Router	TP-Link	2999.00	25%	4.4	30000	TP-Link
69	TP-Link Archer C64	Router	TP-Link	2499.00	30%	4.3	16000	TP-Link
70	D-Link DIR-825	Router	D-Link	2499.00	20%	4.2	9000	TP-Link
71	Tenda AC10	Router	Tenda	1999.00	35%	4.1	7000	Tenda
72	Netgear R6120	Router	Netgear	3999.00	18%	4.3	5000	Netgea
73	Samsung T7 SSD 1TB	SSD	Samsung	8999.00	20%	4.7	12000	Samsu
74	WD My Passport 1TB	Hard Drive	Western Digital	5999.00	25%	4.5	18000	WD Sto
75	Seagate 1TB HDD	Hard Drive	Seagate	5499.00	22%	4.4	15000	Seagat
76	SanDisk Ultra 128GB	Memory Card	SanDisk	999.00	35%	4.5	50000	SanDis
77	Kingston 128GB Pendrive	Pendrive	Kingston	699.00	30%	4.4	28000	Kingst
78	JBL Flip 6	Speaker	JBL	10999.00	20%	4.6	12000	JBL St
79	boAt Stone 350	Speaker	boAt	1499.00	45%	4.3	45000	Imagin

Total rows: 103 | Query complete 00:00:00.149 | CRLF | Ln 125, Col 1

Data Output | Messages | Notifications

Showing rows: 1 to 103 | Page No: 1 of 1

	product_name character varying (255)	product_category character varying (100)	brand character varying (100)	price numeric (10,2)	discount character varying (10)	rating numeric (2,1)	review_count integer	seller charact
79	boAt Stone 350	Speaker	boAt	1499.00	45%	4.3	45000	Imagin
80	Sony SRS-XB100	Speaker	Sony	4990.00	25%	4.5	8000	Sony S
81	Marshall Emberton II	Speaker	Marshall	15999.00	15%	4.7	4200	Marsha
82	Portronics SoundDrum	Speaker	Portronics	1999.00	40%	4.2	11000	Portror
83	Mi Power Bank 10000mAh	Power Bank	Xiaomi	999.00	20%	4.3	32000	Xiaomi
84	boAt Energyshroom PB	Power Bank	boAt	1299.00	35%	4.2	15000	Imagin
85	Ambrane 10000mAh	Power Bank	Ambrane	899.00	40%	4.3	22000	Ambra
86	Portronics Power PRO	Power Bank	Portronics	1099.00	30%	4.2	9000	Portror
87	Samsung 25W Charger	Charger	Samsung	1299.00	15%	4.5	19000	Samsu
88	OnePlus 80W Charger	Charger	OnePlus	1999.00	20%	4.6	11000	OnePlu
89	boAt 20W Charger	Charger	boAt	699.00	35%	4.2	18000	Imagin
90	HP Laptop Backpack	Bag	HP	1499.00	30%	4.4	8500	HP Sto
91	Lenovo Laptop Backpack	Bag	Lenovo	1299.00	25%	4.3	6200	Lenovo
92	American Tourister Laptop Bag	Bag	American Tourister	1999.00	20%	4.5	12000	AT Sto
93	Xiaomi Smart TV 43	Smart TV	Xiaomi	29999.00	25%	4.4	8500	Xiaomi
94	Samsung Crystal 4K 43	Smart TV	Samsung	34999.00	20%	4.5	9200	Samsu
95	LG 43 4K Smart TV	Smart TV	LG	32999.00	22%	4.4	7800	LG Sto

Data Output | Messages | Notifications

Showing rows: 1 to 103 | Page No: 1 of 1

	product_name character varying (255)	product_category character varying (100)	brand character varying (100)	price numeric (10,2)	discount character varying (10)	rating numeric (2,1)	review_count integer	seller charact
87	Samsung 25W Charger	Charger	Samsung	1299.00	15%	4.5	19000	Samsu
88	OnePlus 80W Charger	Charger	OnePlus	1999.00	20%	4.6	11000	OnePlu
89	boAt 20W Charger	Charger	boAt	699.00	35%	4.2	18000	Imagin
90	HP Laptop Backpack	Bag	HP	1499.00	30%	4.4	8500	HP Sto
91	Lenovo Laptop Backpack	Bag	Lenovo	1299.00	25%	4.3	6200	Lenovo
92	American Tourister Laptop Bag	Bag	American Tourister	1999.00	20%	4.5	12000	AT Sto
93	Xiaomi Smart TV 43	Smart TV	Xiaomi	29999.00	25%	4.4	8500	Xiaomi
94	Samsung Crystal 4K 43	Smart TV	Samsung	34999.00	20%	4.5	9200	Samsu
95	LG 43 4K Smart TV	Smart TV	LG	32999.00	22%	4.4	7800	LG Sto
96	OnePlus 43 Y Series	Smart TV	OnePlus	24999.00	30%	4.3	5600	OnePlu
97	Sony Bravia 43	Smart TV	Sony	49999.00	15%	4.6	4200	Sony S
98	Philips Air Fryer 4.1L	Kitchen	Philips	4999.00	30%	4.4	18000	Philips
99	Prestige Air Fryer	Kitchen	Prestige	3999.00	35%	4.3	12000	Prestig
100	Bajaj Mixer Grinder	Kitchen	Bajaj	2999.00	25%	4.2	15000	Bajaj S
101	Philips Electric Kettle	Kitchen	Philips	1799.00	28%	4.4	21000	Philips
102	Prestige Induction Cooktop	Kitchen	Prestige	2499.00	30%	4.3	17500	Prestig
103	lahaba A-line Knee Length Dr...	Dress	lahaba	389.00	68%	3.5	2000	SellerG

Total rows: 103 | Query complete 00:00:00.149 | CRLF | Ln 125, Col 1

Data Output | Messages | Notifications

Showing rows: 1 to 103 | Page No: 1 of 1

	duct_category racter varying (100)	brand character varying (100)	price numeric (10,2)	discount character varying (10)	rating numeric (2,1)	review_count integer	seller character varying (255)	delivery_promise character varying (
1	dphones	Mivi	599.00	80%	3.9	129212	TBL Online	12-Oct
2	dphones	Mivi	699.00	80%	3.9	57094	TBL Online	8-Oct
3	dphones	boAt	1699.00	72%	4.2	3015	BUZZINDIA	9-Oct
4	dphones	JBL	2999.00	40%	4.4	45000	JBL Store	4-Oct
5	dphones	Noise	999.00	50%	4.1	30000	Noise Store	5-Oct
6	dphones	Sony	4490.00	25%	4.5	1800	Sony Store	4-Oct
7	bile	Samsung	31495.00	10%	4.3	2240	BTPLD	3-Oct
8	bile	Apple	69900.00	5%	4.6	8625	GRAHGOODS	4-Oct
9	bile	POCO	29999.00	54%	4.5	243	Flashtech Retail	3-Oct
10	bile	HMD	12999.00	35%	4.5	716	ECOMTAMSINRetail	3-Oct
11	bile	Nothing	44999.00	25%	4.5	6692	IndiFlashMart	30-Sep
12	bile	OPPO	47999.00	10%	4.4	676	XONIGHT E-Commerce	4-Oct
13	bile	vivo	46999.00	24%	4.5	967	Flashstar Commerce	27-Sep
14	bile	OnePlus	35000.00	15%	4.4	820	PETILANTE Online	4-Oct
15	bile	realme	13999.00	33%	4.3	95777	BTPLD	3-Oct
16	bile	LAVA	10499.00	19%	3.9	121	LAVA Retail	4-Oct
17	top	Samsung	65990.00	18%	4.4	820	RetailNet	5-Oct

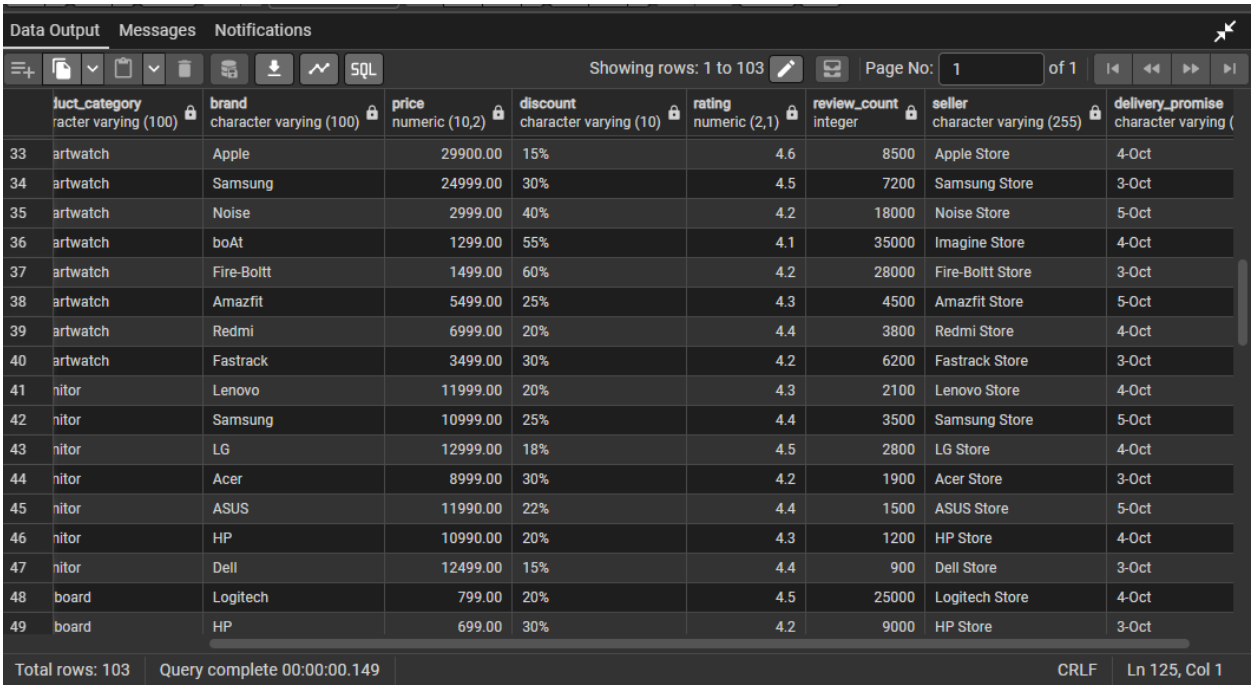
Total rows: 103 | Query complete 00:00:00.149 | CRLF | Ln 125, Col 1

Data Output | Messages | Notifications

Showing rows: 1 to 103 | Page No: 1 of 1

	duct_category racter varying (100)	brand character varying (100)	price numeric (10,2)	discount character varying (10)	rating numeric (2,1)	review_count integer	seller character varying (255)	delivery_promise character varying (
17	top	Samsung	65990.00	18%	4.4	820	RetailNet	5-Oct
18	top	Dell	54990.00	22%	4.3	1250	Dell Store	4-Oct
19	top	Lenovo	47990.00	24%	4.4	2100	Lenovo Store	3-Oct
20	top	Acer	42990.00	28%	4.2	980	Acer Store	5-Oct
21	top	ASUS	49990.00	20%	4.3	1800	ASUS Store	4-Oct
22	top	HP	51990.00	25%	4.4	3200	HP Store	3-Oct
23	top	Lenovo	67990.00	19%	4.5	650	Lenovo Store	5-Oct
24	top	ASUS	47990.00	31%	4.3	625	ASUS Store	4-Oct
25	top	Apple	99990.00	10%	4.8	4200	Apple Store	4-Oct
26	top	Apple	89990.00	12%	4.8	5600	Apple Store	3-Oct
27	phones	OnePlus	1999.00	33%	4.3	25000	OnePlus Store	4-Oct
28	phones	realme	2299.00	42%	4.3	19000	Realme Store	3-Oct
29	buds	boAt	1299.00	57%	4.1	90000	Imagine Store	4-Oct
30	buds	JBL	2499.00	38%	4.2	15000	JBL Store	5-Oct
31	buds	Sony	4999.00	29%	4.4	11000	Sony Store	4-Oct
32	buds	Noise	999.00	50%	4.1	30000	Noise Store	3-Oct
33	artwatch	Apple	29900.00	15%	4.6	8500	Apple Store	4-Oct

Total rows: 103 | Query complete 00:00:00.149 | CRLF | Ln 125, Col 1

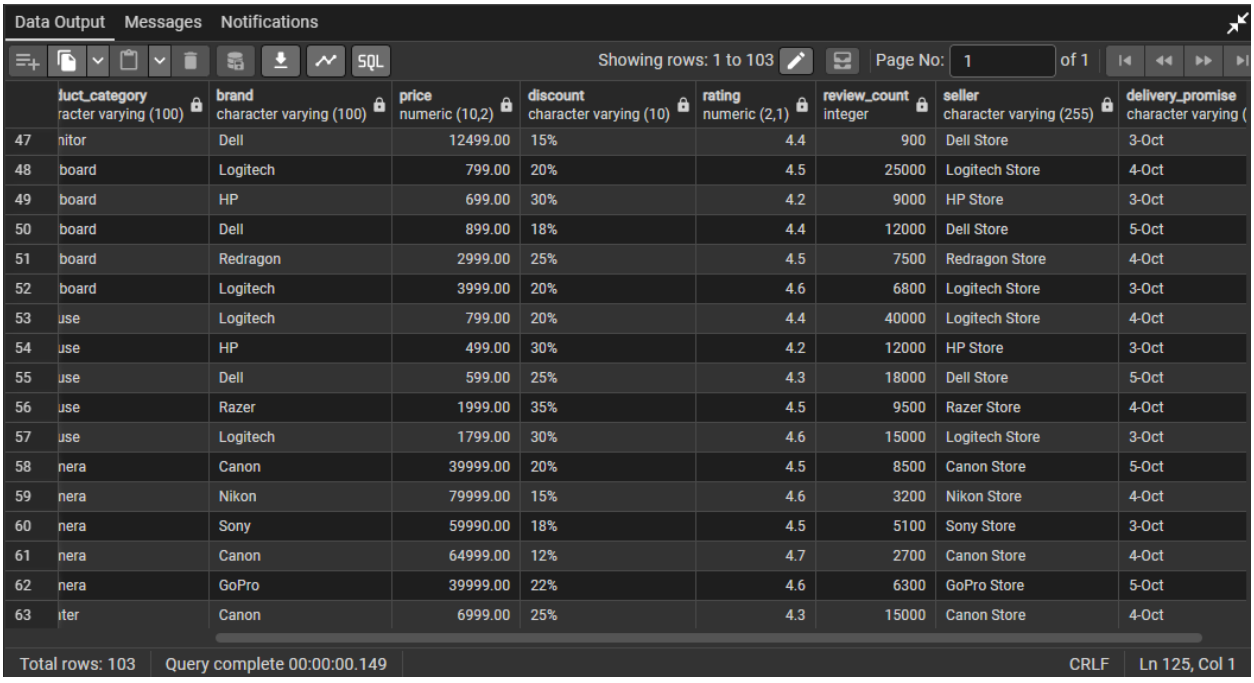


Data Output | Messages | Notifications

Showing rows: 1 to 103 | Page No: 1 of 1

	duct_category racter varying (100)	brand character varying (100)	price numeric (10,2)	discount character varying (10)	rating numeric (2,1)	review_count integer	seller character varying (255)	delivery_promise character varying (
33	artwatch	Apple	29900.00	15%	4.6	8500	Apple Store	4-Oct
34	artwatch	Samsung	24999.00	30%	4.5	7200	Samsung Store	3-Oct
35	artwatch	Noise	2999.00	40%	4.2	18000	Noise Store	5-Oct
36	artwatch	boAt	1299.00	55%	4.1	35000	Imagine Store	4-Oct
37	artwatch	Fire-Boltt	1499.00	60%	4.2	28000	Fire-Boltt Store	3-Oct
38	artwatch	Amazfit	5499.00	25%	4.3	4500	Amazfit Store	5-Oct
39	artwatch	Redmi	6999.00	20%	4.4	3800	Redmi Store	4-Oct
40	artwatch	Fastrack	3499.00	30%	4.2	6200	Fastrack Store	3-Oct
41	nitor	Lenovo	11999.00	20%	4.3	2100	Lenovo Store	4-Oct
42	nitor	Samsung	10999.00	25%	4.4	3500	Samsung Store	5-Oct
43	nitor	LG	12999.00	18%	4.5	2800	LG Store	4-Oct
44	nitor	Acer	8999.00	30%	4.2	1900	Acer Store	3-Oct
45	nitor	ASUS	11990.00	22%	4.4	1500	ASUS Store	5-Oct
46	nitor	HP	10990.00	20%	4.3	1200	HP Store	4-Oct
47	nitor	Dell	12499.00	15%	4.4	900	Dell Store	3-Oct
48	board	Logitech	799.00	20%	4.5	25000	Logitech Store	4-Oct
49	board	HP	699.00	30%	4.2	9000	HP Store	3-Oct

Total rows: 103 | Query complete 00:00:00.149 | CRLF | Ln 125, Col 1

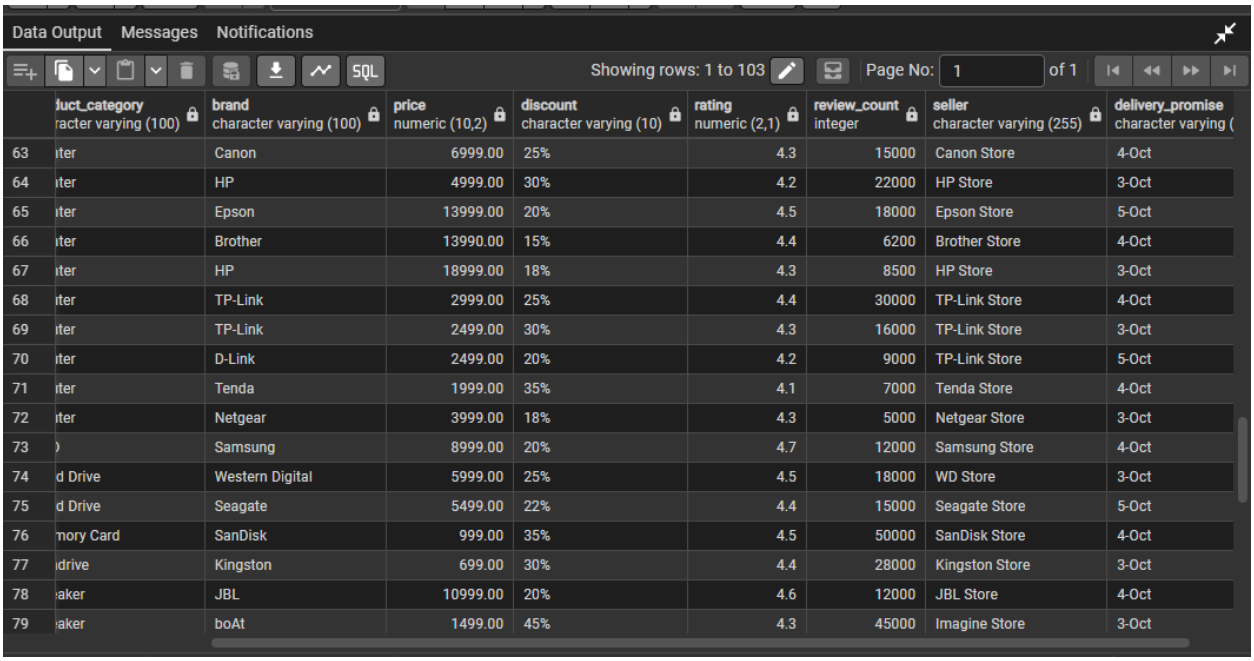


Data Output | Messages | Notifications

Showing rows: 1 to 103 | Page No: 1 of 1

	duct_category racter varying (100)	brand character varying (100)	price numeric (10,2)	discount character varying (10)	rating numeric (2,1)	review_count integer	seller character varying (255)	delivery_promise character varying (
47	nitor	Dell	12499.00	15%	4.4	900	Dell Store	3-Oct
48	board	Logitech	799.00	20%	4.5	25000	Logitech Store	4-Oct
49	board	HP	699.00	30%	4.2	9000	HP Store	3-Oct
50	board	Dell	899.00	18%	4.4	12000	Dell Store	5-Oct
51	board	Redragon	2999.00	25%	4.5	7500	Redragon Store	4-Oct
52	board	Logitech	3999.00	20%	4.6	6800	Logitech Store	3-Oct
53	use	Logitech	799.00	20%	4.4	40000	Logitech Store	4-Oct
54	use	HP	499.00	30%	4.2	12000	HP Store	3-Oct
55	use	Dell	599.00	25%	4.3	18000	Dell Store	5-Oct
56	use	Razer	1999.00	35%	4.5	9500	Razer Store	4-Oct
57	use	Logitech	1799.00	30%	4.6	15000	Logitech Store	3-Oct
58	nera	Canon	39999.00	20%	4.5	8500	Canon Store	5-Oct
59	nera	Nikon	79999.00	15%	4.6	3200	Nikon Store	4-Oct
60	nera	Sony	59990.00	18%	4.5	5100	Sony Store	3-Oct
61	nera	Canon	64999.00	12%	4.7	2700	Canon Store	4-Oct
62	nera	GoPro	39999.00	22%	4.6	6300	GoPro Store	5-Oct
63	ıter	Canon	6999.00	25%	4.3	15000	Canon Store	4-Oct

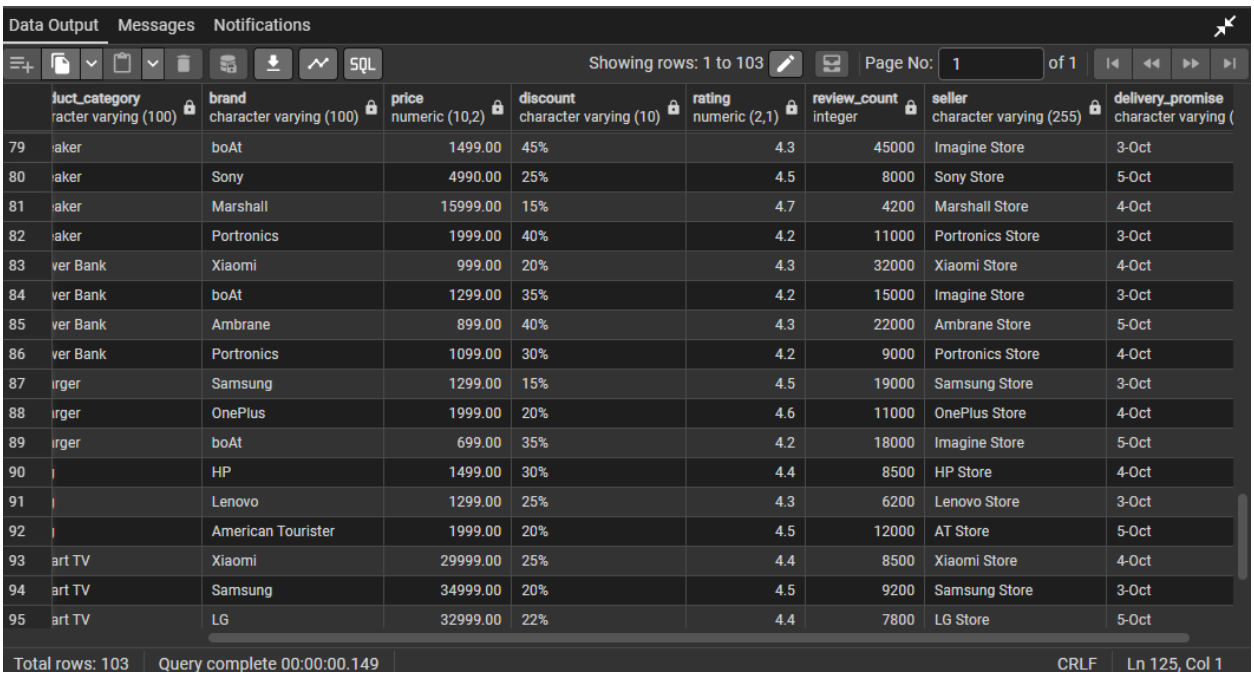
Total rows: 103 | Query complete 00:00:00.149 | CRLF | Ln 125, Col 1



Data Output | Messages | Notifications

Showing rows: 1 to 103 | Page No: 1 of 1

	duct_category racter varying (100)	brand character varying (100)	price numeric (10,2)	discount character varying (10)	rating numeric (2,1)	review_count integer	seller character varying (255)	delivery_promise character varying (
63	ıter	Canon	6999.00	25%	4.3	15000	Canon Store	4-Oct
64	ıter	HP	4999.00	30%	4.2	22000	HP Store	3-Oct
65	ıter	Epson	13999.00	20%	4.5	18000	Epson Store	5-Oct
66	ıter	Brother	13990.00	15%	4.4	6200	Brother Store	4-Oct
67	ıter	HP	18999.00	18%	4.3	8500	HP Store	3-Oct
68	ıter	TP-Link	2999.00	25%	4.4	30000	TP-Link Store	4-Oct
69	ıter	TP-Link	2499.00	30%	4.3	16000	TP-Link Store	3-Oct
70	ıter	D-Link	2499.00	20%	4.2	9000	TP-Link Store	5-Oct
71	ıter	Tenda	1999.00	35%	4.1	7000	Tenda Store	4-Oct
72	ıter	Netgear	3999.00	18%	4.3	5000	Netgear Store	3-Oct
73	)	Samsung	8999.00	20%	4.7	12000	Samsung Store	4-Oct
74	d Drive	Western Digital	5999.00	25%	4.5	18000	WD Store	3-Oct
75	d Drive	Seagate	5499.00	22%	4.4	15000	Seagate Store	5-Oct
76	mory Card	SanDisk	999.00	35%	4.5	50000	SanDisk Store	4-Oct
77	ıdrive	Kingston	699.00	30%	4.4	28000	Kingston Store	3-Oct
78	aker	JBL	10999.00	20%	4.6	12000	JBL Store	4-Oct
79	aker	boAt	1499.00	45%	4.3	45000	Imagine Store	3-Oct

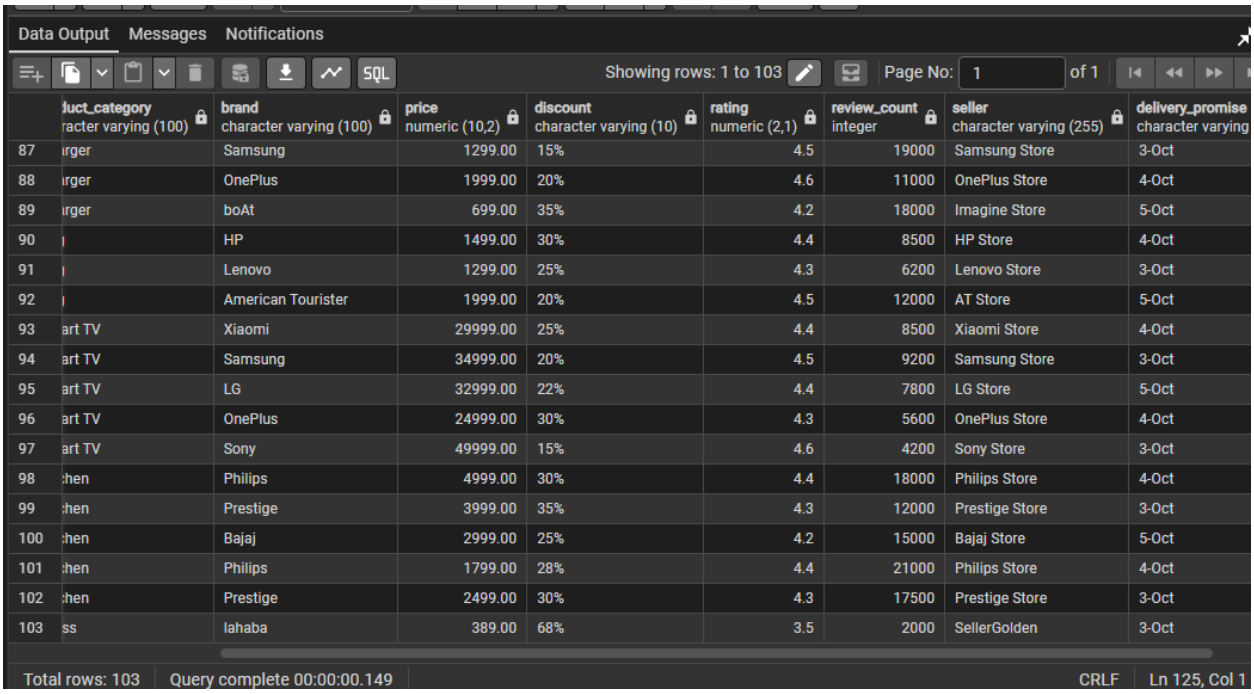


Data Output | Messages | Notifications

Showing rows: 1 to 103 | Page No: 1 of 1

	duct_category racter varying (100)	brand character varying (100)	price numeric (10,2)	discount character varying (10)	rating numeric (2,1)	review_count integer	seller character varying (255)	delivery_promise character varying (
79	aker	boAt	1499.00	45%	4.3	45000	Imagine Store	3-Oct
80	aker	Sony	4990.00	25%	4.5	8000	Sony Store	5-Oct
81	aker	Marshall	15999.00	15%	4.7	4200	Marshall Store	4-Oct
82	aker	Portronics	1999.00	40%	4.2	11000	Portronics Store	3-Oct
83	ver Bank	Xiaomi	999.00	20%	4.3	32000	Xiaomi Store	4-Oct
84	ver Bank	boAt	1299.00	35%	4.2	15000	Imagine Store	3-Oct
85	ver Bank	Ambrane	899.00	40%	4.3	22000	Ambrane Store	5-Oct
86	ver Bank	Portronics	1099.00	30%	4.2	9000	Portronics Store	4-Oct
87	ırger	Samsung	1299.00	15%	4.5	19000	Samsung Store	3-Oct
88	ırger	OnePlus	1999.00	20%	4.6	11000	OnePlus Store	4-Oct
89	ırger	boAt	699.00	35%	4.2	18000	Imagine Store	5-Oct
90	)	HP	1499.00	30%	4.4	8500	HP Store	4-Oct
91	)	Lenovo	1299.00	25%	4.3	6200	Lenovo Store	3-Oct
92	)	American Tourister	1999.00	20%	4.5	12000	AT Store	5-Oct
93	art TV	Xiaomi	29999.00	25%	4.4	8500	Xiaomi Store	4-Oct
94	art TV	Samsung	34999.00	20%	4.5	9200	Samsung Store	3-Oct
95	art TV	LG	32999.00	22%	4.4	7800	LG Store	5-Oct

Total rows: 103 | Query complete 00:00:00.149 | CRLF | Ln 125, Col 1



Data Output | Messages | Notifications

Showing rows: 1 to 103 | Page No: 1 of 1

	luct_category racter varying (100)	brand character varying (100)	price numeric (10,2)	discount character varying (10)	rating numeric (2,1)	review_count integer	seller character varying (255)	delivery_promise character varying
87	irger	Samsung	1299.00	15%	4.5	19000	Samsung Store	3-Oct
88	irger	OnePlus	1999.00	20%	4.6	11000	OnePlus Store	4-Oct
89	irger	boAt	699.00	35%	4.2	18000	Imagine Store	5-Oct
90		HP	1499.00	30%	4.4	8500	HP Store	4-Oct
91		Lenovo	1299.00	25%	4.3	6200	Lenovo Store	3-Oct
92		American Tourister	1999.00	20%	4.5	12000	AT Store	5-Oct
93	art TV	Xiaomi	29999.00	25%	4.4	8500	Xiaomi Store	4-Oct
94	art TV	Samsung	34999.00	20%	4.5	9200	Samsung Store	3-Oct
95	art TV	LG	32999.00	22%	4.4	7800	LG Store	5-Oct
96	art TV	OnePlus	24999.00	30%	4.3	5600	OnePlus Store	4-Oct
97	art TV	Sony	49999.00	15%	4.6	4200	Sony Store	3-Oct
98	hen	Philips	4999.00	30%	4.4	18000	Philips Store	4-Oct
99	hen	Prestige	3999.00	35%	4.3	12000	Prestige Store	3-Oct
100	hen	Bajaj	2999.00	25%	4.2	15000	Bajaj Store	5-Oct
101	hen	Philips	1799.00	28%	4.4	21000	Philips Store	4-Oct
102	hen	Prestige	2499.00	30%	4.3	17500	Prestige Store	3-Oct
103	ss	lahaba	389.00	68%	3.5	2000	SellerGolden	3-Oct

Total rows: 103 | Query complete 00:00:00.149 | CRLF | Ln 125, Col 1

---

8. DATA CLEANING AND PREPROCESSING

Data cleaning is an important step before analysis. It helps improve the quality and consistency of the dataset.

The data preparation process includes:

- Checking column names and data types.

- Checking for missing values.

- Checking for duplicate records.

- Ensuring that price, rating, discount, and review count are represented appropriately.

- Checking product names and category values for consistency.

- Reviewing unavailable seller or delivery information without inventing missing values.

Only the cleaning steps actually performed on the dataset should be reported as completed.

Data cleaning and preprocessing is an important step in data analysis. It helps improve data quality and prepares the dataset for further analysis.

In this project, the collected Flipkart product dataset was checked before performing SQL analysis, Python EDA, and Power BI visualization.

8.1 Missing Values Checking

Missing values were checked in important columns such as product name, product category, brand, price, discount, rating, review count, seller, and delivery promise.

8.2 Duplicate Records Checking

The dataset was checked for duplicate product records to identify repeated entries and maintain data accuracy.

8.3 Data Type Checking

The data types of important columns were checked to ensure that the data was suitable for analysis. Price and rating were stored as numeric values, review count as an integer, and product names and categories as text.

8.4 Importance of Data Cleaning

These checks helped assess the quality of the dataset before further analysis and supported the preparation of data for SQL queries, Python analysis, and Power BI visualizations.

---

9. SQL ANALYSIS

SQL was used to query and analyse the product dataset stored in PostgreSQL.

The analysis focuses on:

- Comparing product prices across categories.

- Counting products associated with each brand.

- Examining product ratings and prices.

- Identifying products with higher review counts.

- Comparing discounts across categories.

- Comparing product counts across categories to study competition.

The SQL query outputs can be included as screenshots in the project report.

```sql
SELECT COUNT(*) AS total_products
FROM ecommerce_products1;

SELECT * FROM ecommerce_products1;
```

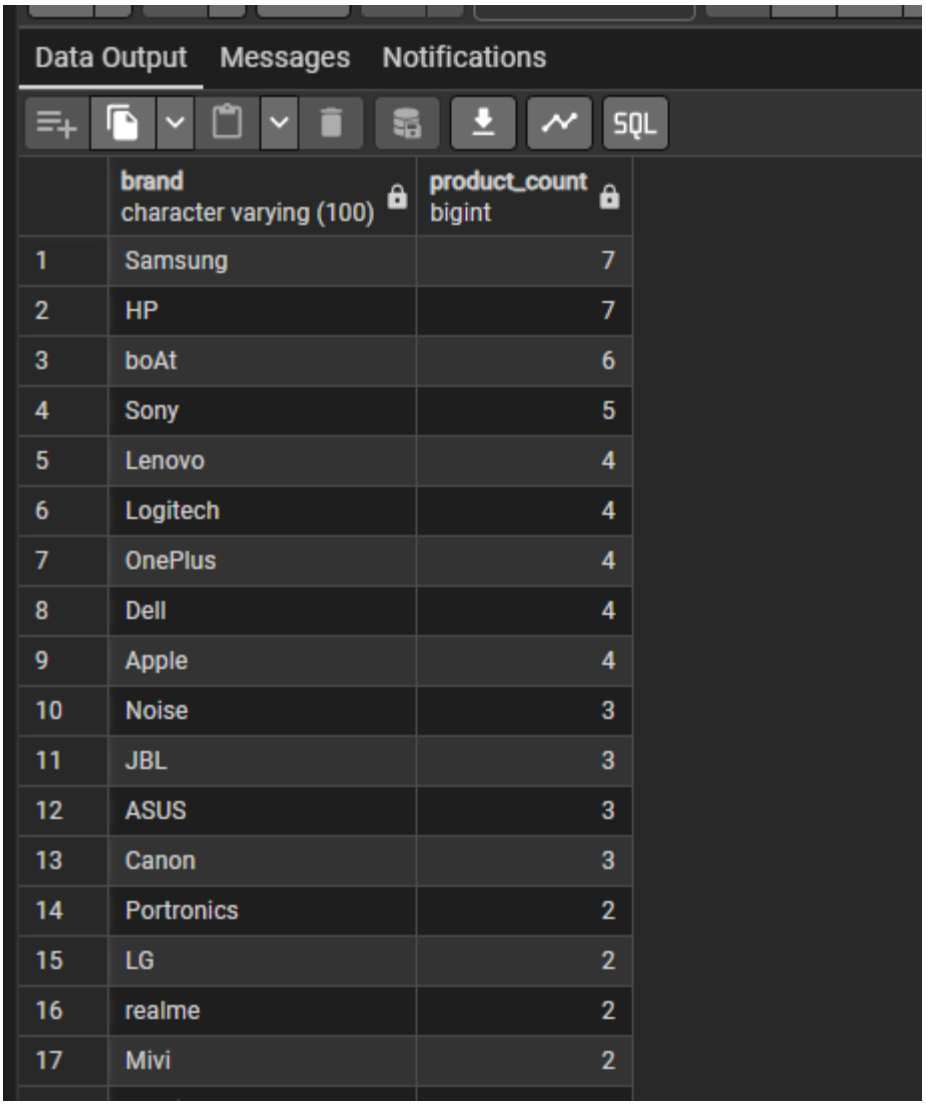
Data Output | Messages | Notifications

	total_products bigint
1	103

```sql
SELECT product_name, brand, COUNT(*) AS duplicate_count
FROM ecommerce_products1
GROUP BY product_name, brand
HAVING COUNT(*) > 1;
```

Data Output | Messages | Notifications

	product_name character varying (255)	brand character varying (100)	duplicate_count bigint



Data Output | Messages | Notifications

SQL

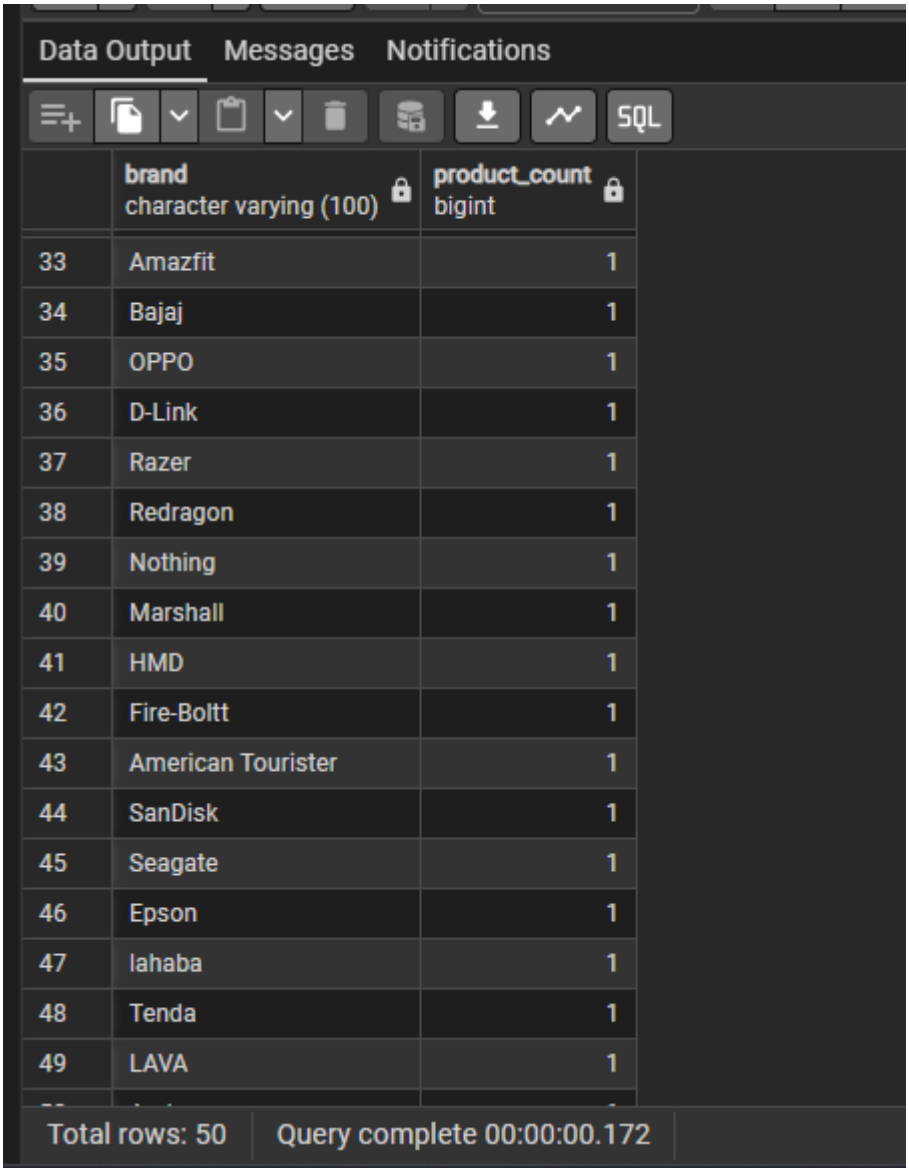
	brand character varying (100)	product_count bigint
1	Samsung	7
2	HP	7
3	boAt	6
4	Sony	5
5	Lenovo	4
6	Logitech	4
7	OnePlus	4
8	Dell	4
9	Apple	4
10	Noise	3
11	JBL	3
12	ASUS	3
13	Canon	3
14	Portronics	2
15	LG	2
16	realme	2
17	Mivi	2



Data Output | Messages | Notifications

	brand character varying (100)	product_count bigint
17	Mivi	2
18	TP-Link	2
19	Philips	2
20	Prestige	2
21	Xiaomi	2
22	Acer	2
23	Fastrack	1
24	Western Digital	1
25	Netgear	1
26	Kingston	1
27	vivo	1
28	GoPro	1
29	POCO	1
30	Redmi	1
31	Nikon	1
32	Brother	1
33	Amazfit	1
34	Bajaj	1

Total rows: 50 | Query complete 00:00:00.172



Data Output | Messages | Notifications

SQL

	brand character varying (100)	product_count bigint
33	Amazfit	1
34	Bajaj	1
35	OPPO	1
36	D-Link	1
37	Razer	1
38	Redragon	1
39	Nothing	1
40	Marshall	1
41	HMD	1
42	Fire-Boltt	1
43	American Tourister	1
44	SanDisk	1
45	Seagate	1
46	Epson	1
47	lahaba	1
48	Tenda	1
49	LAVA	1

Total rows: 50 | Query complete 00:00:00.172

	brand character varying (100)	product_count bigint
34	Bajaj	1
35	OPPO	1
36	D-Link	1
37	Razer	1
38	Redragon	1
39	Nothing	1
40	Marshall	1
41	HMD	1
42	Fire-Boltt	1
43	American Tourister	1
44	SanDisk	1
45	Seagate	1
46	Epson	1
47	lahaba	1
48	Tenda	1
49	LAVA	1
50	Ambrane	1

```
130
131  SELECT brand, COUNT(*) AS product_count
132  FROM ecommerce_products1
133  GROUP BY brand
134  ORDER BY product_count DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 50 | Page No: 1

	brand character varying (100)	product_count bigint
1	Samsung	7
2	HP	7
3	boAt	6
4	Sony	5
5	Lenovo	4
6	Logitech	4
7	OnePlus	4

Total rows: 50 | Query complete 00:00:00.166

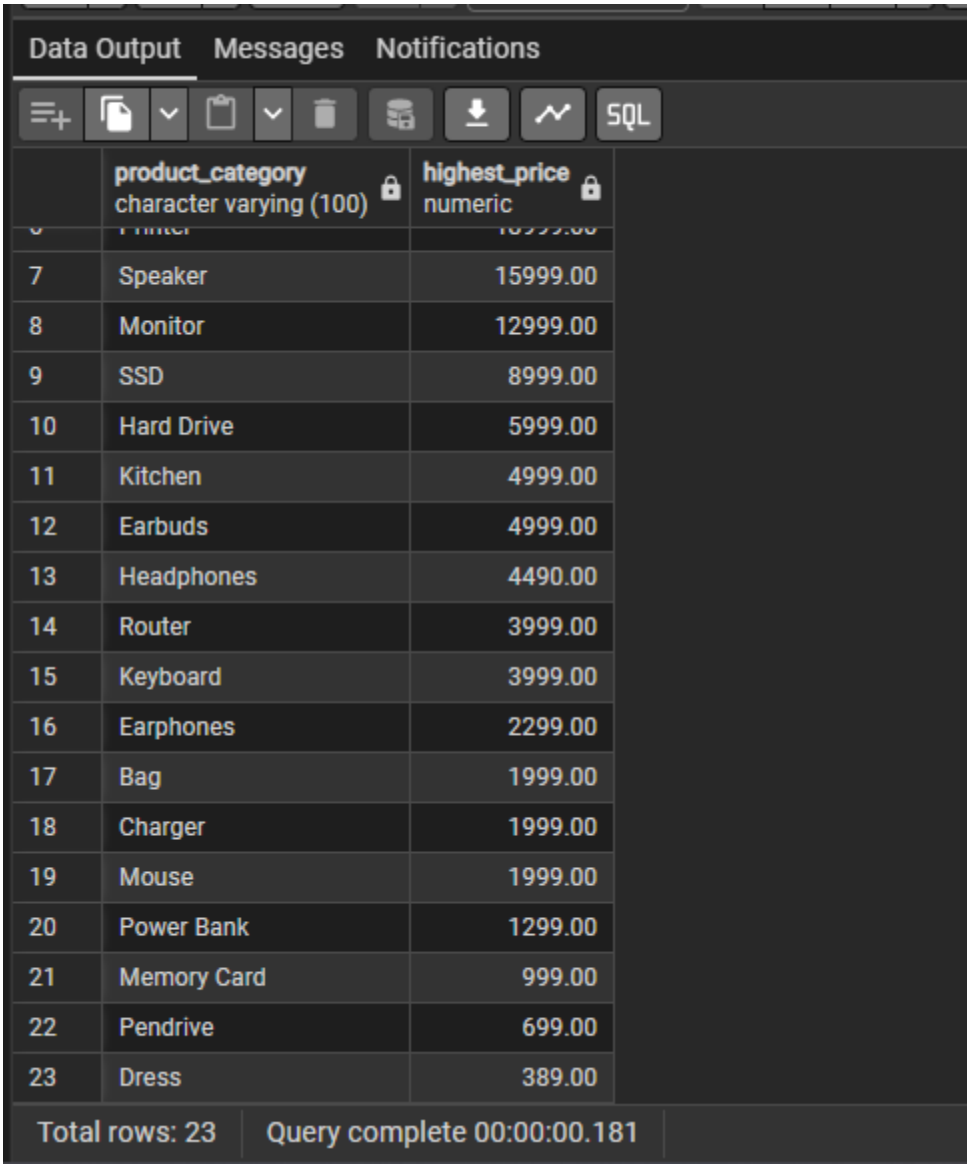
```sql
SELECT product_category, MAX(price) AS highest_price
FROM ecommerce_products1
GROUP BY product_category
ORDER BY highest_price DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 23

	product_category character varying (100)	highest_price numeric
1	Laptop	99990.00
2	Camera	79999.00
3	Mobile	69900.00
4	Smart TV	49999.00
5	Smartwatch	29900.00
6	Printer	18999.00
7	Speaker	15999.00

Total rows: 23 | Query complete 00:00:00.121



Data Output | Messages | Notifications

SQL

	product_category character varying (100)	highest_price numeric
7	Speaker	15999.00
8	Monitor	12999.00
9	SSD	8999.00
10	Hard Drive	5999.00
11	Kitchen	4999.00
12	Earbuds	4999.00
13	Headphones	4490.00
14	Router	3999.00
15	Keyboard	3999.00
16	Earphones	2299.00
17	Bag	1999.00
18	Charger	1999.00
19	Mouse	1999.00
20	Power Bank	1299.00
21	Memory Card	999.00
22	Pendrive	699.00
23	Dress	389.00

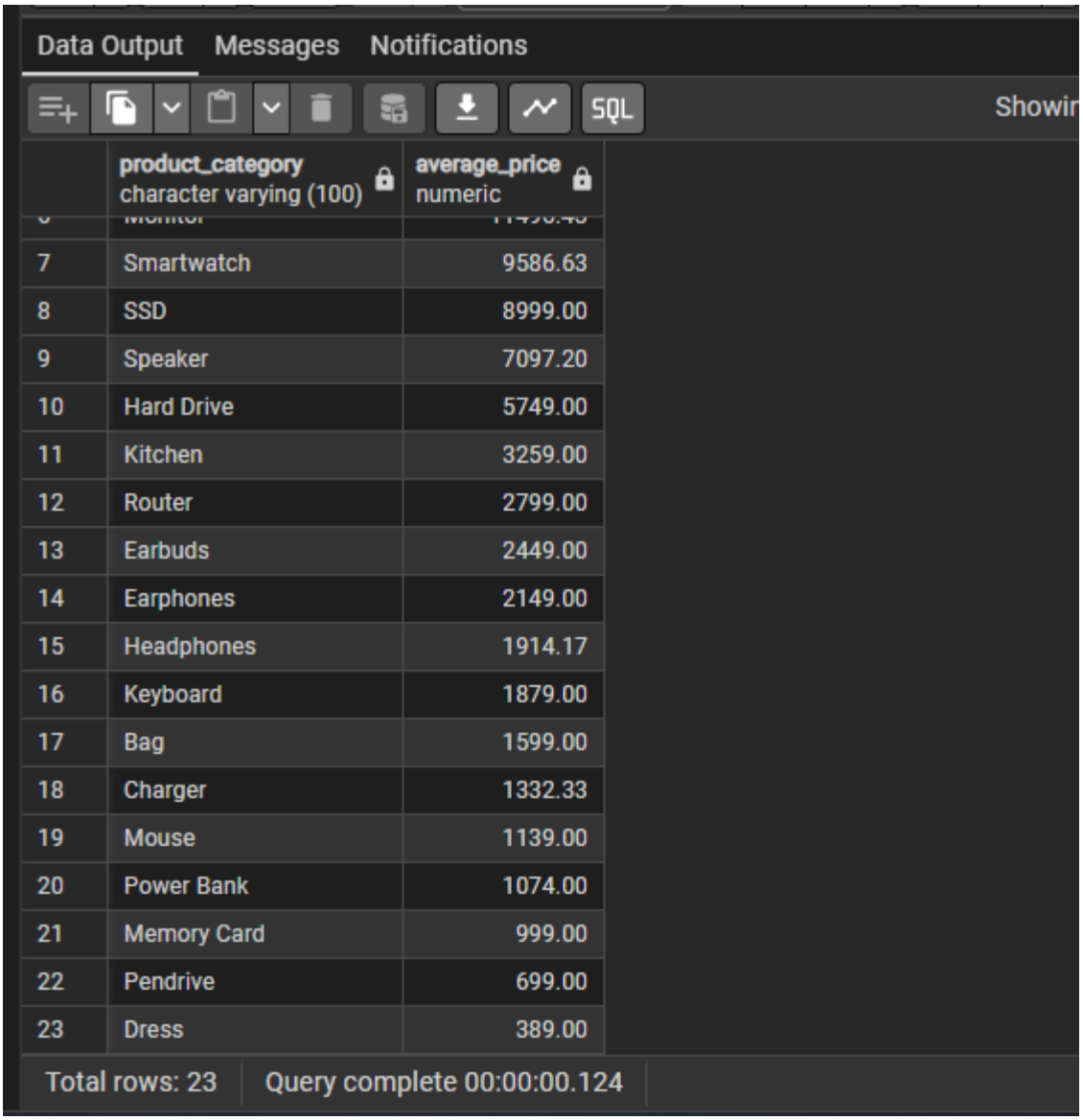
Total rows: 23 | Query complete 00:00:00.181

```
141  SELECT product_category,
142         ROUND(AVG(price), 2) AS average_price
143  FROM ecommerce_products1
144  GROUP BY product_category
145  ORDER BY average_price DESC;
```

Data Output | Messages | Notifications

SQL

	product_category character varying (100)	average_price numeric
1	Laptop	61990.00
2	Camera	56997.20
3	Smart TV	34599.00
4	Mobile	34388.80
5	Printer	11797.20
6	Monitor	11496.43
7	Smartwatch	9586.63



Data Output | Messages | Notifications

Showin

	product_category character varying (100)	average_price numeric
7	Smartwatch	9586.63
8	SSD	8999.00
9	Speaker	7097.20
10	Hard Drive	5749.00
11	Kitchen	3259.00
12	Router	2799.00
13	Earbuds	2449.00
14	Earphones	2149.00
15	Headphones	1914.17
16	Keyboard	1879.00
17	Bag	1599.00
18	Charger	1332.33
19	Mouse	1139.00
20	Power Bank	1074.00
21	Memory Card	999.00
22	Pendrive	699.00
23	Dress	389.00

Total rows: 23 | Query complete 00:00:00.124

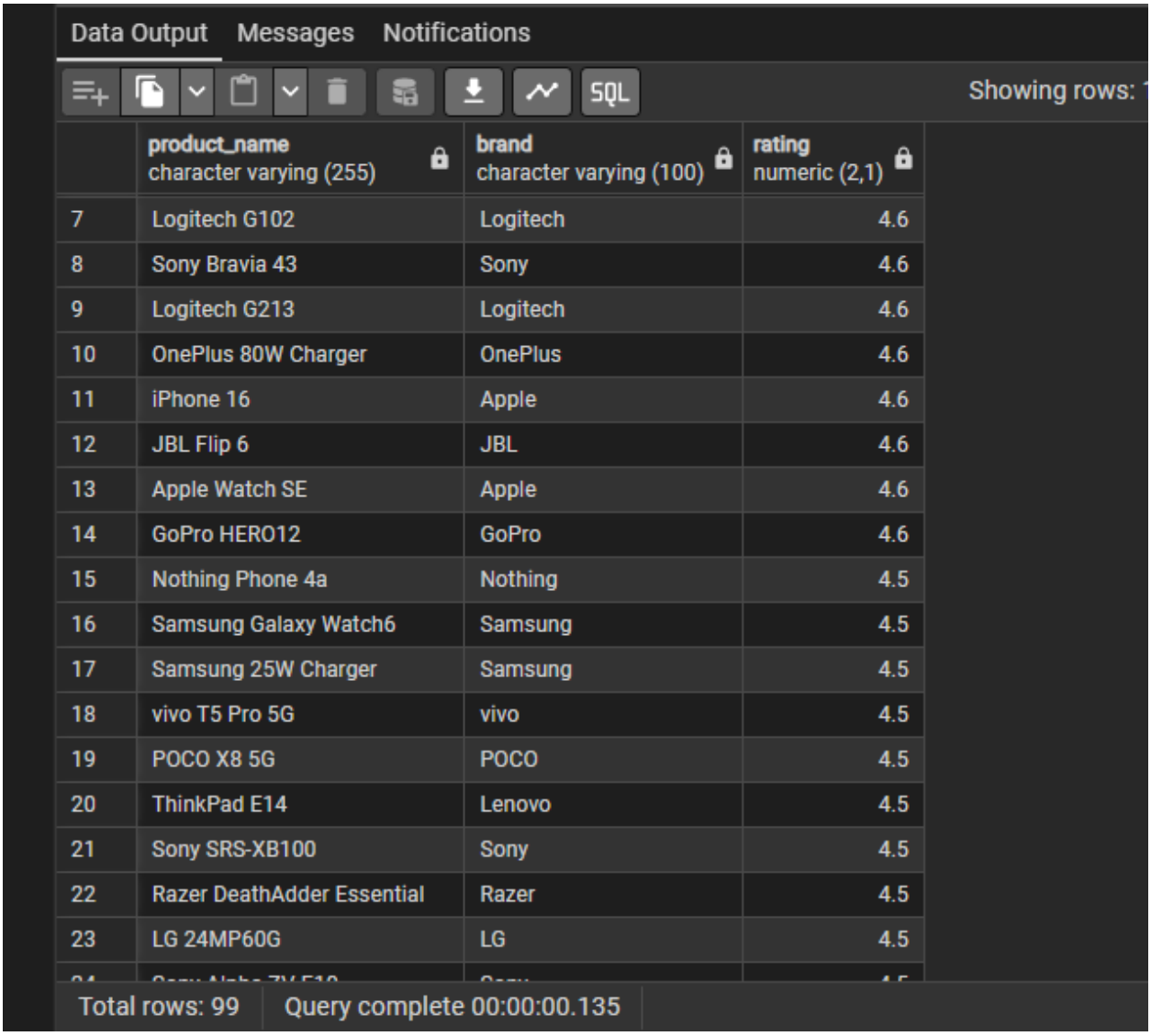
```
146
147  SELECT product_name, brand, rating
148  FROM ecommerce_products1
149  WHERE rating >= 4
150  ORDER BY rating DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 99 | Pag

	product_name character varying (255)	brand character varying (100)	rating numeric (2,1)
1	MacBook Air M2	Apple	4.8
2	MacBook Air M3	Apple	4.8
3	Marshall Emberton II	Marshall	4.7
4	Canon EOS R50	Canon	4.7
5	Samsung T7 SSD 1TB	Samsung	4.7
6	Nikon D7500	Nikon	4.6
7	Logitech G102	Logitech	4.6

Total rows: 99 | Query complete 00:00:00.164

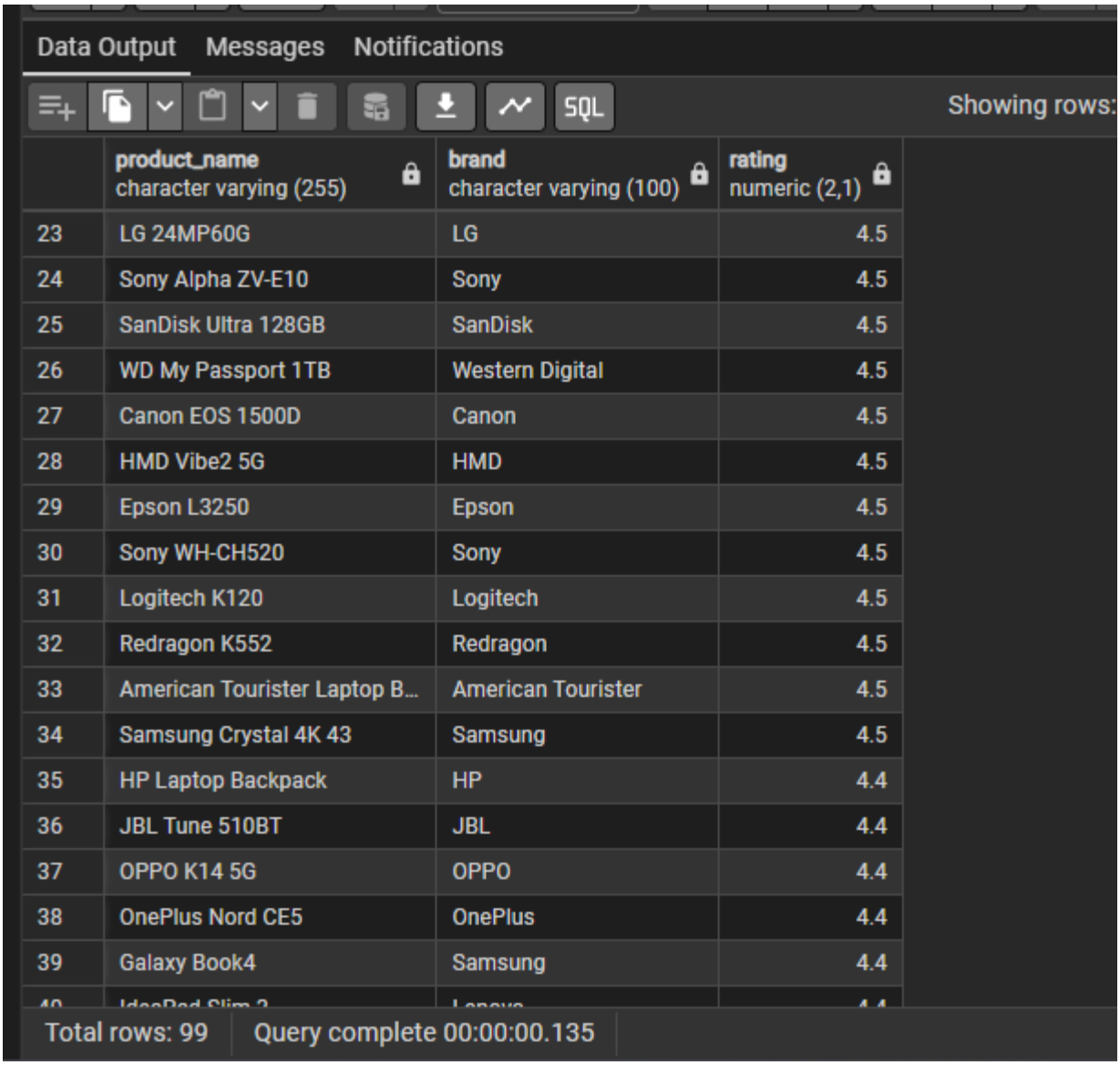


Data Output | Messages | Notifications

Showing rows:

	product_name character varying (255)	brand character varying (100)	rating numeric (2,1)
7	Logitech G102	Logitech	4.6
8	Sony Bravia 43	Sony	4.6
9	Logitech G213	Logitech	4.6
10	OnePlus 80W Charger	OnePlus	4.6
11	iPhone 16	Apple	4.6
12	JBL Flip 6	JBL	4.6
13	Apple Watch SE	Apple	4.6
14	GoPro HERO12	GoPro	4.6
15	Nothing Phone 4a	Nothing	4.5
16	Samsung Galaxy Watch6	Samsung	4.5
17	Samsung 25W Charger	Samsung	4.5
18	vivo T5 Pro 5G	vivo	4.5
19	POCO X8 5G	POCO	4.5
20	ThinkPad E14	Lenovo	4.5
21	Sony SRS-XB100	Sony	4.5
22	Razer DeathAdder Essential	Razer	4.5
23	LG 24MP60G	LG	4.5

Total rows: 99 | Query complete 00:00:00.135

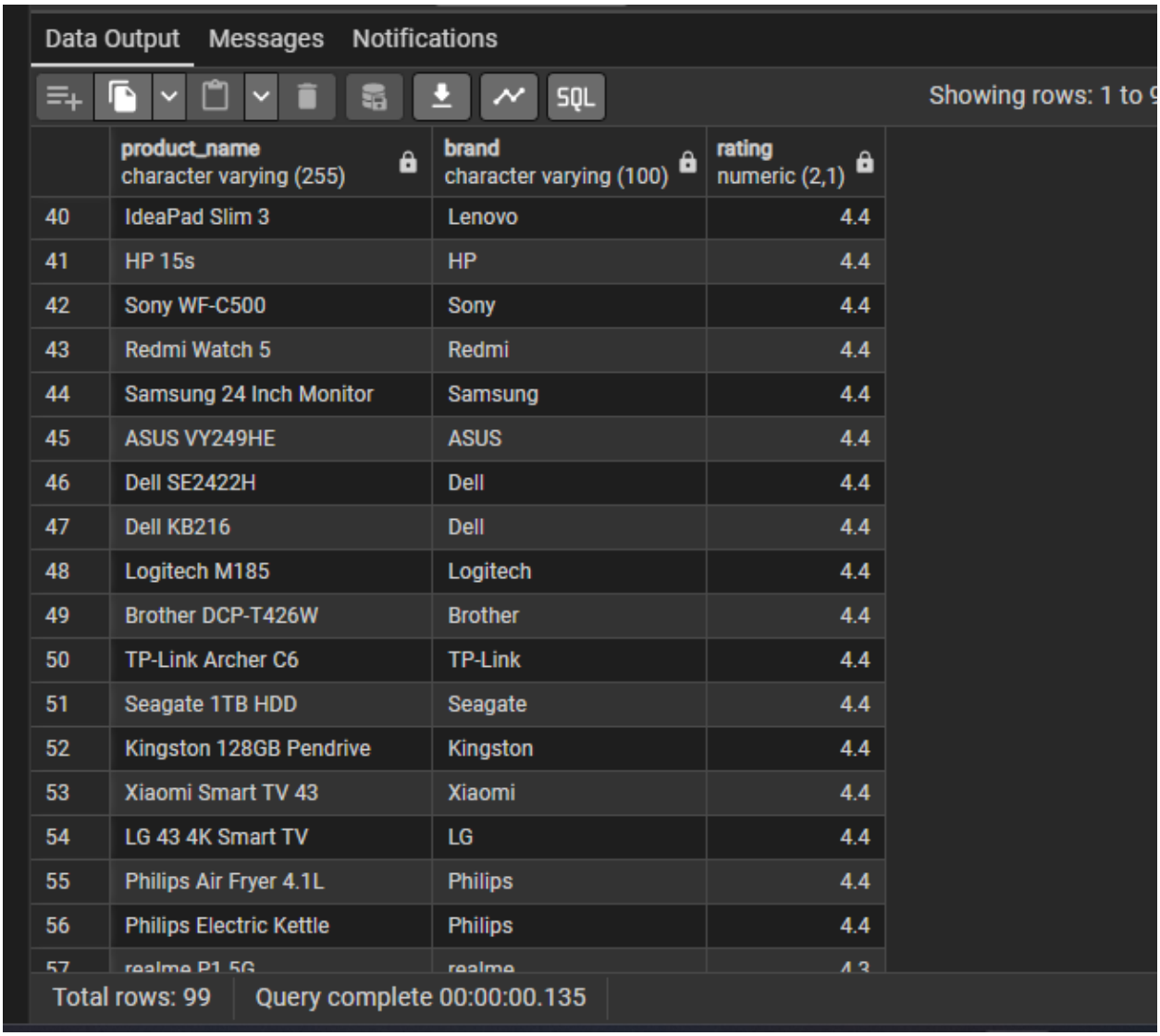


Data Output | Messages | Notifications

Showing rows:

	product_name character varying (255)	brand character varying (100)	rating numeric (2,1)
23	LG 24MP60G	LG	4.5
24	Sony Alpha ZV-E10	Sony	4.5
25	SanDisk Ultra 128GB	SanDisk	4.5
26	WD My Passport 1TB	Western Digital	4.5
27	Canon EOS 1500D	Canon	4.5
28	HMD Vibe2 5G	HMD	4.5
29	Epson L3250	Epson	4.5
30	Sony WH-CH520	Sony	4.5
31	Logitech K120	Logitech	4.5
32	Redragon K552	Redragon	4.5
33	American Tourister Laptop B...	American Tourister	4.5
34	Samsung Crystal 4K 43	Samsung	4.5
35	HP Laptop Backpack	HP	4.4
36	JBL Tune 510BT	JBL	4.4
37	OPPO K14 5G	OPPO	4.4
38	OnePlus Nord CE5	OnePlus	4.4
39	Galaxy Book4	Samsung	4.4

Total rows: 99 | Query complete 00:00:00.135

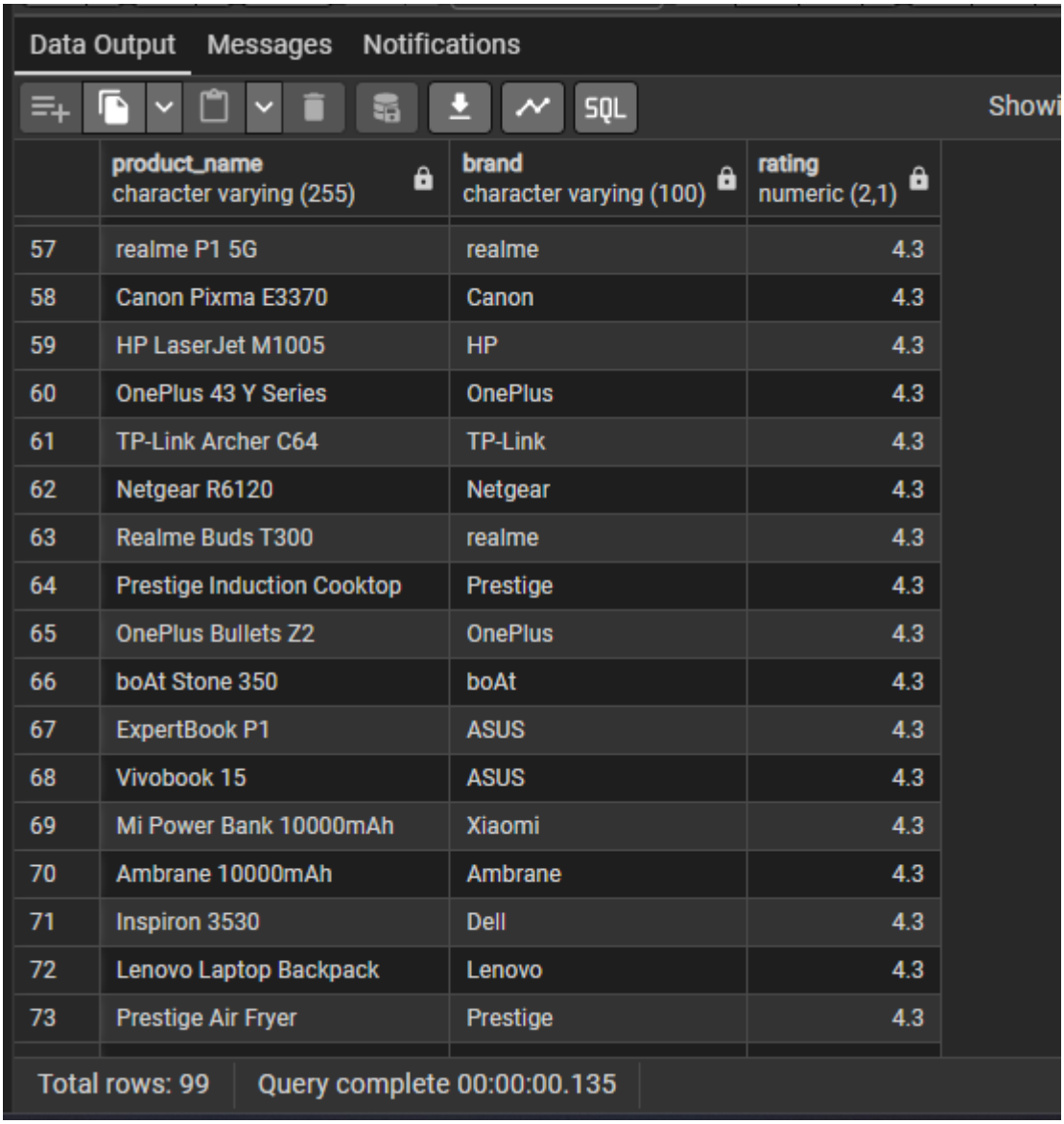


Data Output | Messages | Notifications

Showing rows: 1 to 9

	product_name character varying (255)	brand character varying (100)	rating numeric (2,1)
40	IdeaPad Slim 3	Lenovo	4.4
41	HP 15s	HP	4.4
42	Sony WF-C500	Sony	4.4
43	Redmi Watch 5	Redmi	4.4
44	Samsung 24 Inch Monitor	Samsung	4.4
45	ASUS VY249HE	ASUS	4.4
46	Dell SE2422H	Dell	4.4
47	Dell KB216	Dell	4.4
48	Logitech M185	Logitech	4.4
49	Brother DCP-T426W	Brother	4.4
50	TP-Link Archer C6	TP-Link	4.4
51	Seagate 1TB HDD	Seagate	4.4
52	Kingston 128GB Pendrive	Kingston	4.4
53	Xiaomi Smart TV 43	Xiaomi	4.4
54	LG 43 4K Smart TV	LG	4.4
55	Philips Air Fryer 4.1L	Philips	4.4
56	Philips Electric Kettle	Philips	4.4
57	realme P1 5G	realme	4.3

Total rows: 99 | Query complete 00:00:00.135

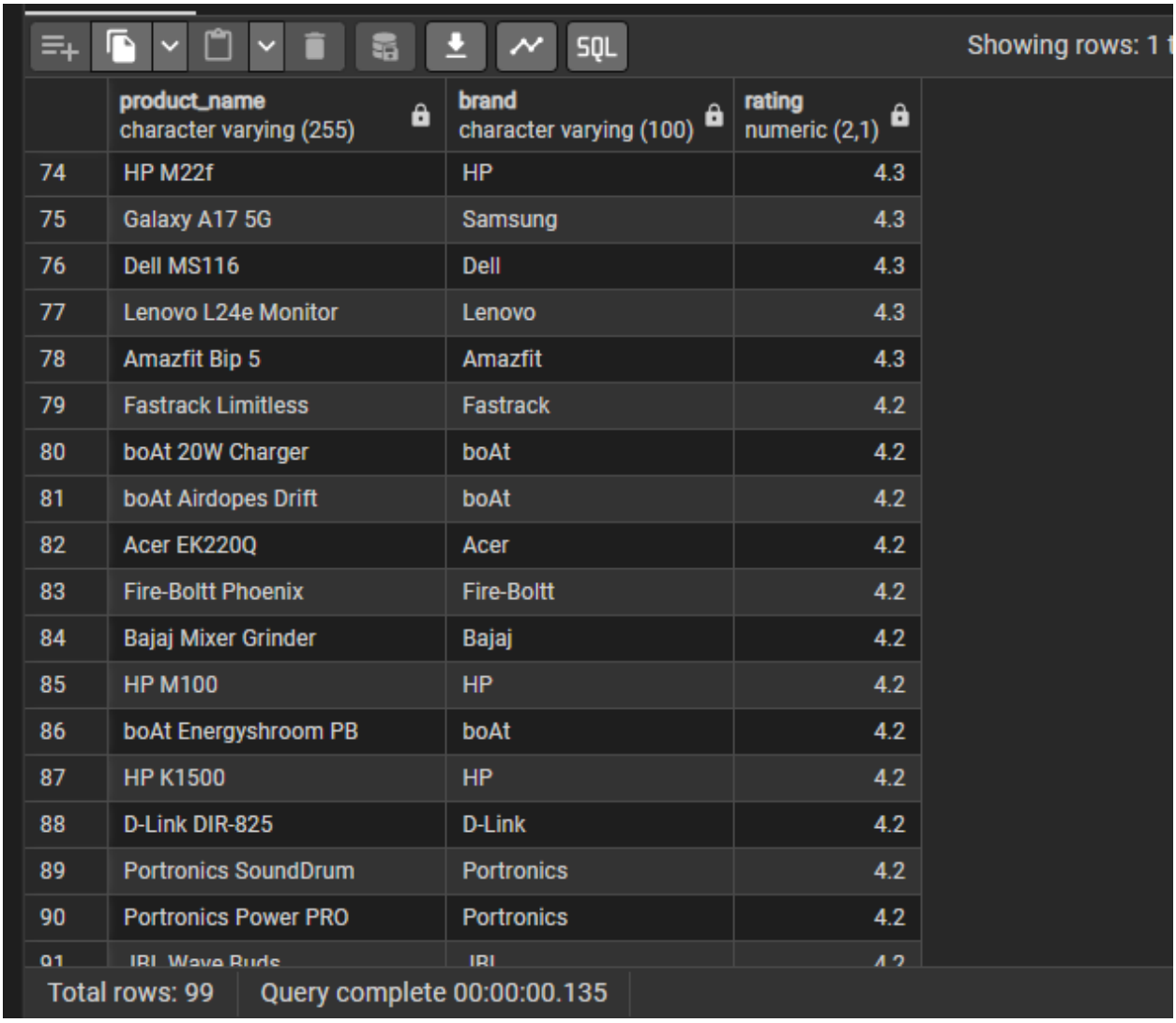


Data Output | Messages | Notifications

SQL | Showi

	product_name character varying (255)	brand character varying (100)	rating numeric (2,1)
57	realme P1 5G	realme	4.3
58	Canon Pixma E3370	Canon	4.3
59	HP LaserJet M1005	HP	4.3
60	OnePlus 43 Y Series	OnePlus	4.3
61	TP-Link Archer C64	TP-Link	4.3
62	Netgear R6120	Netgear	4.3
63	Realme Buds T300	realme	4.3
64	Prestige Induction Cooktop	Prestige	4.3
65	OnePlus Bullets Z2	OnePlus	4.3
66	boAt Stone 350	boAt	4.3
67	ExpertBook P1	ASUS	4.3
68	Vivobook 15	ASUS	4.3
69	Mi Power Bank 10000mAh	Xiaomi	4.3
70	Ambrane 10000mAh	Ambrane	4.3
71	Inspiron 3530	Dell	4.3
72	Lenovo Laptop Backpack	Lenovo	4.3
73	Prestige Air Fryer	Prestige	4.3

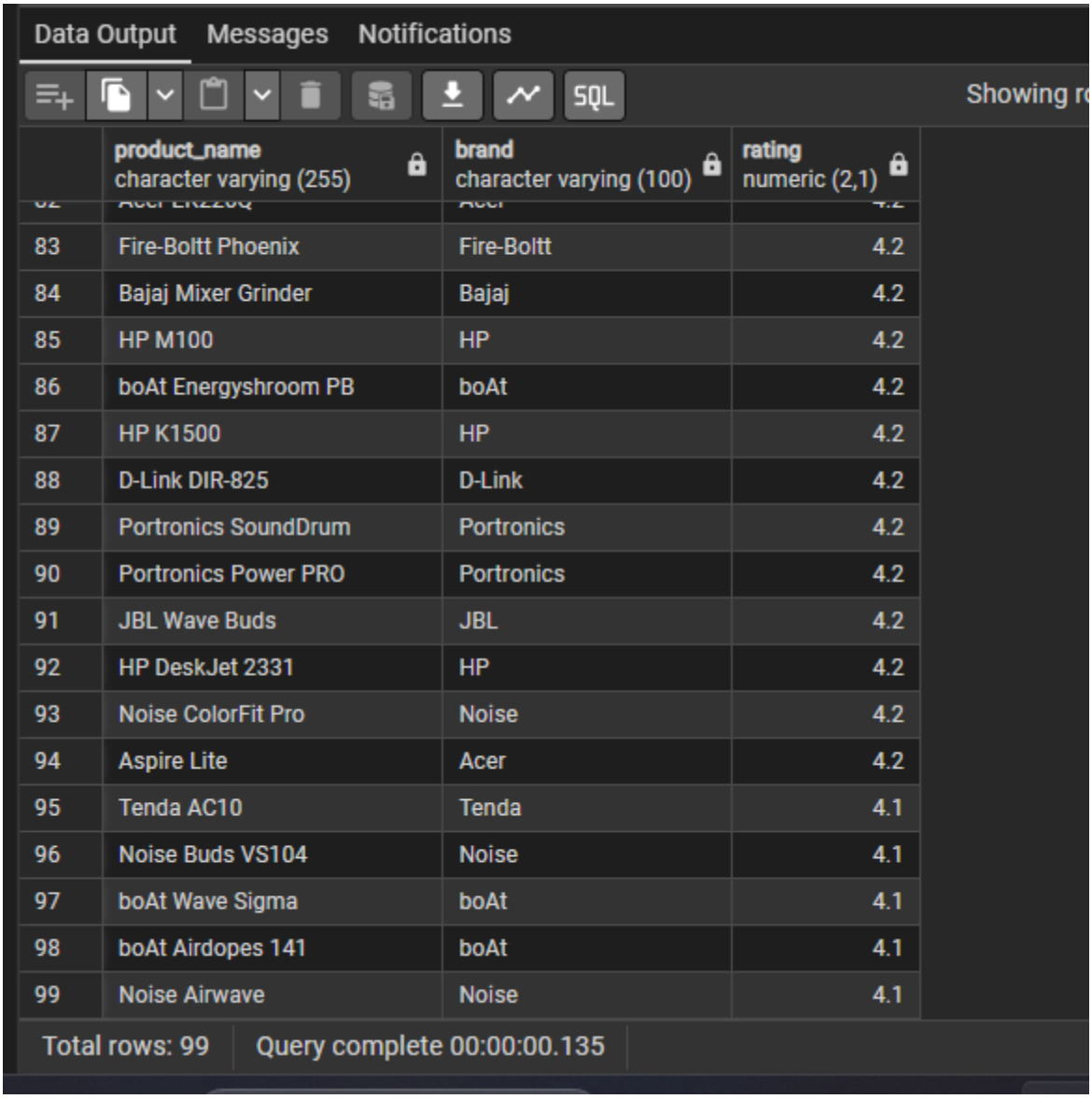
Total rows: 99 | Query complete 00:00:00.135



Showing rows: 1 t

	product_name character varying (255)	brand character varying (100)	rating numeric (2,1)
74	HP M22f	HP	4.3
75	Galaxy A17 5G	Samsung	4.3
76	Dell MS116	Dell	4.3
77	Lenovo L24e Monitor	Lenovo	4.3
78	Amazfit Bip 5	Amazfit	4.3
79	Fastrack Limitless	Fastrack	4.2
80	boAt 20W Charger	boAt	4.2
81	boAt Airdopes Drift	boAt	4.2
82	Acer EK220Q	Acer	4.2
83	Fire-Boltt Phoenix	Fire-Boltt	4.2
84	Bajaj Mixer Grinder	Bajaj	4.2
85	HP M100	HP	4.2
86	boAt Energyshroom PB	boAt	4.2
87	HP K1500	HP	4.2
88	D-Link DIR-825	D-Link	4.2
89	Portronics SoundDrum	Portronics	4.2
90	Portronics Power PRO	Portronics	4.2
91	JBL Wave Buds	JBL	4.2

Total rows: 99 | Query complete 00:00:00.135



Data Output | Messages | Notifications

Showing r

	product_name character varying (255)	brand character varying (100)	rating numeric (2,1)
83	Fire-Boltt Phoenix	Fire-Boltt	4.2
84	Bajaj Mixer Grinder	Bajaj	4.2
85	HP M100	HP	4.2
86	boAt Energyshroom PB	boAt	4.2
87	HP K1500	HP	4.2
88	D-Link DIR-825	D-Link	4.2
89	Portronics SoundDrum	Portronics	4.2
90	Portronics Power PRO	Portronics	4.2
91	JBL Wave Buds	JBL	4.2
92	HP DeskJet 2331	HP	4.2
93	Noise ColorFit Pro	Noise	4.2
94	Aspire Lite	Acer	4.2
95	Tenda AC10	Tenda	4.1
96	Noise Buds VS104	Noise	4.1
97	boAt Wave Sigma	boAt	4.1
98	boAt Airdopes 141	boAt	4.1
99	Noise Airwave	Noise	4.1

Total rows: 99 | Query complete 00:00:00.135

```
151
152  SELECT product_name, brand, review_count
153  FROM ecommerce_products1
154  ORDER BY review_count DESC
155  LIMIT 10;
156
```

Data Output | Messages | Notifications

Showi

	product_name character varying (255)	brand character varying (100)	review_count integer
1	Mivi Duopods K2	Mivi	129212
2	realme P1 5G	realme	95777
3	boAt Airdopes 141	boAt	90000
4	Mivi Duopods I6	Mivi	57094
5	SanDisk Ultra 128GB	SanDisk	50000
6	JBL Tune 510BT	JBL	45000
7	boAt Stone 350	boAt	45000
8	Logitech M185	Logitech	40000
9	boAt Wave Sigma	boAt	35000
10	Mi Power Bank 10000m...	Xiaomi	32000

Total rows: 10 | Query complete 00:00:00.171

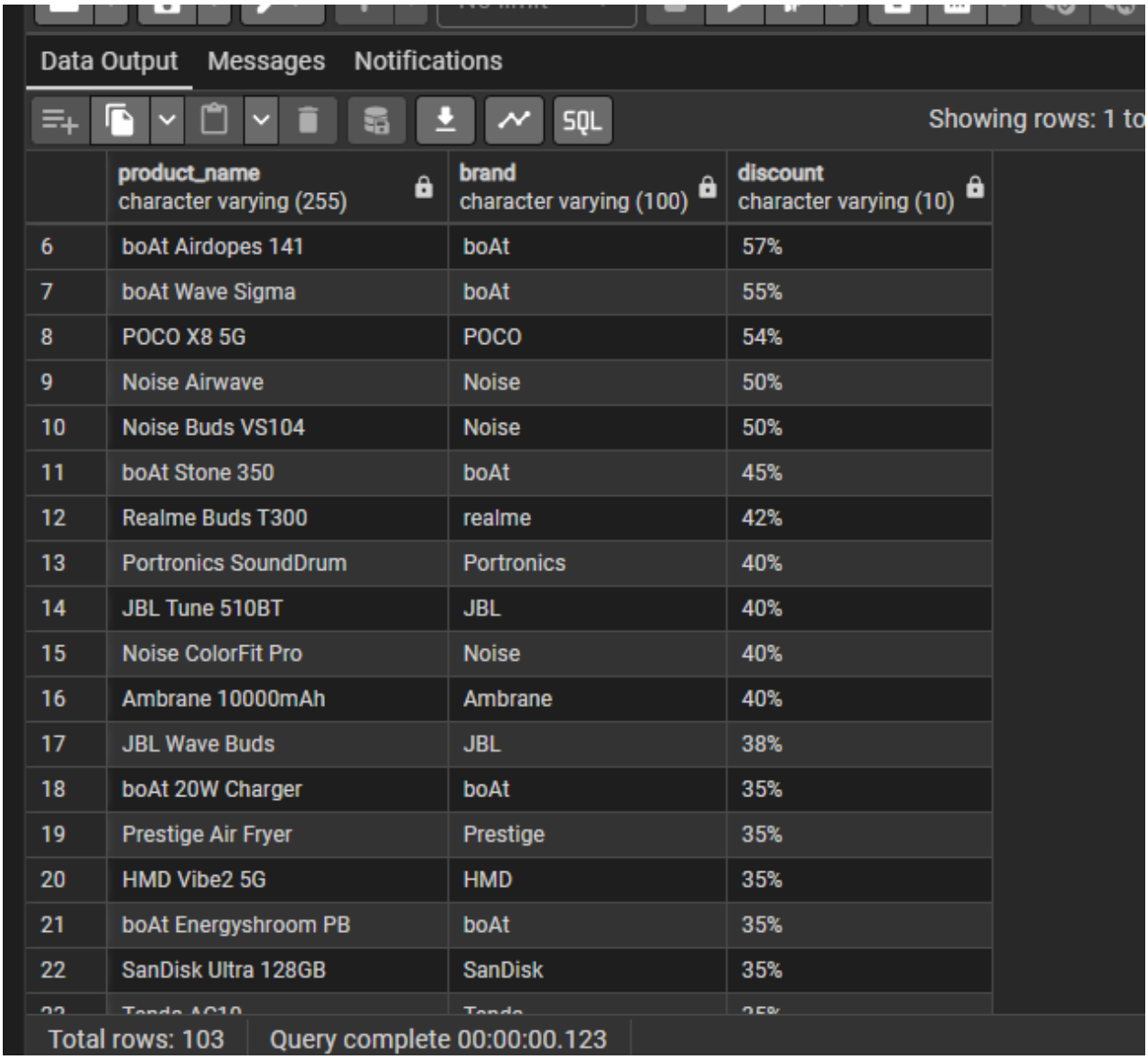
```sql
SELECT product_name, brand, discount
FROM ecommerce_products1
WHERE CAST(REPLACE(discount, '%', '') AS NUMERIC) > 0
ORDER BY CAST(REPLACE(discount, '%', '') AS NUMERIC) DESC;
```

Data Output | Messages | Notifications

Showing rows: 1

	product_name character varying (255)	brand character varying (100)	discount character varying (10)
1	Mivi Duopods K2	Mivi	80%
2	Mivi Duopods I6	Mivi	80%
3	boAt Airdopes Drift	boAt	72%
4	lahaba A-line Knee Length Dr...	lahaba	68%
5	Fire-Boltt Phoenix	Fire-Boltt	60%
6	boAt Airdopes 141	boAt	57%

Total rows: 103 | Query complete 00:00:00.123

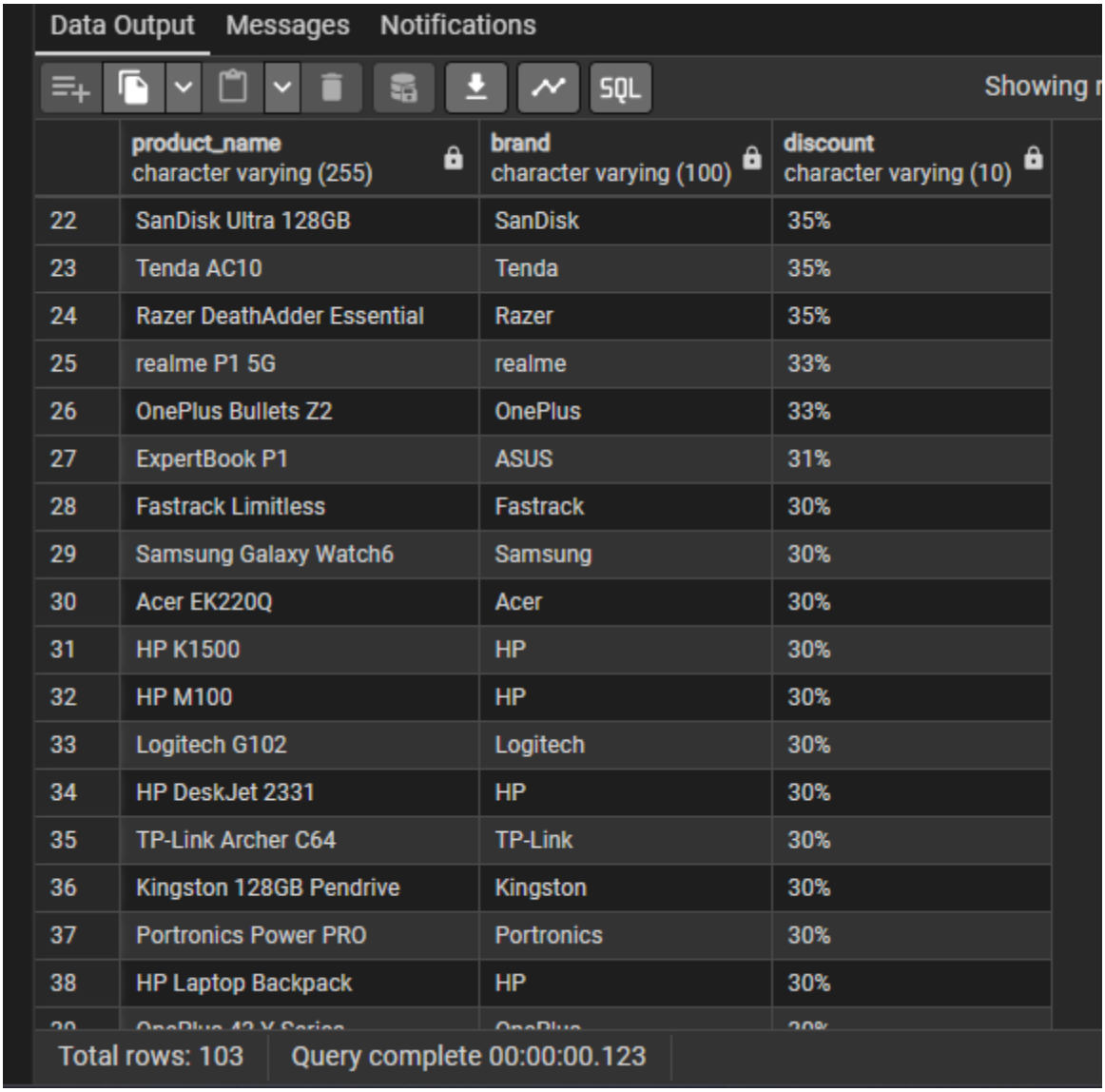


Data Output | Messages | Notifications

Showing rows: 1 to

	product_name character varying (255)	brand character varying (100)	discount character varying (10)
6	boAt Airdopes 141	boAt	57%
7	boAt Wave Sigma	boAt	55%
8	POCO X8 5G	POCO	54%
9	Noise Airwave	Noise	50%
10	Noise Buds VS104	Noise	50%
11	boAt Stone 350	boAt	45%
12	Realme Buds T300	realme	42%
13	Portronics SoundDrum	Portronics	40%
14	JBL Tune 510BT	JBL	40%
15	Noise ColorFit Pro	Noise	40%
16	Ambrane 10000mAh	Ambrane	40%
17	JBL Wave Buds	JBL	38%
18	boAt 20W Charger	boAt	35%
19	Prestige Air Fryer	Prestige	35%
20	HMD Vibe2 5G	HMD	35%
21	boAt Energyshroom PB	boAt	35%
22	SanDisk Ultra 128GB	SanDisk	35%

Total rows: 103 | Query complete 00:00:00.123



Data Output | Messages | Notifications

Showing r

	product_name character varying (255)	brand character varying (100)	discount character varying (10)
22	SanDisk Ultra 128GB	SanDisk	35%
23	Tenda AC10	Tenda	35%
24	Razer DeathAdder Essential	Razer	35%
25	realme P1 5G	realme	33%
26	OnePlus Bullets Z2	OnePlus	33%
27	ExpertBook P1	ASUS	31%
28	Fastrack Limitless	Fastrack	30%
29	Samsung Galaxy Watch6	Samsung	30%
30	Acer EK220Q	Acer	30%
31	HP K1500	HP	30%
32	HP M100	HP	30%
33	Logitech G102	Logitech	30%
34	HP DeskJet 2331	HP	30%
35	TP-Link Archer C64	TP-Link	30%
36	Kingston 128GB Pendrive	Kingston	30%
37	Portronics Power PRO	Portronics	30%
38	HP Laptop Backpack	HP	30%

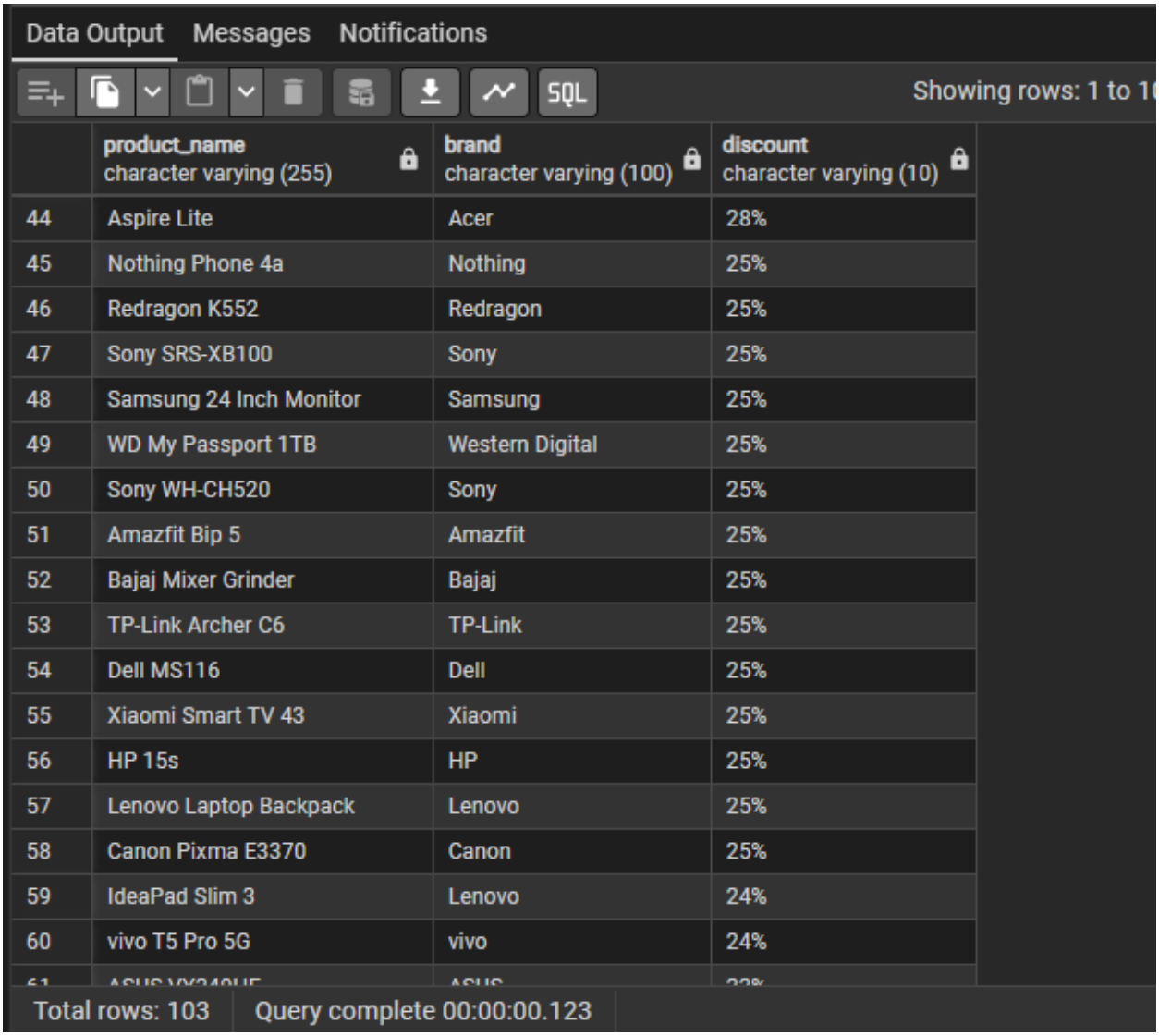
Total rows: 103 | Query complete 00:00:00.123

Data Output | Messages | Notifications

Showing rows: 1 to 103

	product_name character varying (255)	brand character varying (100)	discount character varying (10)
28	Fastrack Limitless	Fastrack	30%
29	Samsung Galaxy Watch6	Samsung	30%
30	Acer EK220Q	Acer	30%
31	HP K1500	HP	30%
32	HP M100	HP	30%
33	Logitech G102	Logitech	30%
34	HP DeskJet 2331	HP	30%
35	TP-Link Archer C64	TP-Link	30%
36	Kingston 128GB Pendrive	Kingston	30%
37	Portronics Power PRO	Portronics	30%
38	HP Laptop Backpack	HP	30%
39	OnePlus 43 Y Series	OnePlus	30%
40	Philips Air Fryer 4.1L	Philips	30%
41	Prestige Induction Cooktop	Prestige	30%
42	Sony WF-C500	Sony	29%
43	Philips Electric Kettle	Philips	28%
44	Aspire Lite	Acer	28%
45	Nothing Phone 4a	Nothing	25%

Total rows: 103 | Query complete 00:00:00.123

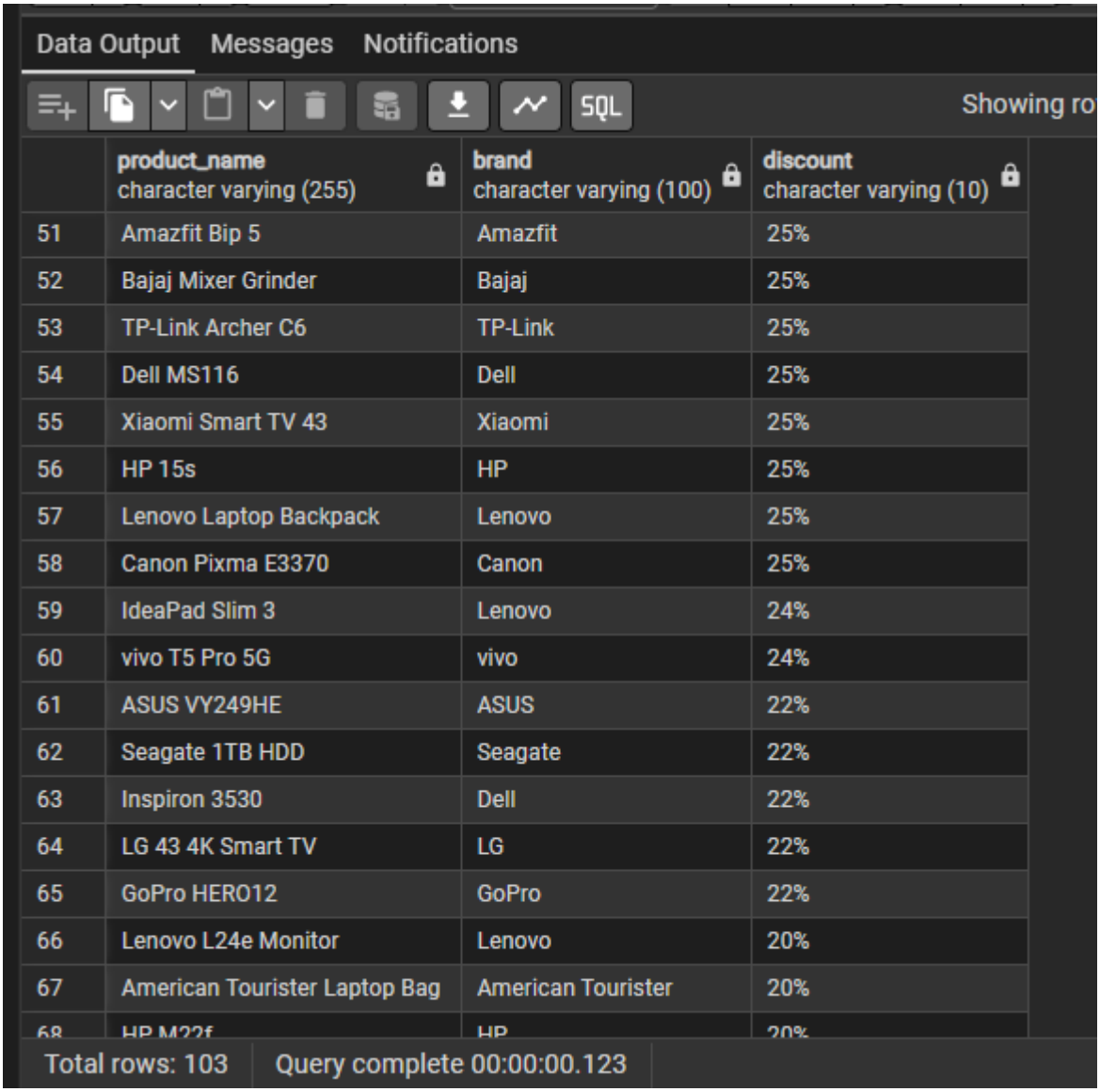


Data Output | Messages | Notifications

Showing rows: 1 to 1

	product_name character varying (255)	brand character varying (100)	discount character varying (10)
44	Aspire Lite	Acer	28%
45	Nothing Phone 4a	Nothing	25%
46	Redragon K552	Redragon	25%
47	Sony SRS-XB100	Sony	25%
48	Samsung 24 Inch Monitor	Samsung	25%
49	WD My Passport 1TB	Western Digital	25%
50	Sony WH-CH520	Sony	25%
51	Amazfit Bip 5	Amazfit	25%
52	Bajaj Mixer Grinder	Bajaj	25%
53	TP-Link Archer C6	TP-Link	25%
54	Dell MS116	Dell	25%
55	Xiaomi Smart TV 43	Xiaomi	25%
56	HP 15s	HP	25%
57	Lenovo Laptop Backpack	Lenovo	25%
58	Canon Pixma E3370	Canon	25%
59	IdeaPad Slim 3	Lenovo	24%
60	vivo T5 Pro 5G	vivo	24%

Total rows: 103 | Query complete 00:00:00.123



Data Output | Messages | Notifications

Showing ro

	product_name character varying (255)	brand character varying (100)	discount character varying (10)
51	Amazfit Bip 5	Amazfit	25%
52	Bajaj Mixer Grinder	Bajaj	25%
53	TP-Link Archer C6	TP-Link	25%
54	Dell MS116	Dell	25%
55	Xiaomi Smart TV 43	Xiaomi	25%
56	HP 15s	HP	25%
57	Lenovo Laptop Backpack	Lenovo	25%
58	Canon Pixma E3370	Canon	25%
59	IdeaPad Slim 3	Lenovo	24%
60	vivo T5 Pro 5G	vivo	24%
61	ASUS VY249HE	ASUS	22%
62	Seagate 1TB HDD	Seagate	22%
63	Inspiron 3530	Dell	22%
64	LG 43 4K Smart TV	LG	22%
65	GoPro HERO12	GoPro	22%
66	Lenovo L24e Monitor	Lenovo	20%
67	American Tourister Laptop Bag	American Tourister	20%
68	HP M22f	HP	20%

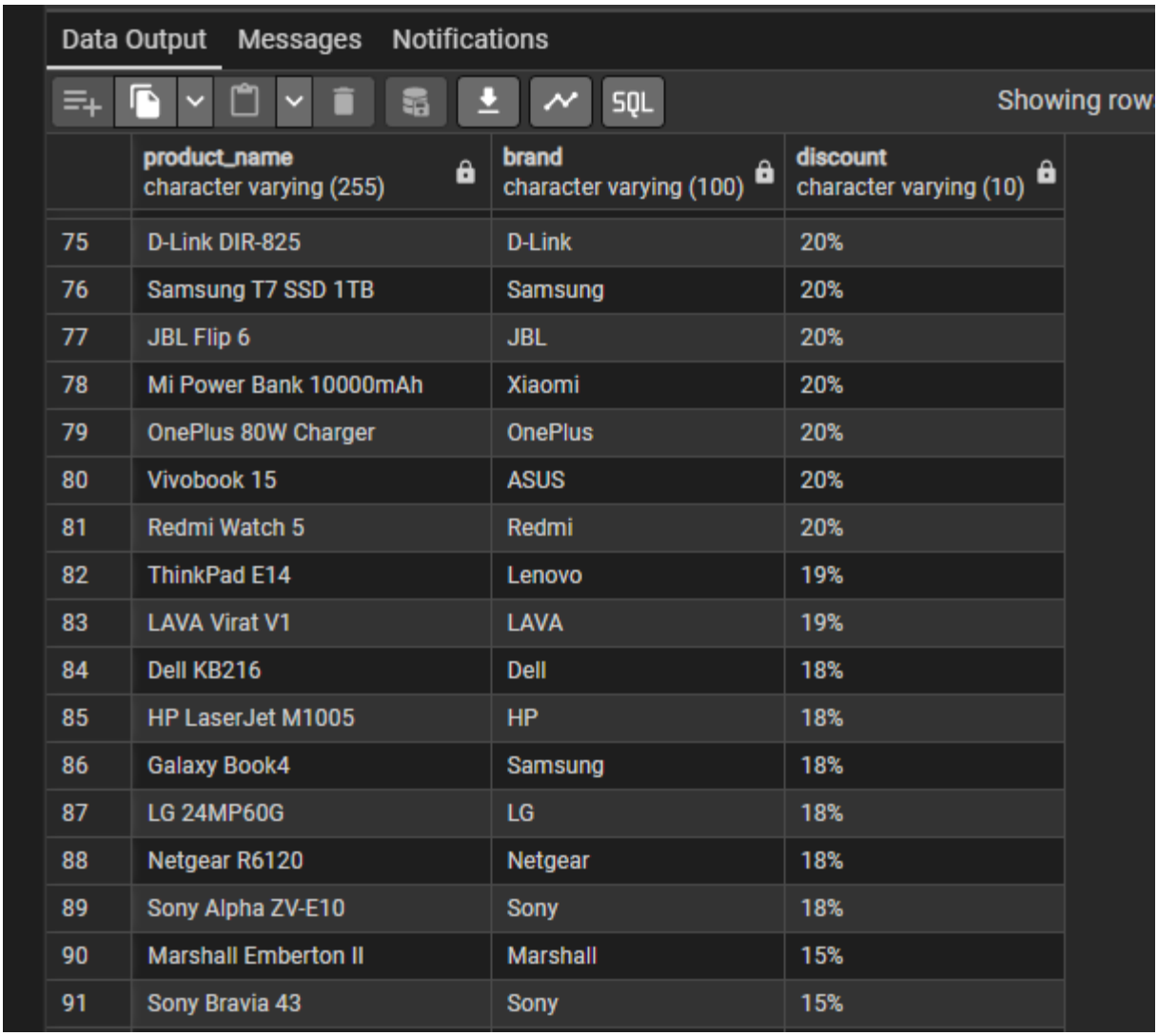
Total rows: 103 | Query complete 00:00:00.123

Data Output | Messages | Notifications

Showing rows: 1

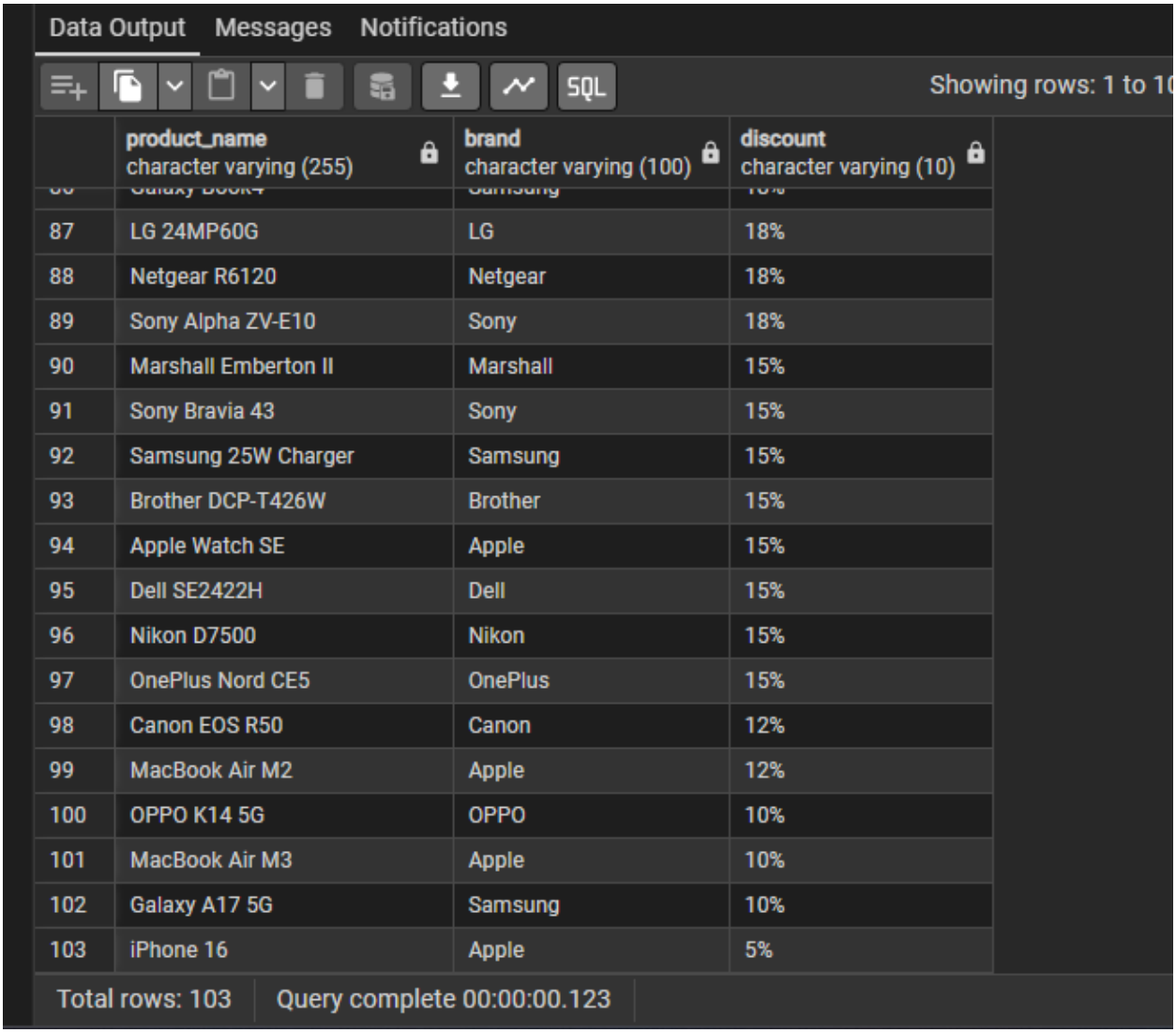
	product_name character varying (255)	brand character varying (100)	discount character varying (10)
68	HP M22f	HP	20%
69	Logitech K120	Logitech	20%
70	Logitech G213	Logitech	20%
71	Logitech M185	Logitech	20%
72	Samsung Crystal 4K 43	Samsung	20%
73	Canon EOS 1500D	Canon	20%
74	Epson L3250	Epson	20%
75	D-Link DIR-825	D-Link	20%
76	Samsung T7 SSD 1TB	Samsung	20%
77	JBL Flip 6	JBL	20%
78	Mi Power Bank 10000mAh	Xiaomi	20%
79	OnePlus 80W Charger	OnePlus	20%
80	Vivobook 15	ASUS	20%
81	Redmi Watch 5	Redmi	20%
82	ThinkPad E14	Lenovo	19%
83	LAVA Virat V1	LAVA	19%
84	Dell KB216	Dell	18%
85	HP Laser Jet M1005	HP	18%

Total rows: 103 | Query complete 00:00:00.123


Data Output Messages Notifications

SQL Showing row

	product_name character varying (255)	brand character varying (100)	discount character varying (10)
75	D-Link DIR-825	D-Link	20%
76	Samsung T7 SSD 1TB	Samsung	20%
77	JBL Flip 6	JBL	20%
78	Mi Power Bank 10000mAh	Xiaomi	20%
79	OnePlus 80W Charger	OnePlus	20%
80	Vivobook 15	ASUS	20%
81	Redmi Watch 5	Redmi	20%
82	ThinkPad E14	Lenovo	19%
83	LAVA Virat V1	LAVA	19%
84	Dell KB216	Dell	18%
85	HP LaserJet M1005	HP	18%
86	Galaxy Book4	Samsung	18%
87	LG 24MP60G	LG	18%
88	Netgear R6120	Netgear	18%
89	Sony Alpha ZV-E10	Sony	18%
90	Marshall Emberton II	Marshall	15%
91	Sony Bravia 43	Sony	15%



Data Output | Messages | Notifications

Showing rows: 1 to 10

	product_name character varying (255)	brand character varying (100)	discount character varying (10)
87	LG 24MP60G	LG	18%
88	Netgear R6120	Netgear	18%
89	Sony Alpha ZV-E10	Sony	18%
90	Marshall Emberton II	Marshall	15%
91	Sony Bravia 43	Sony	15%
92	Samsung 25W Charger	Samsung	15%
93	Brother DCP-T426W	Brother	15%
94	Apple Watch SE	Apple	15%
95	Dell SE2422H	Dell	15%
96	Nikon D7500	Nikon	15%
97	OnePlus Nord CE5	OnePlus	15%
98	Canon EOS R50	Canon	12%
99	MacBook Air M2	Apple	12%
100	OPPO K14 5G	OPPO	10%
101	MacBook Air M3	Apple	10%
102	Galaxy A17 5G	Samsung	10%
103	iPhone 16	Apple	5%

Total rows: 103 | Query complete 00:00:00.123

```
161
162  SELECT product_category, COUNT(*) AS total_product
163  FROM ecommerce_products1
164  GROUP BY product_category
165  ORDER BY total_products DESC;
```

Data Output | Messages | Notifications

	product_category character varying (100)	total_products bigint
1	Mobile	10
2	Laptop	10
3	Smartwatch	8
4	Monitor	7
5	Headphones	6
6	Mouse	5

Total rows: 23 | Query complete 00:00:00.125

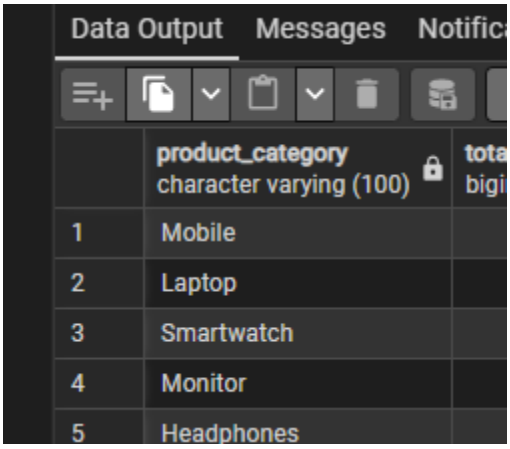
```
162  SELECT product_category, COUNT(*) AS total_products
163  FROM ecommerce_products1
164  GROUP BY product_category
165  ORDER BY total_products DESC;
```

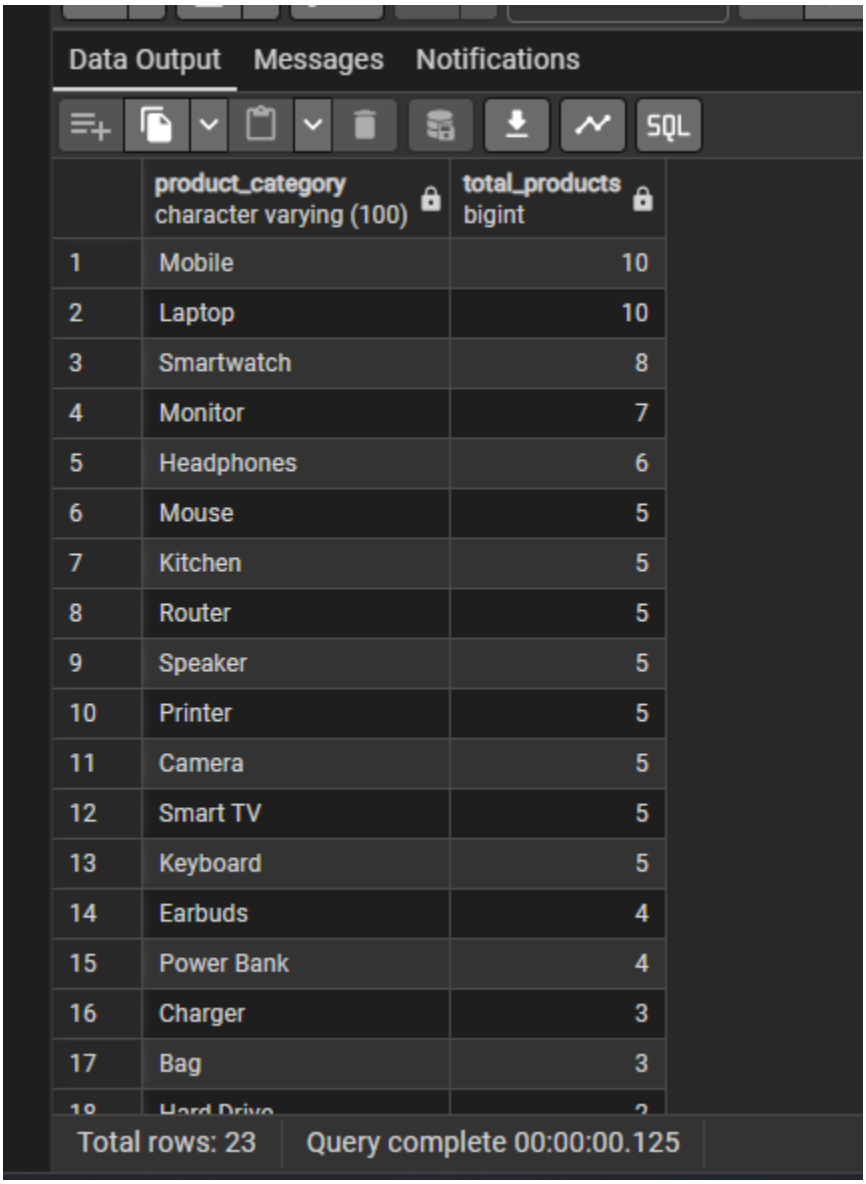
Data Output | Messages | Notifications

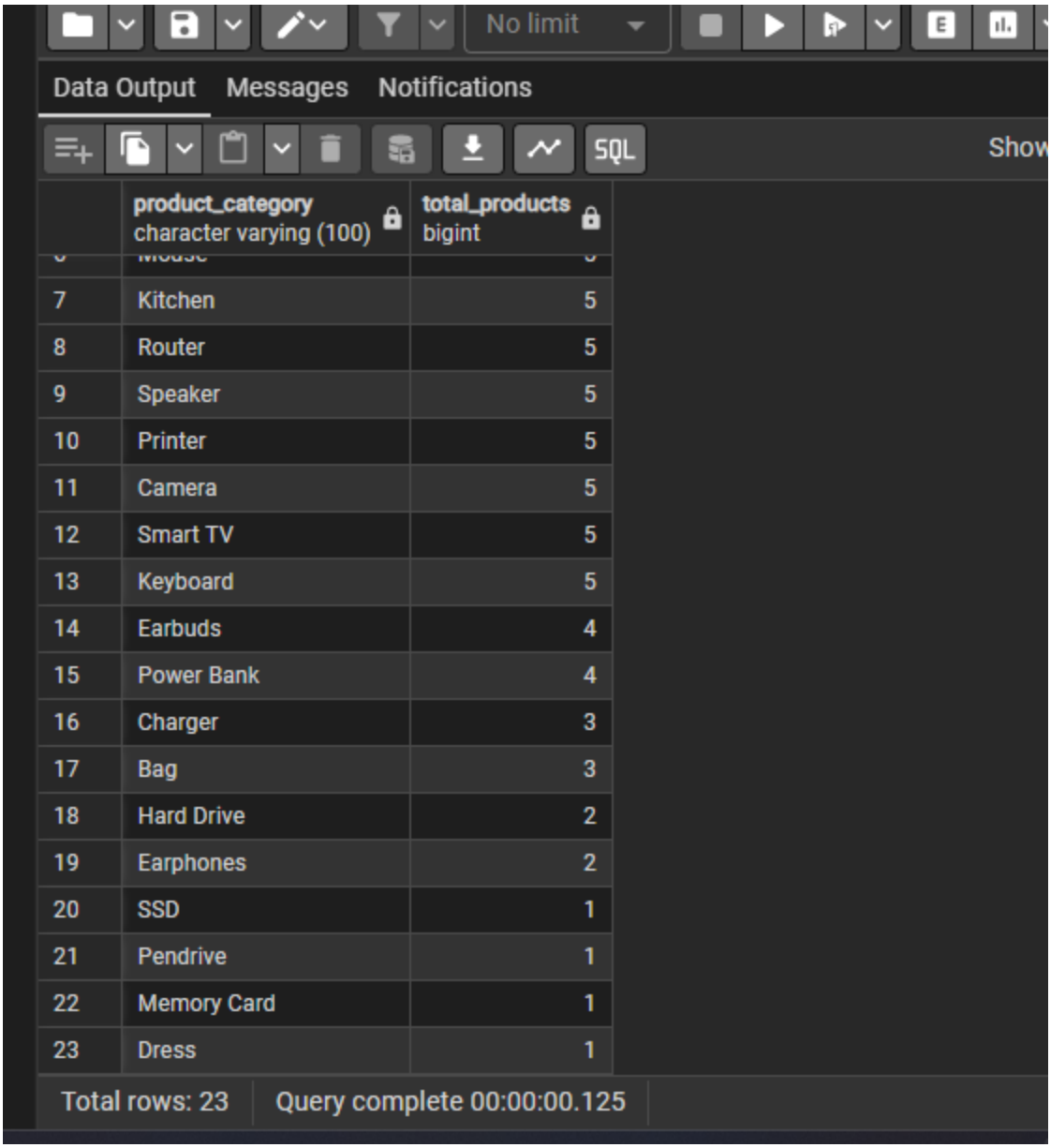
Showing rows: 1 t

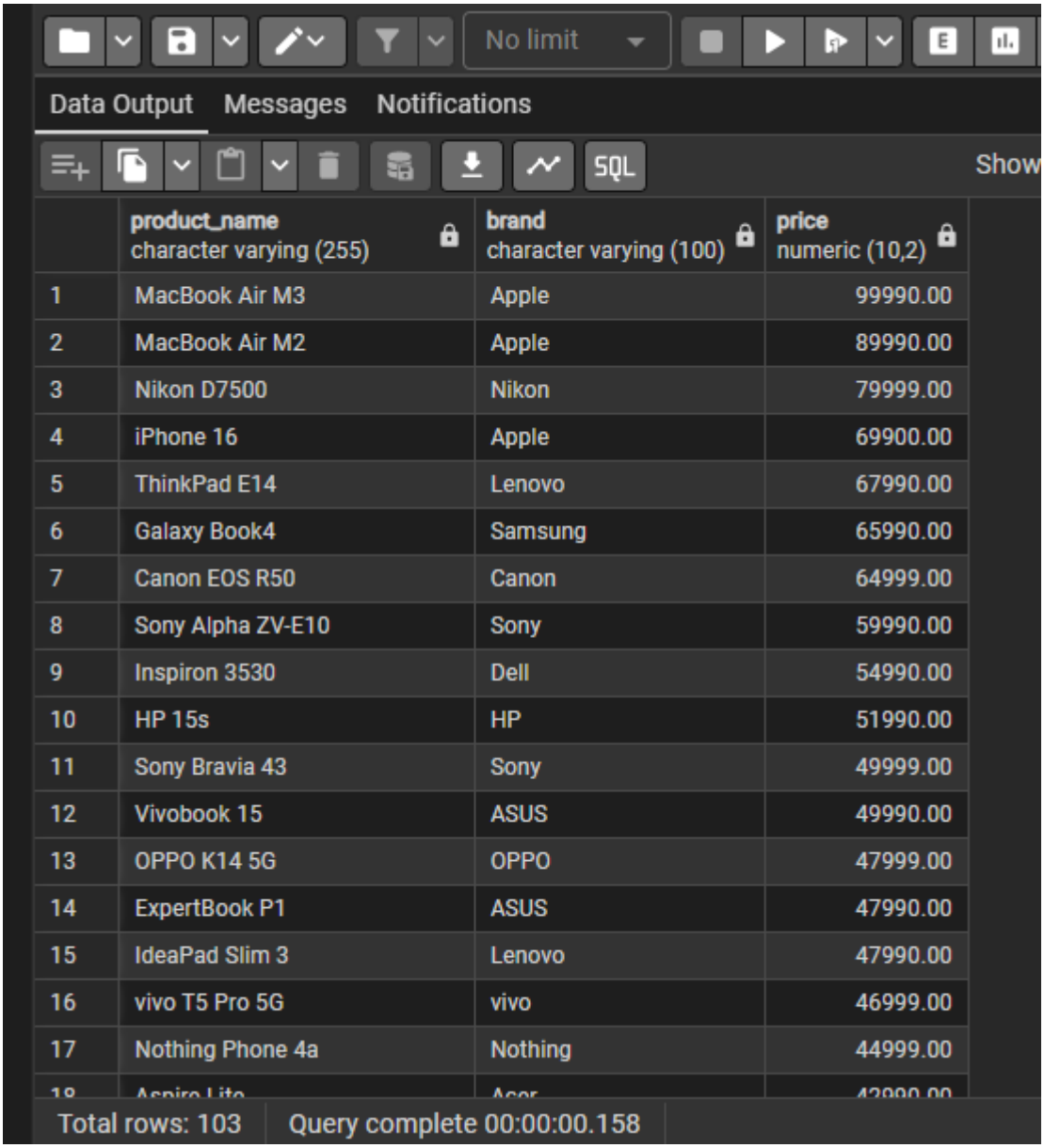
	product_category character varying (100)	total_products bigint
1	Mobile	10
2	Laptop	10
3	Smartwatch	8
4	Monitor	7
5	Headphones	6
6	Mouse	5

Total rows: 23 | Query complete 00:00:00.125


Data Output
Messages
Notific
product_category
character varying (100)
1 Mobile
2 Laptop
3 Smartwatch
4 Monitor
5 Headphones


Data Output
Messages
Notifications
SQL
product_category
character varying (100)
total_products
bigint
1 Mobile 10
2 Laptop 10
3 Smartwatch 8
4 Monitor 7
5 Headphones 6
6 Mouse 5
7 Kitchen 5
8 Router 5
9 Speaker 5
10 Printer 5
11 Camera 5
12 Smart TV 5
13 Keyboard 5
14 Earbuds 4
15 Power Bank 4
16 Charger 3
17 Bag 3
Total rows: 23
Query complete 00:00:00.125


No limit
Data Output
Messages
Notifications
SQL
Show
product_category
character varying (100)
total_products
bigint
7 Kitchen 5
8 Router 5
9 Speaker 5
10 Printer 5
11 Camera 5
12 Smart TV 5
13 Keyboard 5
14 Earbuds 4
15 Power Bank 4
16 Charger 3
17 Bag 3
18 Hard Drive 2
19 Earphones 2
20 SSD 1
21 Pendrive 1
22 Memory Card 1
23 Dress 1
Total rows: 23
Query complete 00:00:00.125



	product_name character varying (255)	brand character varying (100)	price numeric (10,2)
1	MacBook Air M3	Apple	99990.00
2	MacBook Air M2	Apple	89990.00
3	Nikon D7500	Nikon	79999.00
4	iPhone 16	Apple	69900.00
5	ThinkPad E14	Lenovo	67990.00
6	Galaxy Book4	Samsung	65990.00
7	Canon EOS R50	Canon	64999.00
8	Sony Alpha ZV-E10	Sony	59990.00
9	Inspiron 3530	Dell	54990.00
10	HP 15s	HP	51990.00
11	Sony Bravia 43	Sony	49999.00
12	Vivobook 15	ASUS	49990.00
13	OPPO K14 5G	OPPO	47999.00
14	ExpertBook P1	ASUS	47990.00
15	IdeaPad Slim 3	Lenovo	47990.00
16	vivo T5 Pro 5G	vivo	46999.00
17	Nothing Phone 4a	Nothing	44999.00
18	Aspire Lite	Acer	42990.00

Data Output | Messages | Notifications

Showing rows: 1 to 103

	product_name character varying (255)	brand character varying (100)	price numeric (10,2)
18	Aspire Lite	Acer	42990.00
19	Canon EOS 1500D	Canon	39999.00
20	GoPro HERO12	GoPro	39999.00
21	OnePlus Nord CE5	OnePlus	35000.00
22	Samsung Crystal 4K 43	Samsung	34999.00
23	LG 43 4K Smart TV	LG	32999.00
24	Galaxy A17 5G	Samsung	31495.00
25	Xiaomi Smart TV 43	Xiaomi	29999.00
26	POCO X8 5G	POCO	29999.00
27	Apple Watch SE	Apple	29900.00
28	OnePlus 43 Y Series	OnePlus	24999.00
29	Samsung Galaxy Watch6	Samsung	24999.00
30	HP LaserJet M1005	HP	18999.00
31	Marshall Emberton II	Marshall	15999.00
32	realme P1 5G	realme	13999.00
33	Epson L3250	Epson	13999.00
34	Brother DCP-T426W	Brother	13990.00
35	LG 24MP60G	LG	12999.00

Total rows: 103 | Query complete 00:00:00.158

Data Output | Messages | Notifications

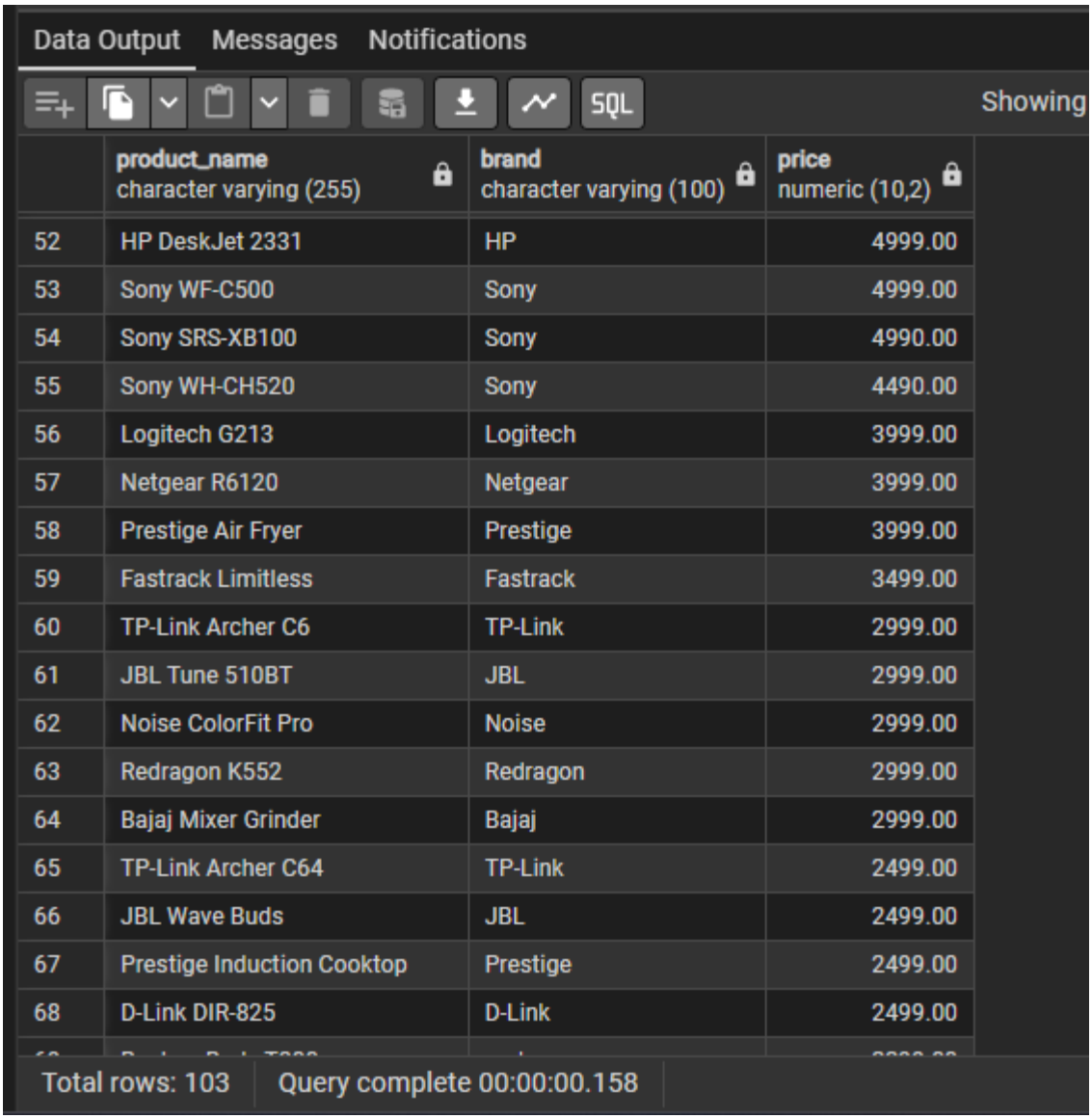
SQL | Showing row

	product_name character varying (255)	brand character varying (100)	price numeric (10,2)
35	LG 24MP60G	LG	12999.00
36	HMD Vibe2 5G	HMD	12999.00
37	Dell SE2422H	Dell	12499.00
38	Lenovo L24e Monitor	Lenovo	11999.00
39	ASUS VY249HE	ASUS	11990.00
40	JBL Flip 6	JBL	10999.00
41	Samsung 24 Inch Monitor	Samsung	10999.00
42	HP M22f	HP	10990.00
43	LAVA Virat V1	LAVA	10499.00
44	Samsung T7 SSD 1TB	Samsung	8999.00
45	Acer EK220Q	Acer	8999.00
46	Redmi Watch 5	Redmi	6999.00
47	Canon Pixma E3370	Canon	6999.00
48	WD My Passport 1TB	Western Digital	5999.00
49	Seagate 1TB HDD	Seagate	5499.00
50	Amazfit Bip 5	Amazfit	5499.00
51	Philips Air Fryer 4.1L	Philips	4999.00

Total rows: 103 | Query complete 00:00:00.158

Data Output | Messages | Noti

	product_name character varying (255)
52	HP DeskJet 2331
53	Sony WF-C500
54	Sony SRS-XB100
55	Sony WH-CH520
56	Logitech G213
57	Netgear R6120

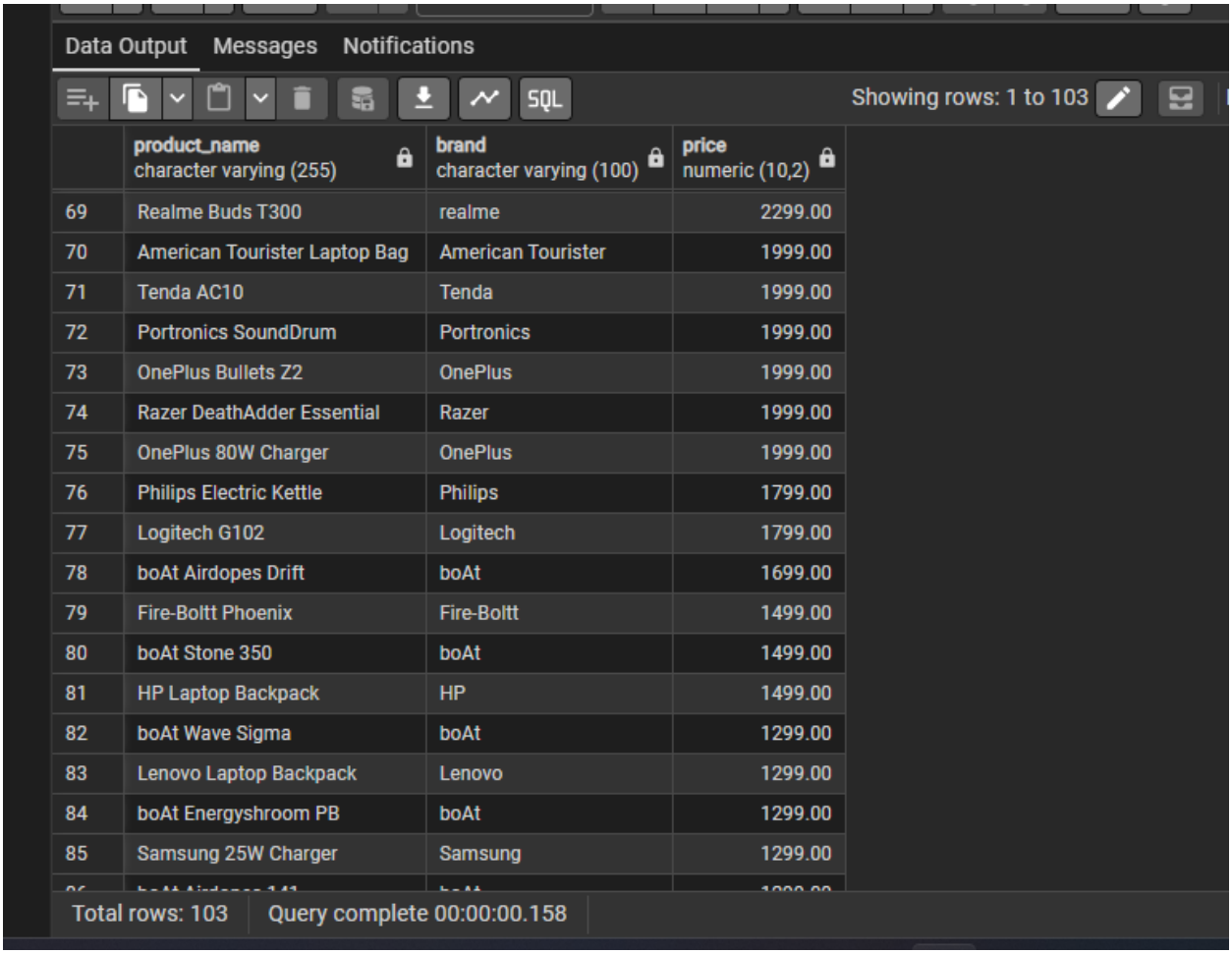


Data Output | Messages | Notifications

Showing

	product_name character varying (255)	brand character varying (100)	price numeric (10,2)
52	HP DeskJet 2331	HP	4999.00
53	Sony WF-C500	Sony	4999.00
54	Sony SRS-XB100	Sony	4990.00
55	Sony WH-CH520	Sony	4490.00
56	Logitech G213	Logitech	3999.00
57	Netgear R6120	Netgear	3999.00
58	Prestige Air Fryer	Prestige	3999.00
59	Fastrack Limitless	Fastrack	3499.00
60	TP-Link Archer C6	TP-Link	2999.00
61	JBL Tune 510BT	JBL	2999.00
62	Noise ColorFit Pro	Noise	2999.00
63	Redragon K552	Redragon	2999.00
64	Bajaj Mixer Grinder	Bajaj	2999.00
65	TP-Link Archer C64	TP-Link	2499.00
66	JBL Wave Buds	JBL	2499.00
67	Prestige Induction Cooktop	Prestige	2499.00
68	D-Link DIR-825	D-Link	2499.00

Total rows: 103 | Query complete 00:00:00.158

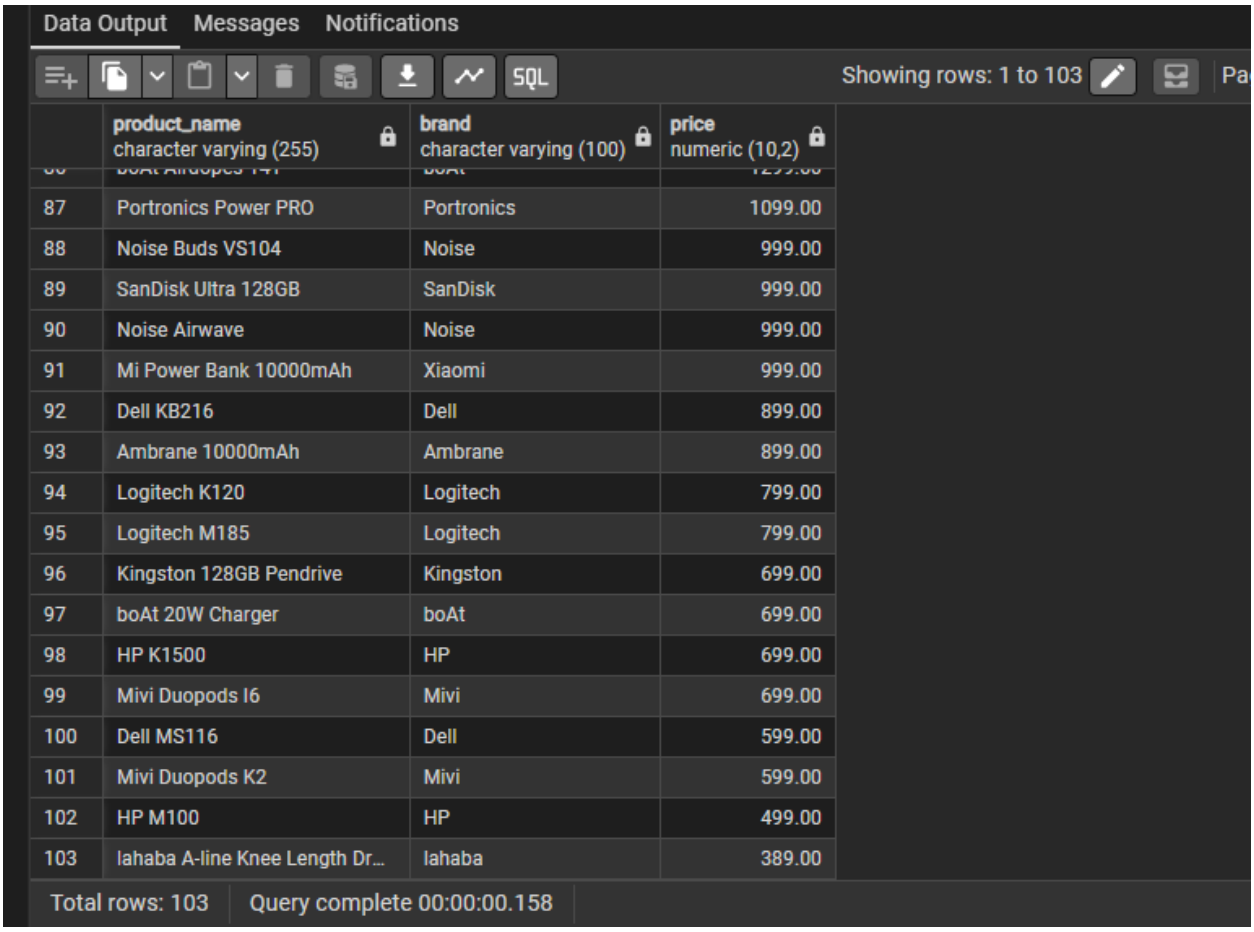


Data Output | Messages | Notifications

Showing rows: 1 to 103

	product_name character varying (255)	brand character varying (100)	price numeric (10,2)
69	Realme Buds T300	realme	2299.00
70	American Tourister Laptop Bag	American Tourister	1999.00
71	Tenda AC10	Tenda	1999.00
72	Portronics SoundDrum	Portronics	1999.00
73	OnePlus Bullets Z2	OnePlus	1999.00
74	Razer DeathAdder Essential	Razer	1999.00
75	OnePlus 80W Charger	OnePlus	1999.00
76	Philips Electric Kettle	Philips	1799.00
77	Logitech G102	Logitech	1799.00
78	boAt Airdopes Drift	boAt	1699.00
79	Fire-Boltt Phoenix	Fire-Boltt	1499.00
80	boAt Stone 350	boAt	1499.00
81	HP Laptop Backpack	HP	1499.00
82	boAt Wave Sigma	boAt	1299.00
83	Lenovo Laptop Backpack	Lenovo	1299.00
84	boAt Energyshroom PB	boAt	1299.00
85	Samsung 25W Charger	Samsung	1299.00

Total rows: 103 | Query complete 00:00:00.158



Data Output | Messages | Notifications

Showing rows: 1 to 103

	product_name character varying (255)	brand character varying (100)	price numeric (10,2)
87	Portronics Power PRO	Portronics	1099.00
88	Noise Buds VS104	Noise	999.00
89	SanDisk Ultra 128GB	SanDisk	999.00
90	Noise Airwave	Noise	999.00
91	Mi Power Bank 10000mAh	Xiaomi	999.00
92	Dell KB216	Dell	899.00
93	Ambrane 10000mAh	Ambrane	899.00
94	Logitech K120	Logitech	799.00
95	Logitech M185	Logitech	799.00
96	Kingston 128GB Pendrive	Kingston	699.00
97	boAt 20W Charger	boAt	699.00
98	HP K1500	HP	699.00
99	Mivi Duopods I6	Mivi	699.00
100	Dell MS116	Dell	599.00
101	Mivi Duopods K2	Mivi	599.00
102	HP M100	HP	499.00
103	lahaba A-line Knee Length Dr...	lahaba	389.00

Total rows: 103 | Query complete 00:00:00.158

```
170
171  SELECT column_name, data_type
172  FROM information_schema.columns
173  WHERE table_name = 'ecommerce_products1';
```

Data Output | Messages | Notifications

	column_name name	data_type character varying
1	review_count	integer
2	price	numeric
3	rating	numeric
4	discount	character varying
5	seller	character varying
6	product_name	character varying
7	delivery_promise	character varying
8	product_categ...	character varying
9	brand	character varying

```
174
175  SELECT *
176  FROM ecommerce_products1
177  WHERE product_name IS NULL;
```

Data Output | Messages | Notifications

	product_name character varying (255)	product_category character varying (100)	brand character varying (100)	price numeric (10,2)	discount character varying (10)	rating numeric (2,1)	review_count integer	seller character varyin

```sql
SELECT
    COUNT(*) AS total_records,
    COUNT(*) - COUNT(product_name) AS missing_product_name,
    COUNT(*) - COUNT(product_category) AS missing_category,
    COUNT(*) - COUNT(brand) AS missing_brand,
    COUNT(*) - COUNT(price) AS missing_price,
    COUNT(*) - COUNT(discount) AS missing_discount,
    COUNT(*) - COUNT(rating) AS missing_rating,
    COUNT(*) - COUNT(review_count) AS missing_review_count,
    COUNT(*) - COUNT(seller) AS missing_seller,
    COUNT(*) - COUNT(delivery_promise) AS missing_delivery_promise
FROM ecommerce_products1;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page No: 1 of 1

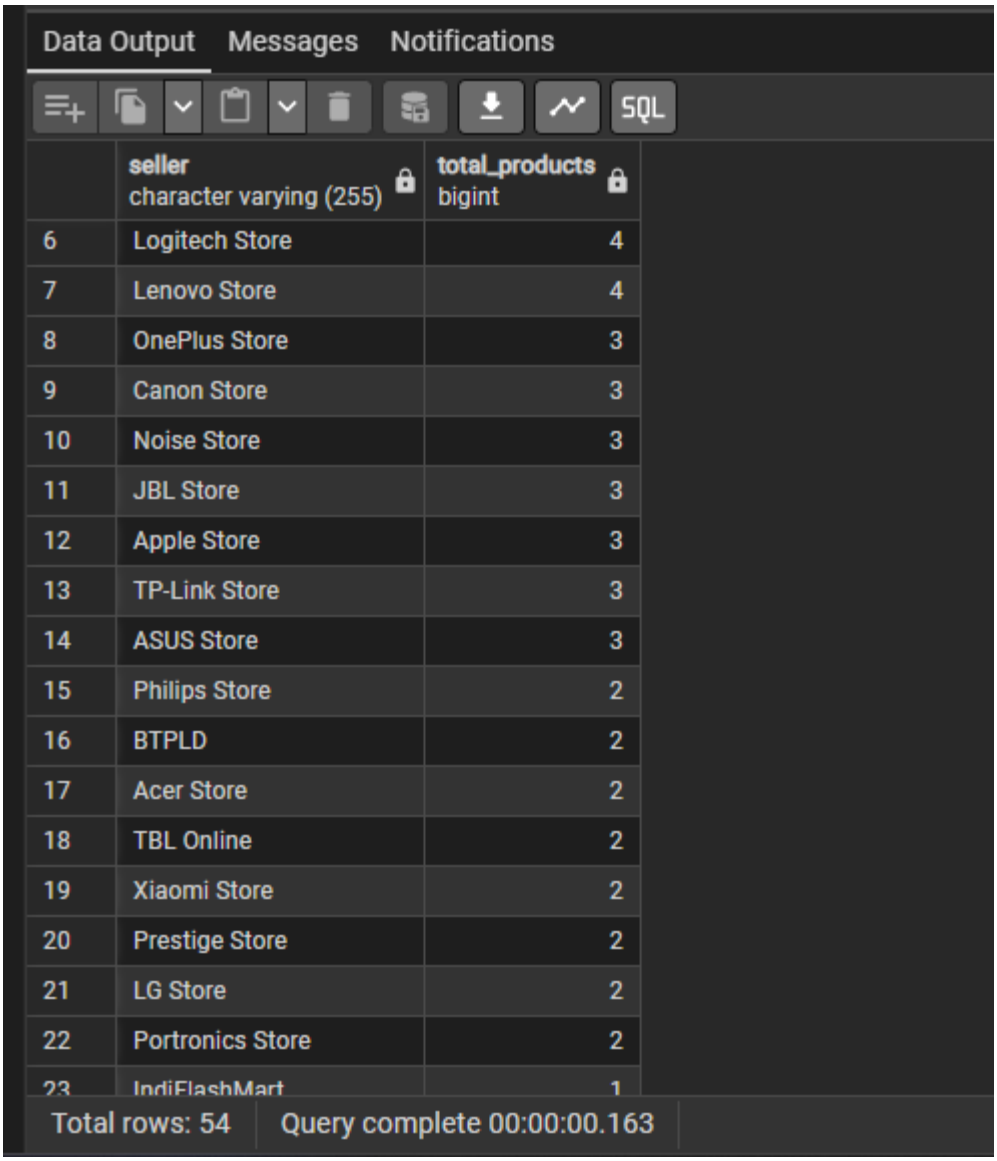
	total_records bigint	missing_product_name bigint	missing_category bigint	missing_brand bigint	missing_price bigint	missing_discount bigint	missing_rating bigint	missing_review_count bigint	missi bigint
1	103	0	0	0	0	0	0	0	

Total rows: 1 | Query complete 00:00:00.091 | CRLF | Ln 192, Col 1

```sql
SELECT seller, COUNT(*) AS total_products
FROM ecommerce_products1
GROUP BY seller
ORDER BY total_products DESC;
```

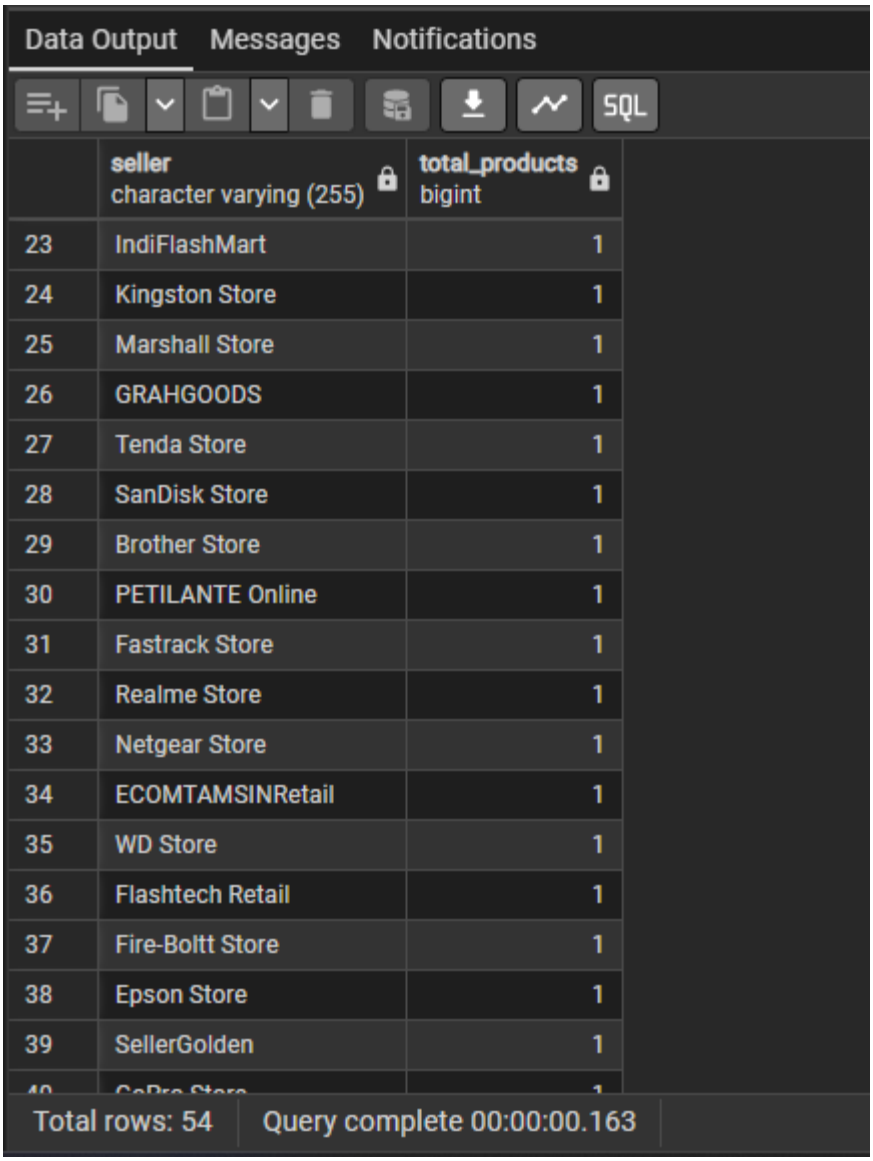
Data Output | Messages | Notifications

	seller character varying (255)	total_products bigint
1	HP Store	7
2	Samsung Store	5
3	Sony Store	5
4	Imagine Store	5
5	Dell Store	4
6	Logitech Store	4



	seller character varying (255)	total_products bigint
6	Logitech Store	4
7	Lenovo Store	4
8	OnePlus Store	3
9	Canon Store	3
10	Noise Store	3
11	JBL Store	3
12	Apple Store	3
13	TP-Link Store	3
14	ASUS Store	3
15	Philips Store	2
16	BTPLD	2
17	Acer Store	2
18	TBL Online	2
19	Xiaomi Store	2
20	Prestige Store	2
21	LG Store	2
22	Portronics Store	2
23	IndiFlashMart	1

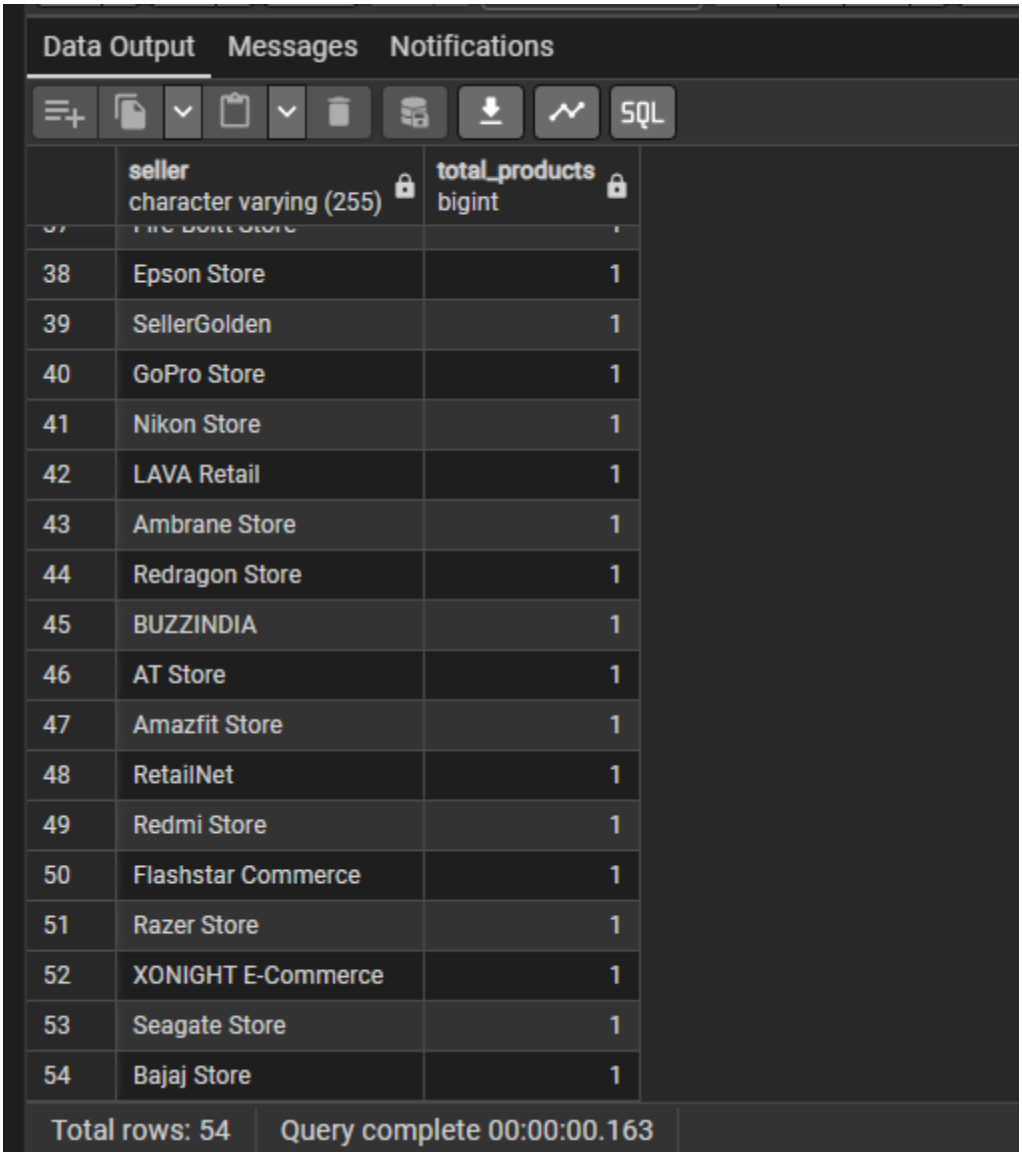
Total rows: 54 | Query complete 00:00:00.163



Data Output | Messages | Notifications

	seller character varying (255)	total_products bigint
23	IndiFlashMart	1
24	Kingston Store	1
25	Marshall Store	1
26	GRAHGOODS	1
27	Tenda Store	1
28	SanDisk Store	1
29	Brother Store	1
30	PETILANTE Online	1
31	Fastrack Store	1
32	Realme Store	1
33	Netgear Store	1
34	ECOMTAMSINRetail	1
35	WD Store	1
36	Flashtech Retail	1
37	Fire-Boltt Store	1
38	Epson Store	1
39	SellerGolden	1

Total rows: 54 | Query complete 00:00:00.163



Data Output | Messages | Notifications

SQL

	seller character varying (255)	total_products bigint
38	Epson Store	1
39	SellerGolden	1
40	GoPro Store	1
41	Nikon Store	1
42	LAVA Retail	1
43	Ambrane Store	1
44	Redragon Store	1
45	BUZZINDIA	1
46	AT Store	1
47	Amazfit Store	1
48	RetailNet	1
49	Redmi Store	1
50	Flashstar Commerce	1
51	Razer Store	1
52	XONIGHT E-Commerce	1
53	Seagate Store	1
54	Bajaj Store	1

Total rows: 54 | Query complete 00:00:00.163

```
197
198  SELECT product_name, product_category, brand, price
199  FROM ecommerce_products1
200  ORDER BY price ASC
```

Data Output | Messages | Notifications

Showing rows: 1 to 10 | Page No:

	product_name character varying (255)	product_category character varying (100)	brand character varying (100)	price numeric (10,2)
1	lahaba A-line Knee Length Dr...	Dress	lahaba	389.00
2	HP M100	Mouse	HP	499.00
3	Mivi Duopods K2	Headphones	Mivi	599.00
4	Dell MS116	Mouse	Dell	599.00
5	Mivi Duopods I6	Headphones	Mivi	699.00
6	Kingston 128GB Pendrive	Pendrive	Kingston	699.00
7	HP K1500	Keyboard	HP	699.00
8	boAt 20W Charger	Charger	boAt	699.00
9	Logitech M185	Mouse	Logitech	799.00
10	Logitech K120	Keyboard	Logitech	799.00

Total rows: 10 | Query complete 00:00:00.161

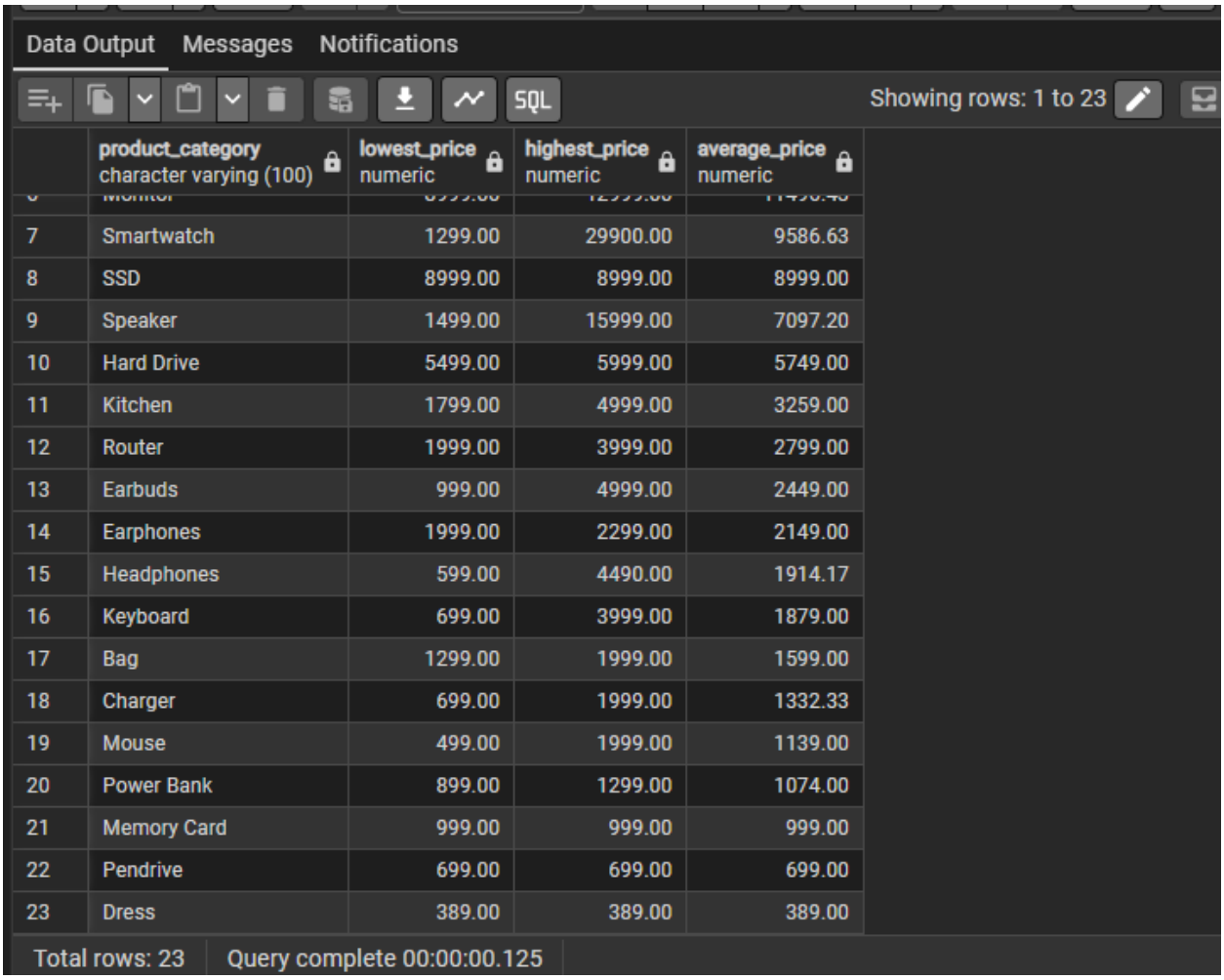
```sql
SELECT
    product_category,
    MIN(price) AS lowest_price,
    MAX(price) AS highest_price,
    ROUND(AVG(price), 2) AS average_price
FROM ecommerce_products1
GROUP BY product_category
ORDER BY average_price DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 23

	product_category character varying (100)	lowest_price numeric	highest_price numeric	average_price numeric
1	Laptop	42990.00	99990.00	61990.00
2	Camera	39999.00	79999.00	56997.20
3	Smart TV	24999.00	49999.00	34599.00
4	Mobile	10499.00	69900.00	34388.80
5	Printer	4999.00	18999.00	11797.20
6	Monitor	8999.00	12999.00	11496.43
7	Smartwatch	1299.00	29900.00	9586.63
8	SSD	8999.00	8999.00	8999.00

Total rows: 23 | Query complete 00:00:00.125



Data Output | Messages | Notifications

Showing rows: 1 to 23

	product_category character varying (100)	lowest_price numeric	highest_price numeric	average_price numeric
7	Smartwatch	1299.00	29900.00	9586.63
8	SSD	8999.00	8999.00	8999.00
9	Speaker	1499.00	15999.00	7097.20
10	Hard Drive	5499.00	5999.00	5749.00
11	Kitchen	1799.00	4999.00	3259.00
12	Router	1999.00	3999.00	2799.00
13	Earbuds	999.00	4999.00	2449.00
14	Earphones	1999.00	2299.00	2149.00
15	Headphones	599.00	4490.00	1914.17
16	Keyboard	699.00	3999.00	1879.00
17	Bag	1299.00	1999.00	1599.00
18	Charger	699.00	1999.00	1332.33
19	Mouse	499.00	1999.00	1139.00
20	Power Bank	899.00	1299.00	1074.00
21	Memory Card	999.00	999.00	999.00
22	Pendrive	699.00	699.00	699.00
23	Dress	389.00	389.00	389.00

Total rows: 23 | Query complete 00:00:00.125

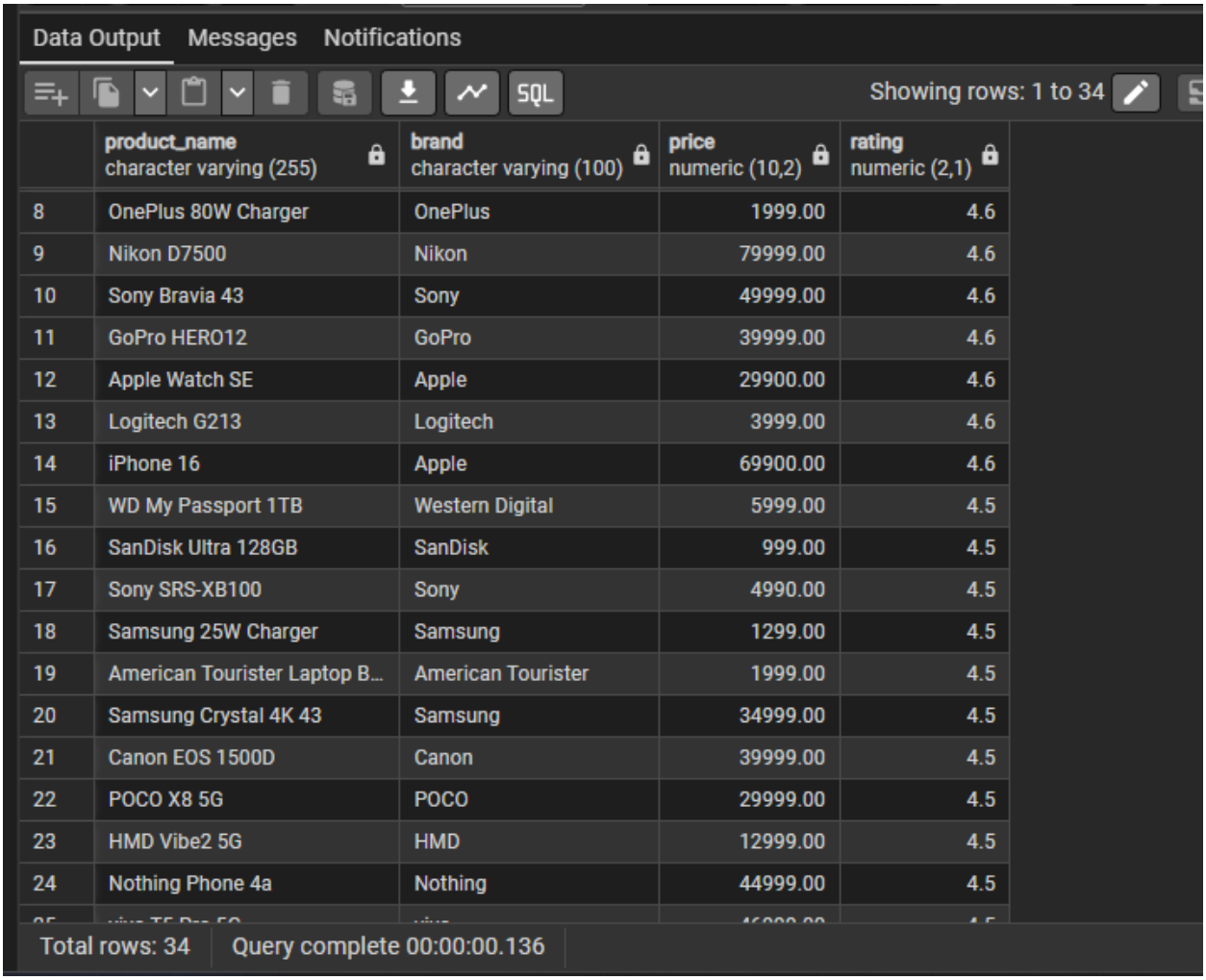
```
211
212  SELECT product_name, brand, price, rating
213  FROM ecommerce_products1
214  WHERE rating >= 4.5
215  ORDER BY rating DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 34

	product_name character varying (255)	brand character varying (100)	price numeric (10,2)	rating numeric (2,1)
1	MacBook Air M3	Apple	99990.00	4.8
2	MacBook Air M2	Apple	89990.00	4.8
3	Samsung T7 SSD 1TB	Samsung	8999.00	4.7
4	Marshall Emberton II	Marshall	15999.00	4.7
5	Canon EOS R50	Canon	64999.00	4.7
6	JBL Flip 6	JBL	10999.00	4.6
7	Logitech G102	Logitech	1799.00	4.6
8	OnePlus 80W Charger	OnePlus	1999.00	4.6

Total rows: 34 | Query complete 00:00:00.136

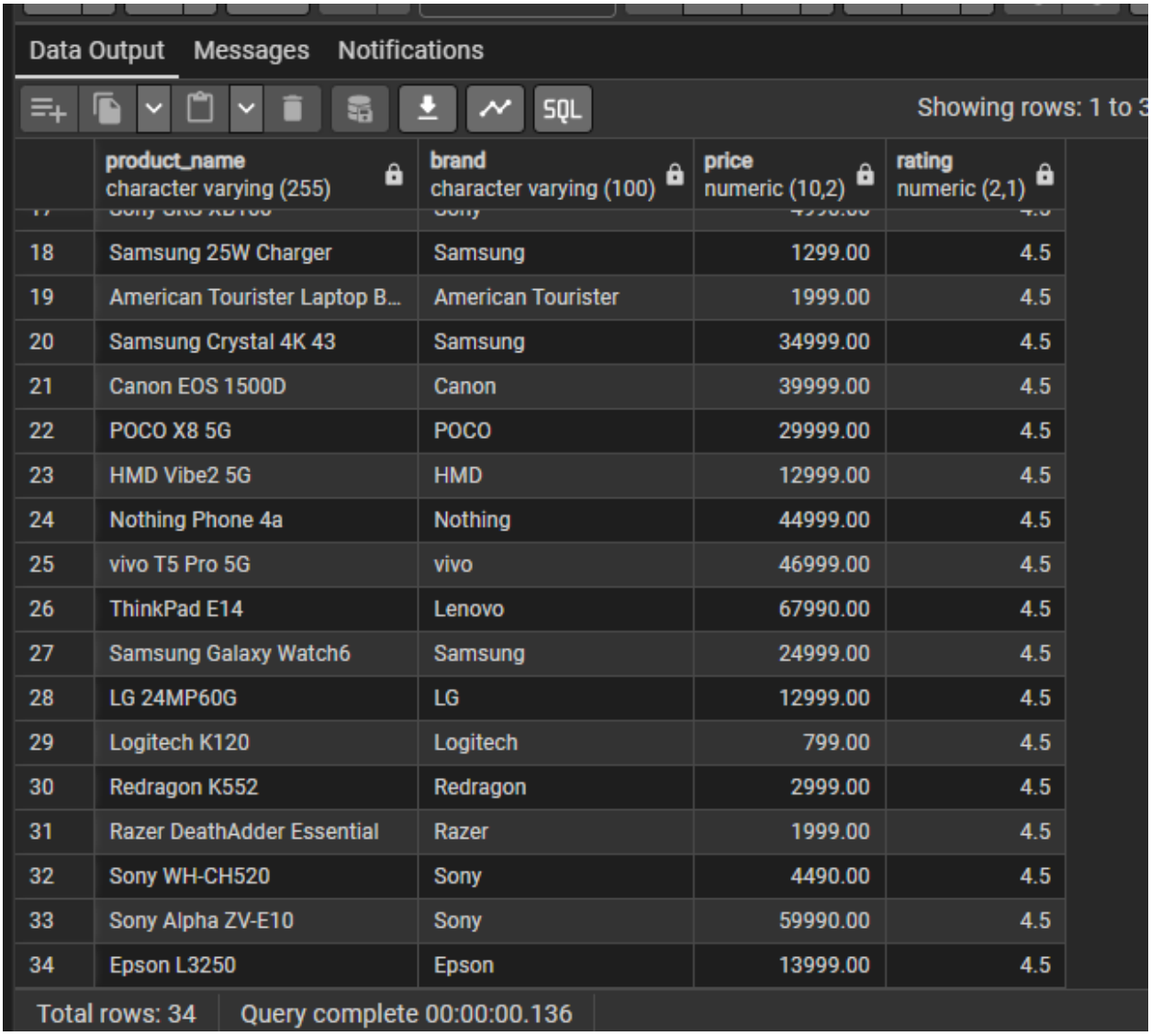


Data Output | Messages | Notifications

Showing rows: 1 to 34

	product_name character varying (255)	brand character varying (100)	price numeric (10,2)	rating numeric (2,1)
8	OnePlus 80W Charger	OnePlus	1999.00	4.6
9	Nikon D7500	Nikon	79999.00	4.6
10	Sony Bravia 43	Sony	49999.00	4.6
11	GoPro HERO12	GoPro	39999.00	4.6
12	Apple Watch SE	Apple	29900.00	4.6
13	Logitech G213	Logitech	3999.00	4.6
14	iPhone 16	Apple	69900.00	4.6
15	WD My Passport 1TB	Western Digital	5999.00	4.5
16	SanDisk Ultra 128GB	SanDisk	999.00	4.5
17	Sony SRS-XB100	Sony	4990.00	4.5
18	Samsung 25W Charger	Samsung	1299.00	4.5
19	American Tourister Laptop B...	American Tourister	1999.00	4.5
20	Samsung Crystal 4K 43	Samsung	34999.00	4.5
21	Canon EOS 1500D	Canon	39999.00	4.5
22	POCO X8 5G	POCO	29999.00	4.5
23	HMD Vibe2 5G	HMD	12999.00	4.5
24	Nothing Phone 4a	Nothing	44999.00	4.5

Total rows: 34 | Query complete 00:00:00.136



Data Output | Messages | Notifications

Showing rows: 1 to 3

	product_name character varying (255)	brand character varying (100)	price numeric (10,2)	rating numeric (2,1)
18	Samsung 25W Charger	Samsung	1299.00	4.5
19	American Tourister Laptop B...	American Tourister	1999.00	4.5
20	Samsung Crystal 4K 43	Samsung	34999.00	4.5
21	Canon EOS 1500D	Canon	39999.00	4.5
22	POCO X8 5G	POCO	29999.00	4.5
23	HMD Vibe2 5G	HMD	12999.00	4.5
24	Nothing Phone 4a	Nothing	44999.00	4.5
25	vivo T5 Pro 5G	vivo	46999.00	4.5
26	ThinkPad E14	Lenovo	67990.00	4.5
27	Samsung Galaxy Watch6	Samsung	24999.00	4.5
28	LG 24MP60G	LG	12999.00	4.5
29	Logitech K120	Logitech	799.00	4.5
30	Redragon K552	Redragon	2999.00	4.5
31	Razer DeathAdder Essential	Razer	1999.00	4.5
32	Sony WH-CH520	Sony	4490.00	4.5
33	Sony Alpha ZV-E10	Sony	59990.00	4.5
34	Epson L3250	Epson	13999.00	4.5

Total rows: 34 | Query complete 00:00:00.136

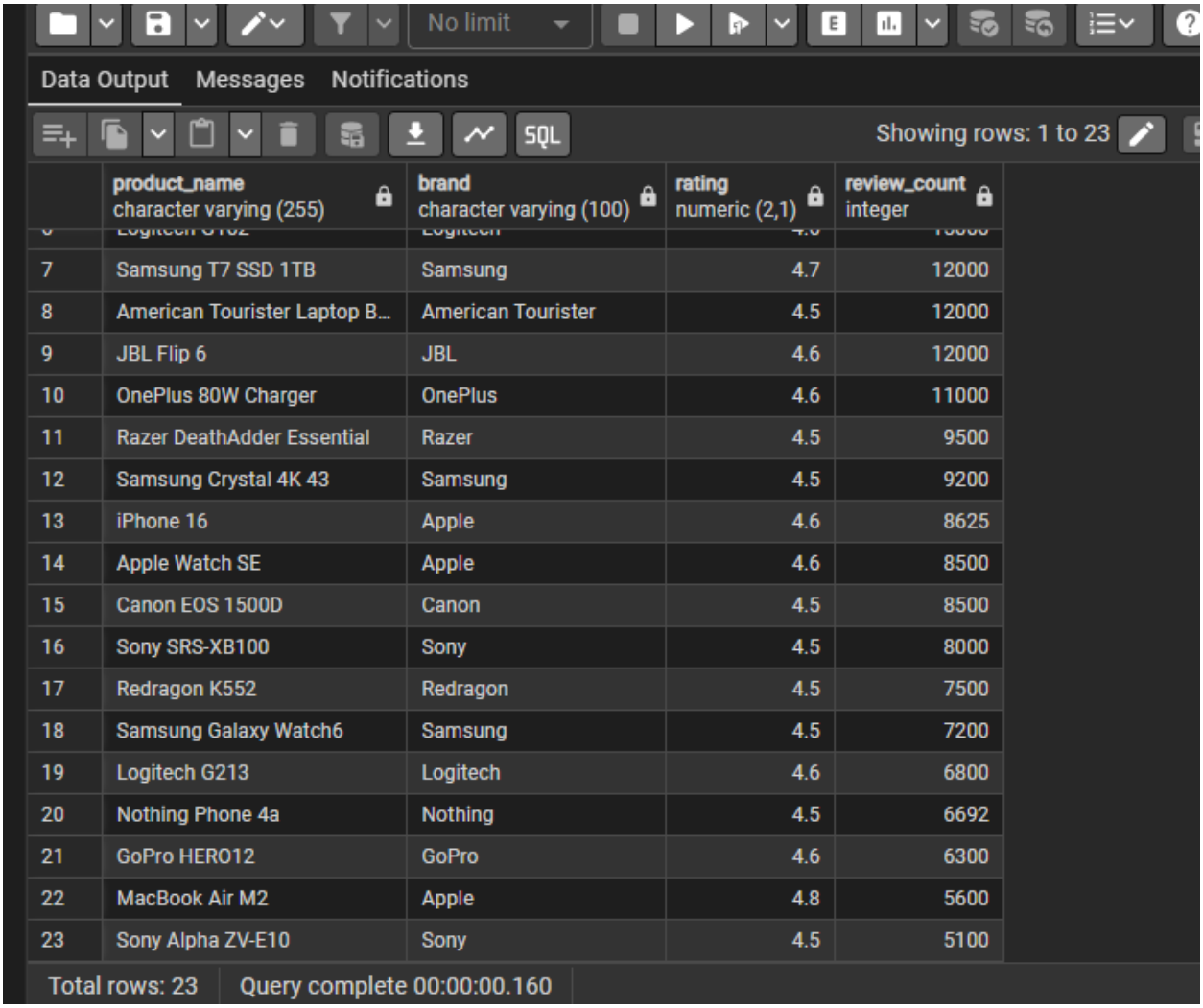
```sql
SELECT product_name, brand, rating, review_count
FROM ecommerce_products1
WHERE rating >= 4.5
  AND review_count >= 5000
ORDER BY review_count DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 23

	product_name character varying (255)	brand character varying (100)	rating numeric (2,1)	review_count integer
1	SanDisk Ultra 128GB	SanDisk	4.5	50000
2	Logitech K120	Logitech	4.5	25000
3	Samsung 25W Charger	Samsung	4.5	19000
4	Epson L3250	Epson	4.5	18000
5	WD My Passport 1TB	Western Digital	4.5	18000
6	Logitech G102	Logitech	4.6	15000
7	Samsung T7 SSD 1TB	Samsung	4.7	12000
8	American Tourister Laptop B...	American Tourister	4.5	12000

Total rows: 23 | Query complete 00:00:00.160



	product_name character varying (255)	brand character varying (100)	rating numeric (2,1)	review_count integer
7	Samsung T7 SSD 1TB	Samsung	4.7	12000
8	American Tourister Laptop B...	American Tourister	4.5	12000
9	JBL Flip 6	JBL	4.6	12000
10	OnePlus 80W Charger	OnePlus	4.6	11000
11	Razer DeathAdder Essential	Razer	4.5	9500
12	Samsung Crystal 4K 43	Samsung	4.5	9200
13	iPhone 16	Apple	4.6	8625
14	Apple Watch SE	Apple	4.6	8500
15	Canon EOS 1500D	Canon	4.5	8500
16	Sony SRS-XB100	Sony	4.5	8000
17	Redragon K552	Redragon	4.5	7500
18	Samsung Galaxy Watch6	Samsung	4.5	7200
19	Logitech G213	Logitech	4.6	6800
20	Nothing Phone 4a	Nothing	4.5	6692
21	GoPro HERO12	GoPro	4.6	6300
22	MacBook Air M2	Apple	4.8	5600
23	Sony Alpha ZV-E10	Sony	4.5	5100

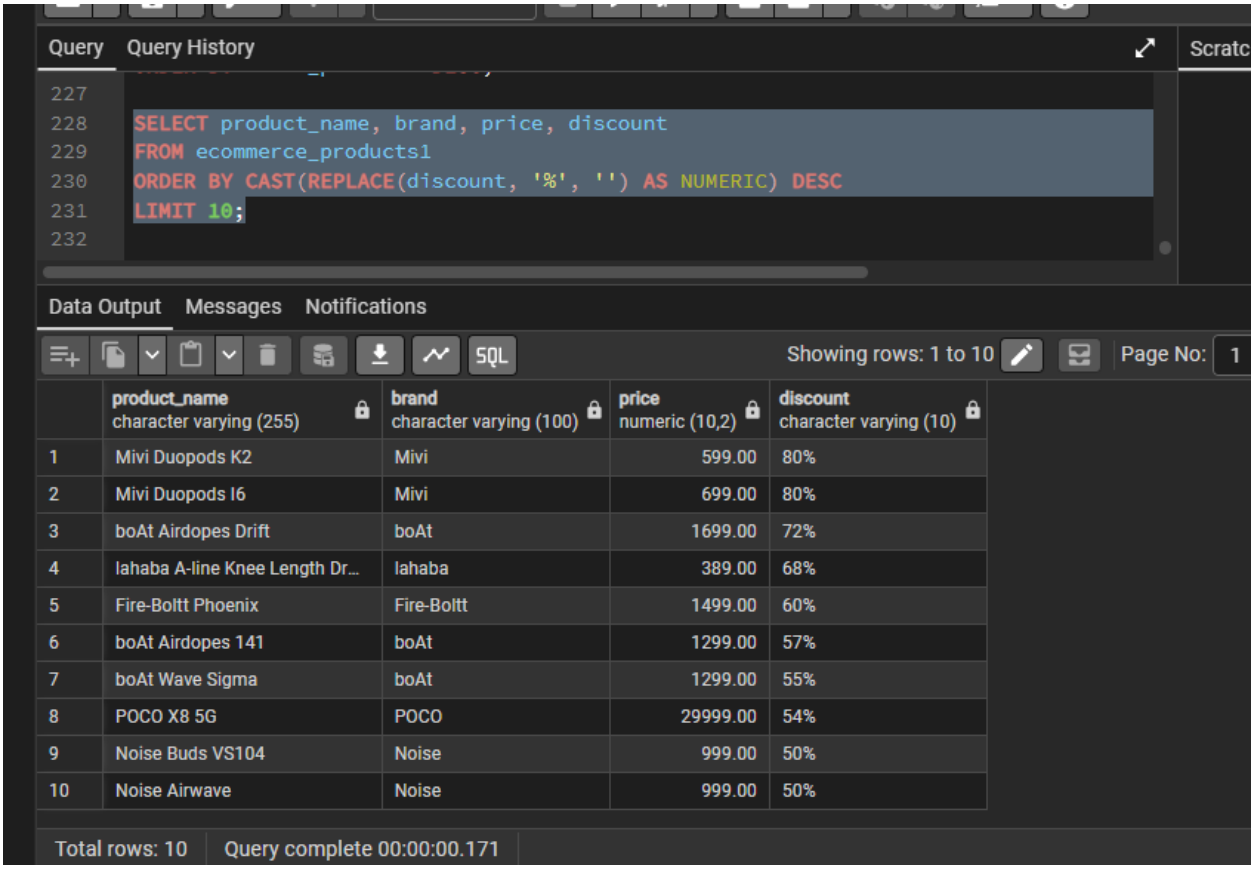
```
222
223  SELECT delivery_promise, COUNT(*) AS total_products
224  FROM ecommerce_products1
225  GROUP BY delivery_promise
226  ORDER BY total_products DESC;
227
```

Data Output | Messages | Notifications

Showing rows: 1 to 8

	delivery_promise character varying (20)	total_products bigint
1	4-Oct	41
2	3-Oct	35
3	5-Oct	22
4	30-Sep	1
5	27-Sep	1
6	12-Oct	1
7	8-Oct	1
8	9-Oct	1

Total rows: 8 | Query complete 00:00:00.128


Query
Query History
Scratc
227
228 SELECT product_name, brand, price, discount
229 FROM ecommerce_products1
230 ORDER BY CAST(REPLACE(discount, '%', '') AS NUMERIC) DESC
231 LIMIT 10;
232
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 10
Page No: 1
product_name character varying (255)
brand character varying (100)
price numeric (10,2)
discount character varying (10)
1 Mivi Duopods K2 Mivi 599.00 80%
2 Mivi Duopods I6 Mivi 699.00 80%
3 boAt Airdopes Drift boAt 1699.00 72%
4 lahaba A-line Knee Length Dr... lahaba 389.00 68%
5 Fire-Boltt Phoenix Fire-Boltt 1499.00 60%
6 boAt Airdopes 141 boAt 1299.00 57%
7 boAt Wave Sigma boAt 1299.00 55%
8 POCO X8 5G POCO 29999.00 54%
9 Noise Buds VS104 Noise 999.00 50%
10 Noise Airwave Noise 999.00 50%
Total rows: 10
Query complete 00:00:00.171

Query | Query History

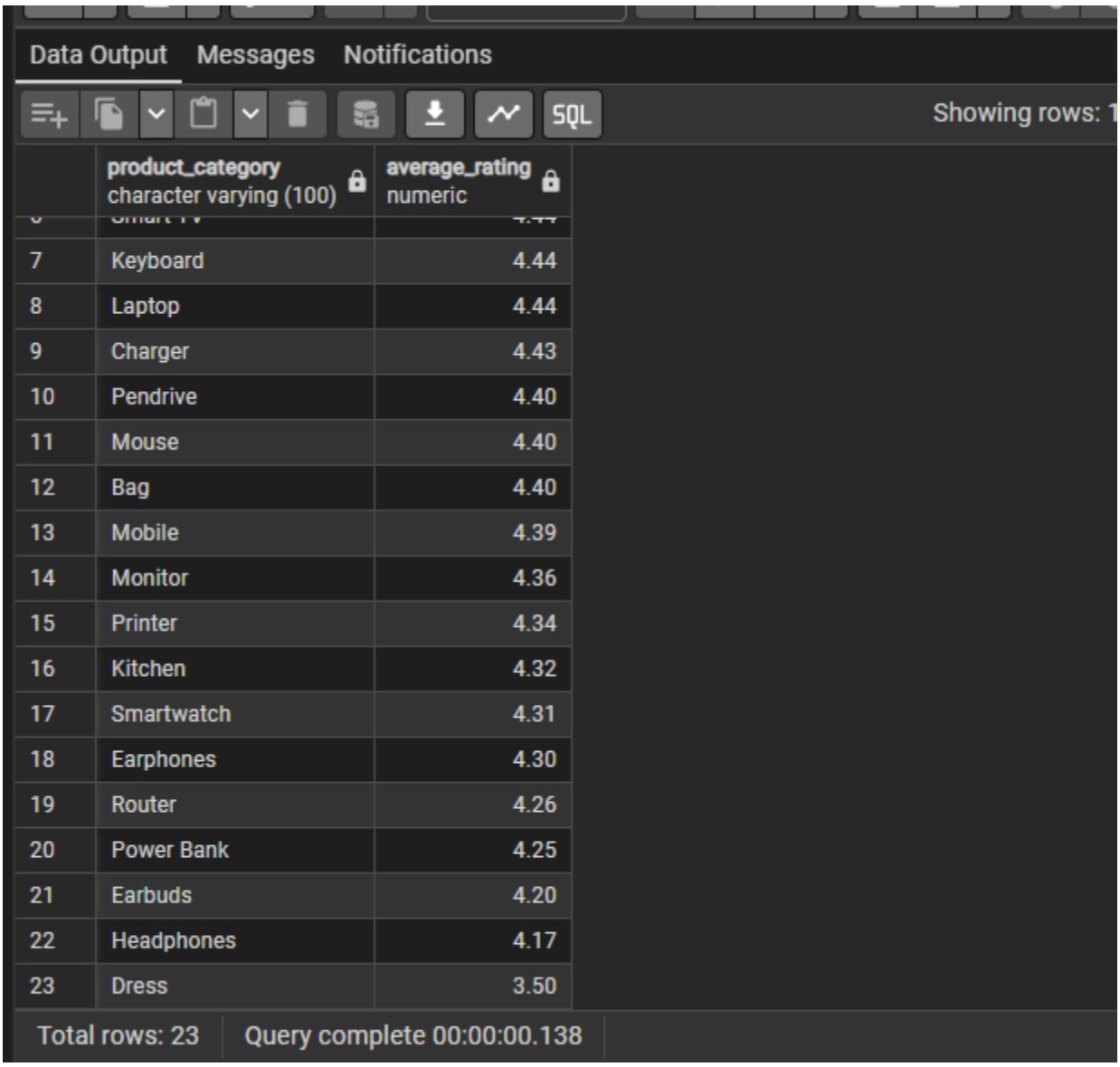
```
232
233 SELECT
234     product_category,
235     ROUND(AVG(rating), 2) AS average_rating
236 FROM ecommerce_products1
237 GROUP BY product_category
238 ORDER BY average_rating DESC;
239
```

Data Output | Messages | Notifications

Showing rows: 1 to 23

	product_category character varying (100)	average_rating numeric
1	SSD	4.70
2	Camera	4.58
3	Memory Card	4.50
4	Speaker	4.46
5	Hard Drive	4.45
6	Smart TV	4.44
7	Keyboard	4.44
8	Laptop	4.44
9	Charger	4.43

Total rows: 23 | Query complete 00:00:00.138

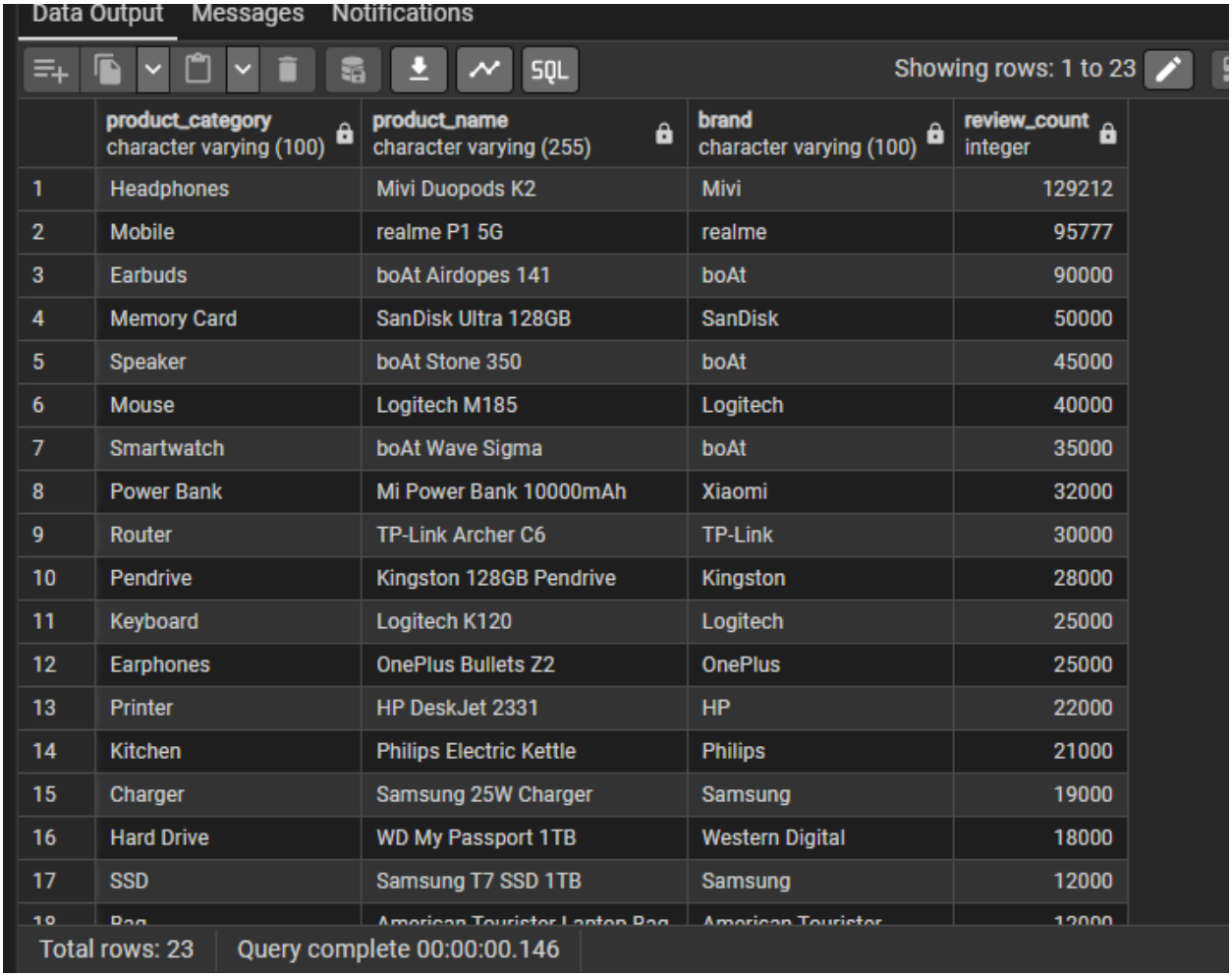


Data Output | Messages | Notifications

Showing rows: 1

	product_category character varying (100)	average_rating numeric
7	Keyboard	4.44
8	Laptop	4.44
9	Charger	4.43
10	Pendrive	4.40
11	Mouse	4.40
12	Bag	4.40
13	Mobile	4.39
14	Monitor	4.36
15	Printer	4.34
16	Kitchen	4.32
17	Smartwatch	4.31
18	Earphones	4.30
19	Router	4.26
20	Power Bank	4.25
21	Earbuds	4.20
22	Headphones	4.17
23	Dress	3.50

Total rows: 23 | Query complete 00:00:00.138

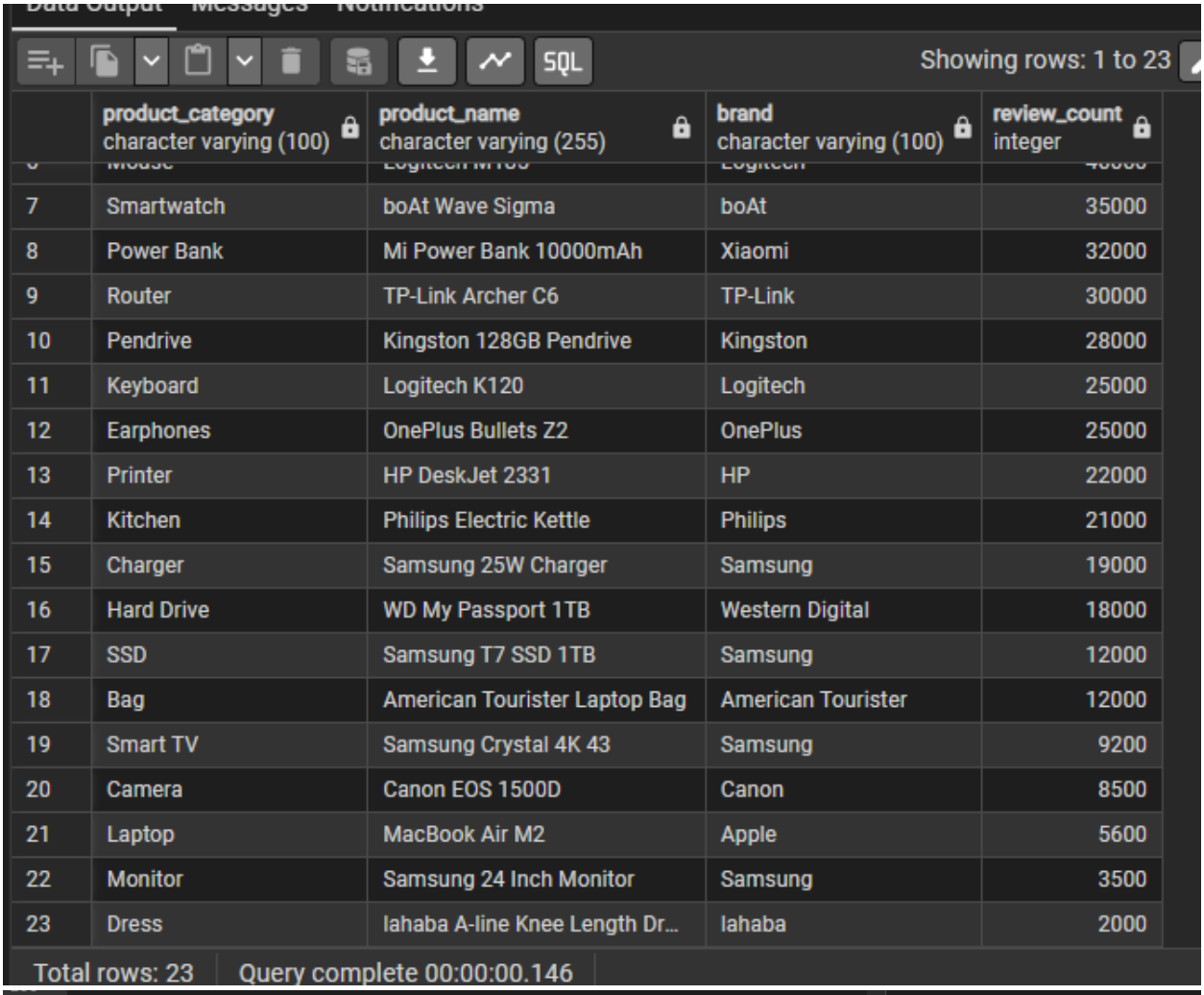


Data Output | Messages | Notifications

SQL | Showing rows: 1 to 23

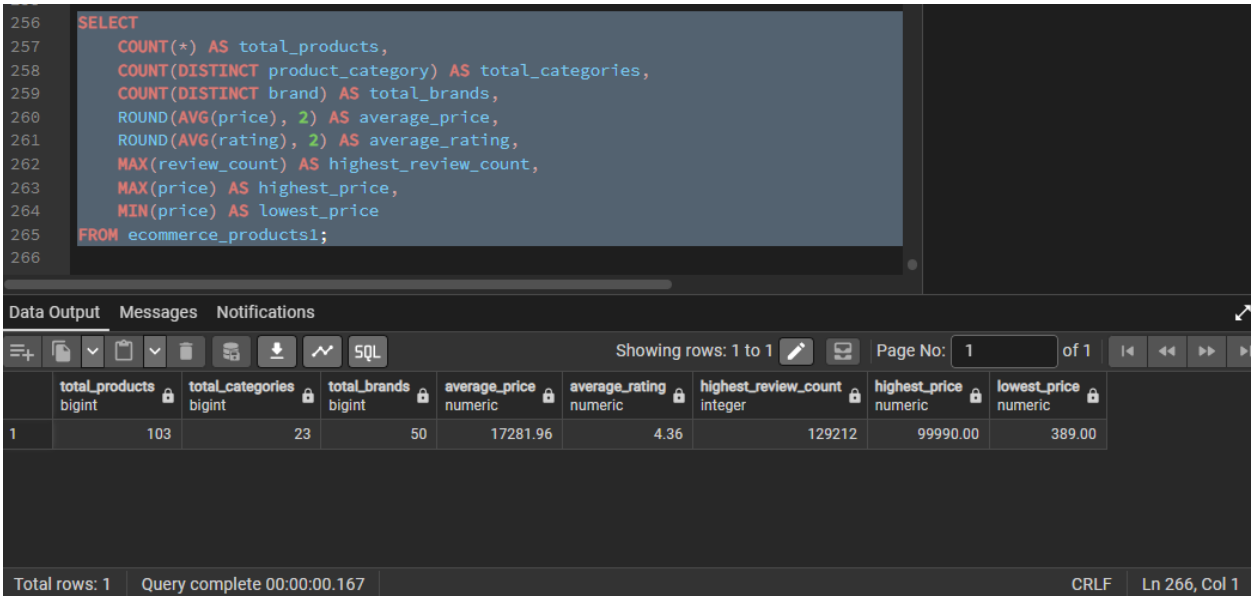
	product_category character varying (100)	product_name character varying (255)	brand character varying (100)	review_count integer
1	Headphones	Mivi Duopods K2	Mivi	129212
2	Mobile	realme P1 5G	realme	95777
3	Earbuds	boAt Airdopes 141	boAt	90000
4	Memory Card	SanDisk Ultra 128GB	SanDisk	50000
5	Speaker	boAt Stone 350	boAt	45000
6	Mouse	Logitech M185	Logitech	40000
7	Smartwatch	boAt Wave Sigma	boAt	35000
8	Power Bank	Mi Power Bank 10000mAh	Xiaomi	32000
9	Router	TP-Link Archer C6	TP-Link	30000
10	Pendrive	Kingston 128GB Pendrive	Kingston	28000
11	Keyboard	Logitech K120	Logitech	25000
12	Earphones	OnePlus Bullets Z2	OnePlus	25000
13	Printer	HP DeskJet 2331	HP	22000
14	Kitchen	Philips Electric Kettle	Philips	21000
15	Charger	Samsung 25W Charger	Samsung	19000
16	Hard Drive	WD My Passport 1TB	Western Digital	18000
17	SSD	Samsung T7 SSD 1TB	Samsung	12000
18	Bag	American Tourister Laptop Bag	American Tourister	12000

Total rows: 23 | Query complete 00:00:00.146



	product_category character varying (100)	product_name character varying (255)	brand character varying (100)	review_count integer
7	Smartwatch	boAt Wave Sigma	boAt	35000
8	Power Bank	Mi Power Bank 10000mAh	Xiaomi	32000
9	Router	TP-Link Archer C6	TP-Link	30000
10	Pendrive	Kingston 128GB Pendrive	Kingston	28000
11	Keyboard	Logitech K120	Logitech	25000
12	Earphones	OnePlus Bullets Z2	OnePlus	25000
13	Printer	HP DeskJet 2331	HP	22000
14	Kitchen	Philips Electric Kettle	Philips	21000
15	Charger	Samsung 25W Charger	Samsung	19000
16	Hard Drive	WD My Passport 1TB	Western Digital	18000
17	SSD	Samsung T7 SSD 1TB	Samsung	12000
18	Bag	American Tourister Laptop Bag	American Tourister	12000
19	Smart TV	Samsung Crystal 4K 43	Samsung	9200
20	Camera	Canon EOS 1500D	Canon	8500
21	Laptop	MacBook Air M2	Apple	5600
22	Monitor	Samsung 24 Inch Monitor	Samsung	3500
23	Dress	lahaba A-line Knee Length Dr...	lahaba	2000

Total rows: 23 | Query complete 00:00:00.146



```sql
SELECT
    COUNT(*) AS total_products,
    COUNT(DISTINCT product_category) AS total_categories,
    COUNT(DISTINCT brand) AS total_brands,
    ROUND(AVG(price), 2) AS average_price,
    ROUND(AVG(rating), 2) AS average_rating,
    MAX(review_count) AS highest_review_count,
    MAX(price) AS highest_price,
    MIN(price) AS lowest_price
FROM ecommerce_products1;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page No: 1 of 1

	total_products bigint	total_categories bigint	total_brands bigint	average_price numeric	average_rating numeric	highest_review_count integer	highest_price numeric	lowest_price numeric
1	103	23	50	17281.96	4.36	129212	99990.00	389.00

Total rows: 1 | Query complete 00:00:00.167 | CRLF | Ln 266, Col 1

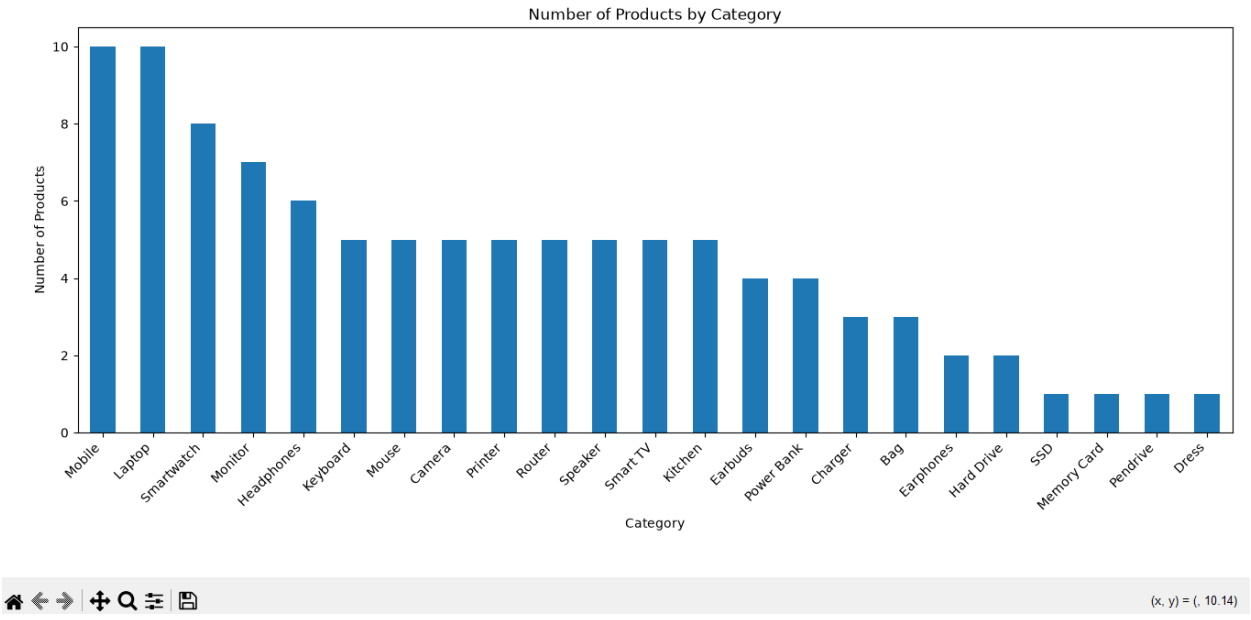
10. PYTHON EXPLORATORY DATA ANALYSIS (EDA)

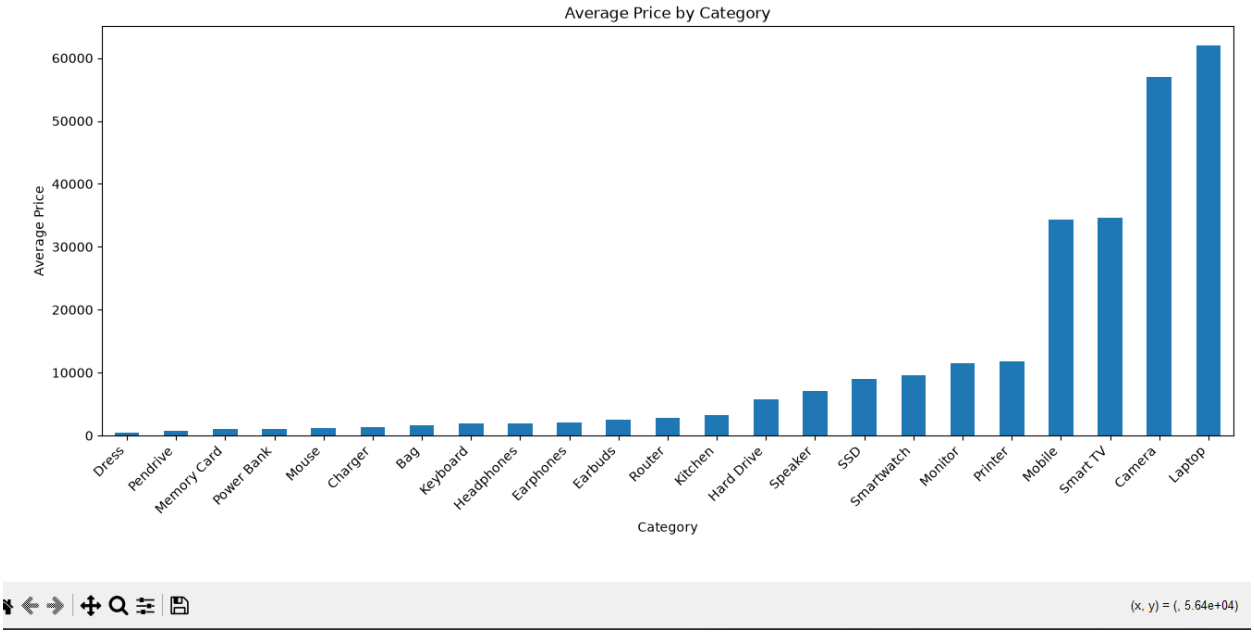
Python was used to perform Exploratory Data Analysis on the collected product dataset.

Pandas and NumPy can be used for data handling and numerical analysis, while Matplotlib can be used for visualizations.

The analysis focuses on understanding the structure of the dataset, examining product prices and ratings, studying review counts and discounts, and identifying patterns in the collected data.

Actual Python outputs and charts can be included in this section.


Number of Products by Category
Number of Products
Category
0
2
4
6
8
10
Mobile
Laptop
Smartwatch
Monitor
Headphones
Keyboard
Mouse
Camera
Printer
Router
Speaker
Smart TV
Kitchen
Earbuds
Power Bank
Charger
Bag
Earphones
Hard Drive
SSD
Memory Card
Pendrive
Dress
(x, y) = (, 10.14)


Average Price by Category
Average Price
Category
0
10000
20000
30000
40000
50000
60000
Dress
Pendrive
Memory Card
Power Bank
Mouse
Charger
Bag
Keyboard
Headphones
Earphones
Earbuds
Router
Kitchen
Hard Drive
Speaker
SSD
Smartwatch
Monitor
Printer
Mobile
Smart TV
Camera
Laptop
(x, y) = (, 5.64e+04)

```
========== RESTART: C:/Users/HP/Desktop/Ecommerce_EDA/ecommerce_eda.py =
Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 103 entries, 0 to 102
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   product_name      103 non-null    str
 1   product_category  103 non-null    str
 2   brand             103 non-null    str
 3   price             103 non-null    float64
 4   discount          103 non-null    str
 5   rating            103 non-null    float64
 6   review_count      103 non-null    int64
 7   seller            103 non-null    str
 8   delivery_promise  103 non-null    str
dtypes: float64(2), int64(1), str(6)
memory usage: 7.4 KB
None

Duplicate Records:
0

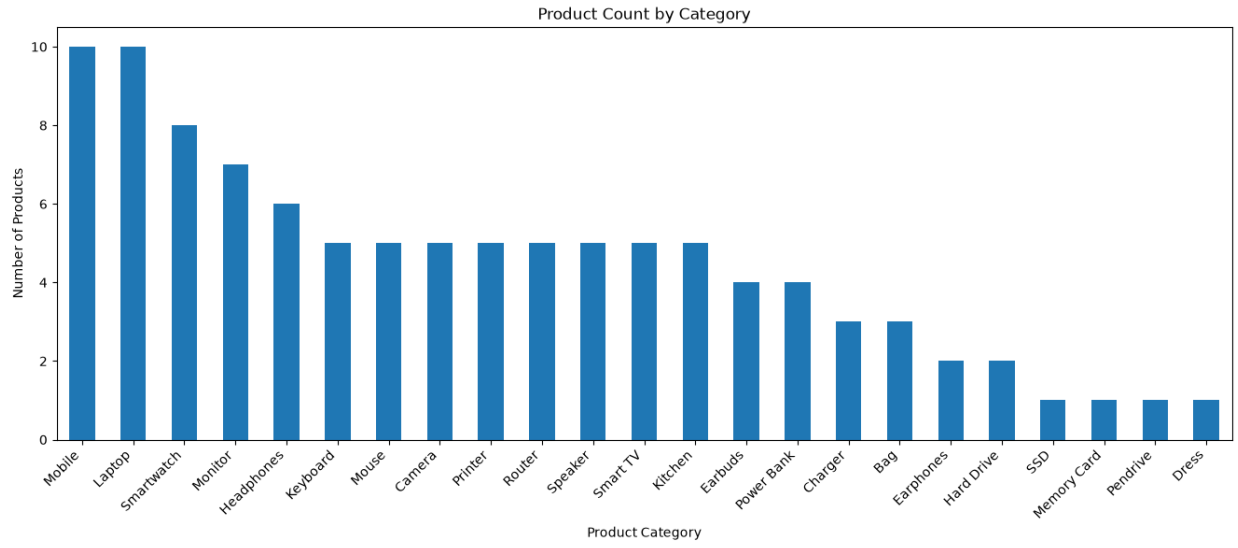
Average Price by Category:
product_category
Bag              1599.00
Camera          56997.20
Charger          1332.33
Dress             389.00
Earbuds          2449.00
Earphones        2149.00
Hard Drive       5749.00
Headphones       1914.17
Keyboard         1879.00
Kitchen          3259.00
Laptop          61990.00
Memory Card       999.00
Mobile          34388.80
```

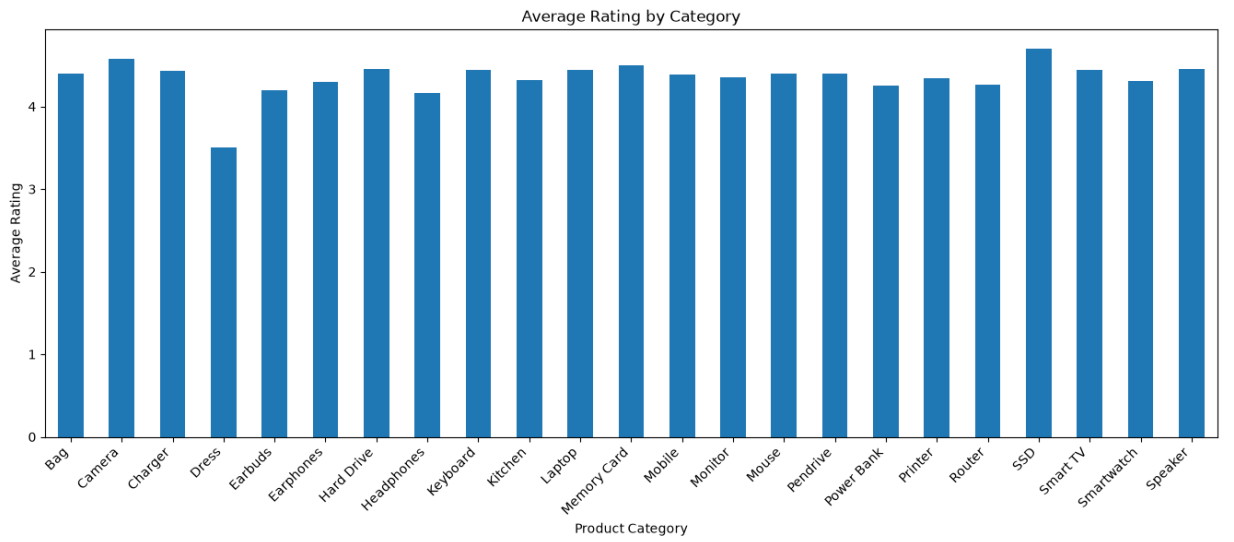
```
Memory Card      999.00
Mobile         34388.80
Monitor        11496.43
Mouse           1139.00
Pendrive         699.00
Power Bank      1074.00
Printer        11797.20
Router          2799.00
SSD             8999.00
Smart TV       34599.00
Smartwatch      9586.62
Speaker         7097.20
Name: price, dtype: float64

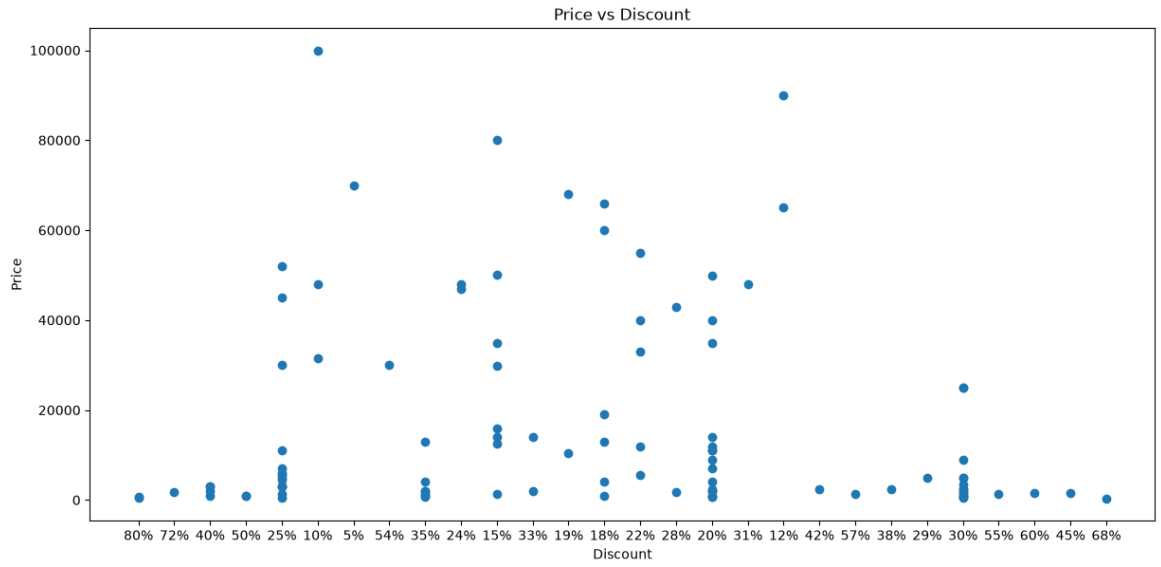
Average Rating by Category:
product_category
Bag             4.40
Camera          4.58
Charger         4.43
Dress           3.50
Earbuds         4.20
Earphones       4.30
Hard Drive      4.45
Headphones      4.17
Keyboard        4.44
Kitchen         4.32
Laptop          4.44
Memory Card     4.50
Mobile          4.39
Monitor         4.36
Mouse           4.40
Pendrive        4.40
Power Bank      4.25
Printer         4.34
Router          4.26
SSD             4.70
Smart TV        4.44
Smartwatch      4.31
Speaker         4.46
Name: rating, dtype: float64
```

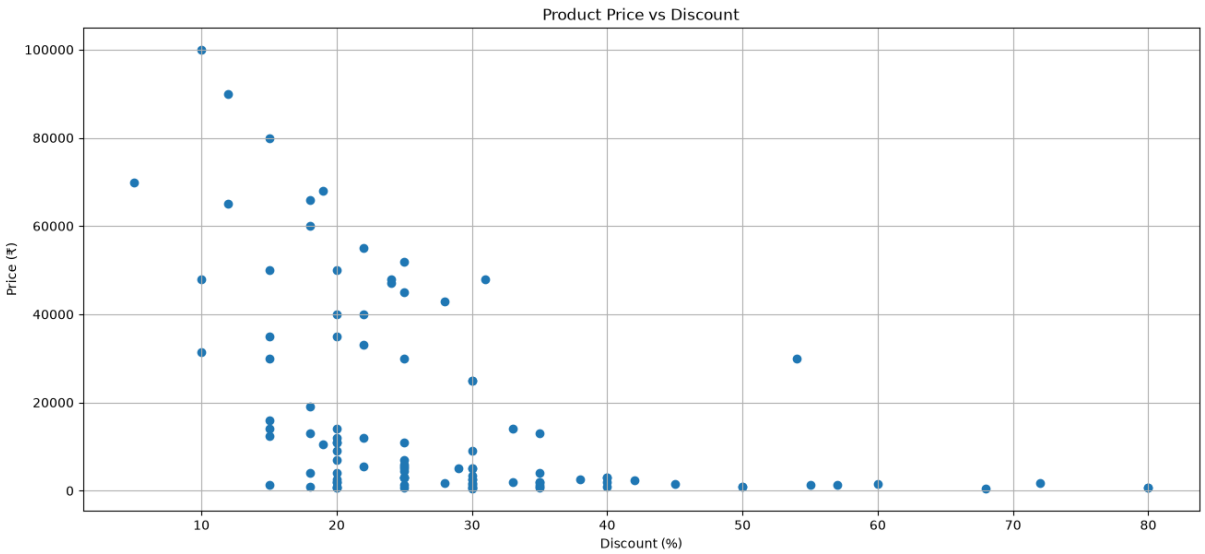
```
Top 10 Expensive Products:
         product_name    price
24     MacBook Air M3  99990.0
25     MacBook Air M2  89990.0
58        Nikon D7500  79999.0
7           iPhone 16  69900.0
22       ThinkPad E14  67990.0
16       Galaxy Book4  65990.0
60      Canon EOS R50  64999.0
59  Sony Alpha ZV-E10  59990.0
17      Inspiron 3530  54990.0
21              HP 15s  51990.0

Top 10 Most Reviewed Products:
              product_name  review_count  rating
0          Mivi Duopods K2        129212     3.9
14            realme P1 5G         95777     4.3
28       boAt Airdopes 141         90000     4.1
1          Mivi Duopods I6         57094     3.9
75     SanDisk Ultra 128GB         50000     4.5
3            JBL Tune 510BT        45000     4.4
78           boAt Stone 350        45000     4.3
52            Logitech M185        40000     4.4
35         boAt Wave Sigma         35000     4.1
82  Mi Power Bank 10000mAh         32000     4.3
>>>
```




Average Rating by Category
Average Rating
0
1
2
3
4
Bag
Camera
Charger
Dress
Earbuds
Earphones
Hard Drive
Headphones
Keyboard
Kitchen
Laptop
Memory Card
Mobile
Monitor
Mouse
Pendrive
Power Bank
Printer
Router
SSD
Smart TV
Smartwatch
Speaker
Product Category


Price vs Discount
Price
0
20000
40000
60000
80000
100000
80% 72% 40% 50% 25% 10% 5% 54% 35% 24% 15% 33% 19% 18% 22% 28% 20% 31% 12% 42% 57% 38% 29% 30% 55% 60% 45% 68%
Discount




```
--- BUSINESS INSIGHTS ---
Total Products: 103
Total Categories: 23
Products by Category:
product_category
Mobile          10
Laptop          10
Smartwatch       8
Monitor          7
Headphones       6
Keyboard         5
Mouse            5
Camera           5
Printer          5
Router           5
Speaker          5
Smart TV         5
Kitchen          5
Earbuds          4
Power Bank       4
Charger          3
Bag              3
Earphones        2
Hard Drive       2
SSD              1
Memory Card      1
Pendrive         1
Dress            1
Name: count, dtype: int64

Highest Priced Product:
```

```
Highest Priced Product:
product_name        MacBook Air M3
product_category            Laptop
price                      99990.0
Name: 24, dtype: object

Lowest Priced Product:
product_name        lahaba A-line Knee Length Dress
product_category                              Dress
price                                         389.0
Name: 102, dtype: object

Highest Rated Product:
product_name    MacBook Air M3
rating                     4.8
review_count              4200
Name: 24, dtype: object

Most Reviewed Product:
product_name    Mivi Duopods K2
rating                      3.9
review_count             129212
Name: 0, dtype: object

Average Price: 17281.96
Average Rating: 4.36
>
```

```
Earphones       2
Hard Drive      2
SSD             1
Memory Card     1
Pendrive        1
Dress           1
Name: count, dtype: int64

Highest Priced Product:
product_name        MacBook Air M3
product_category            Laptop
price                      99990.0
Name: 24, dtype: object

Lowest Priced Product:
product_name        lahaba A-line Knee Length Dres
product_category                              Dres
price                                         389.
Name: 102, dtype: object

Highest Rated Product:
product_name    MacBook Air M3
rating                     4.8
review_count              4200
Name: 24, dtype: object

Most Reviewed Product:
product_name    Mivi Duopods K2
rating                      3.9
review_count             129212
Name: 0, dtype: object

Average Price: 17281.96
Average Rating: 4.36
>>
```

```
--- BUSINESS INSIGHTS ---
Total Products: 103
Total Categories: 23
Products by Category:
product_category
Mobile          10
Laptop          10
Smartwatch       8
Monitor          7
Headphones       6
Keyboard         5
Mouse            5
Camera           5
Printer          5
Router           5
Speaker          5
Smart TV         5
Kitchen          5
Earbuds          4
Power Bank       4
Charger          3
Bag              3
Earphones        2
Hard Drive       2
SSD              1
Memory Card      1
Pendrive         1
Dress            1
Name: count, dtype: int64

Highest Priced Product:
product_name        MacBook Air M3
product_category            Laptop
price                      99990.0
Name: 24, dtype: object
```

---

11. POWER BI DASHBOARD

Microsoft Power BI was used to create a dashboard for visualizing the collected e-commerce product data.

The dashboard includes visual analysis of:

- Category-wise Price Analysis

- Brand-wise Average Rating

- Brand-wise Product Count

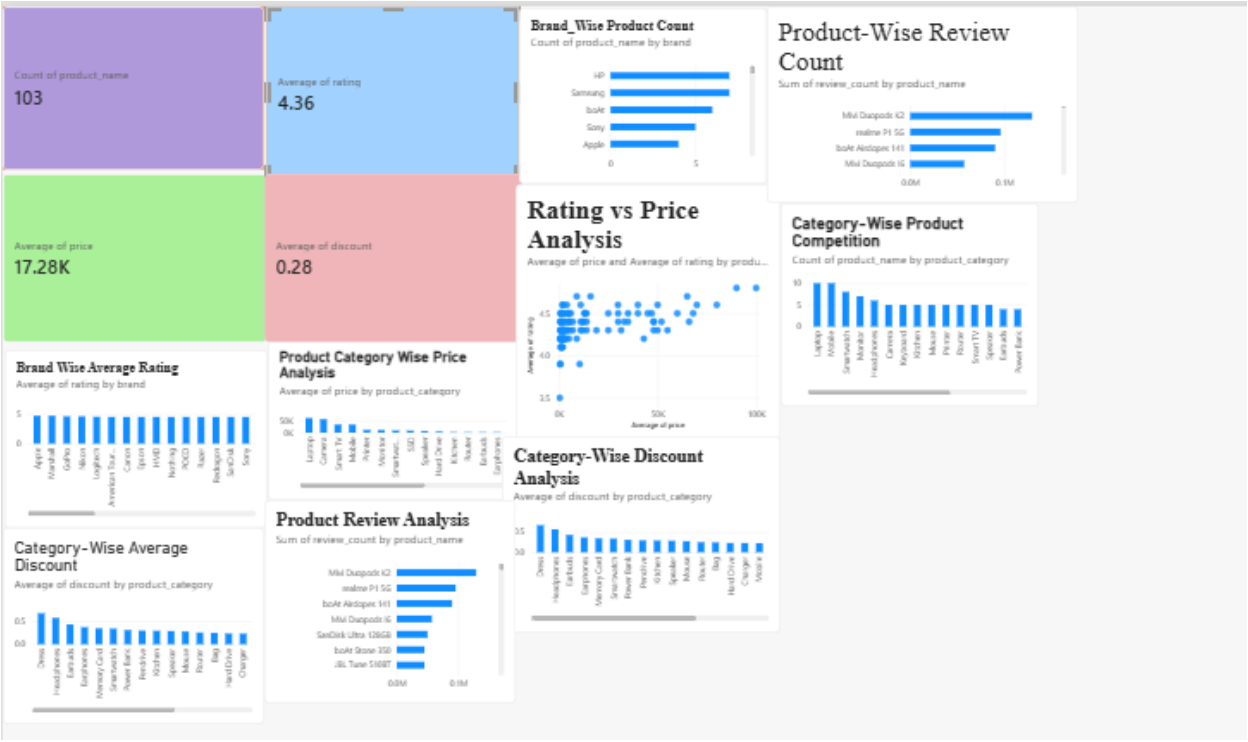
- Rating vs Price Analysis

- Product Review Analysis

- Category-wise Average Discount

- Category-wise Product Competition

These visualizations help present the analysis in an understandable graphical format.



12.BUSINESS INSIGHTS

Project Title :- E-Commerce Product Price & Review Analysis

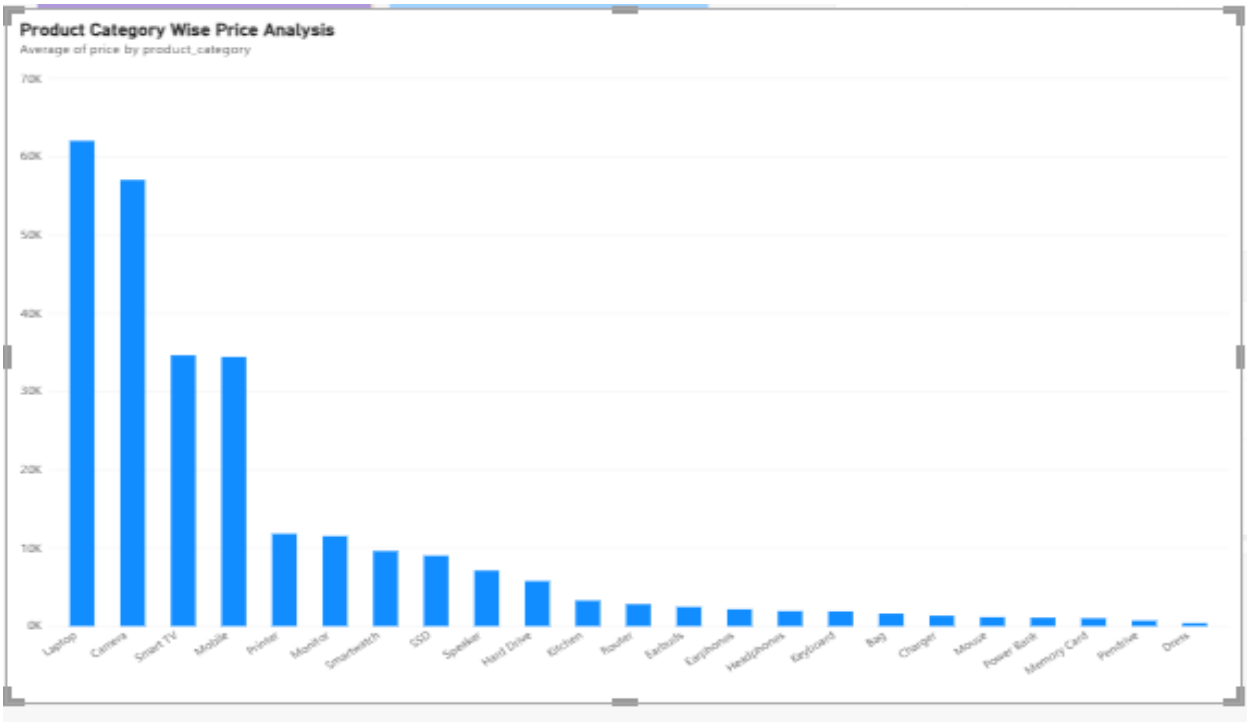
Question 1: Which category has the highest average price?

Answer:

Based on the Product Category Wise Price Analysis chart, **Laptops** have the highest average price at approximately 62,000, followed by **Cameras** at approximately 57,000. Smart TVs and Mobiles have average prices of around 34,000.

Business Insight:

Laptops and cameras are the highest-priced product categories in the dataset. Businesses can use this information to identify high-value product categories, plan pricing strategies, and focus on products that may generate higher revenue per sale.



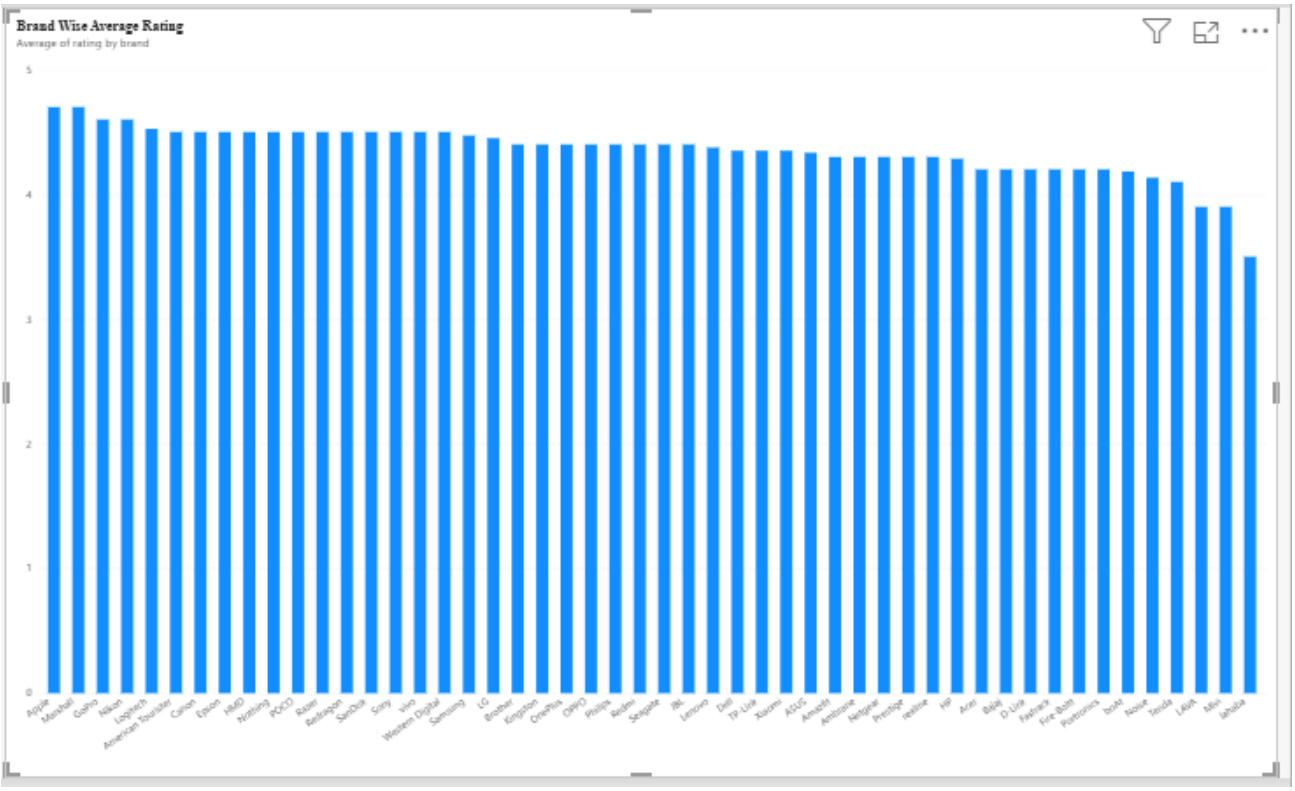
Question 2: Which brand has the highest average product rating?

Answer:

Based on the Brand Wise Average Rating chart, **boAt and Motorola** have the highest average ratings, at approximately 4.7 out of 5. Several other brands also have average ratings above 4.5.

Business Insight:

Brands with higher average ratings may have stronger customer satisfaction. Businesses can use this information to identify well-rated brands, compare product quality, and make better product selection and marketing decisions.



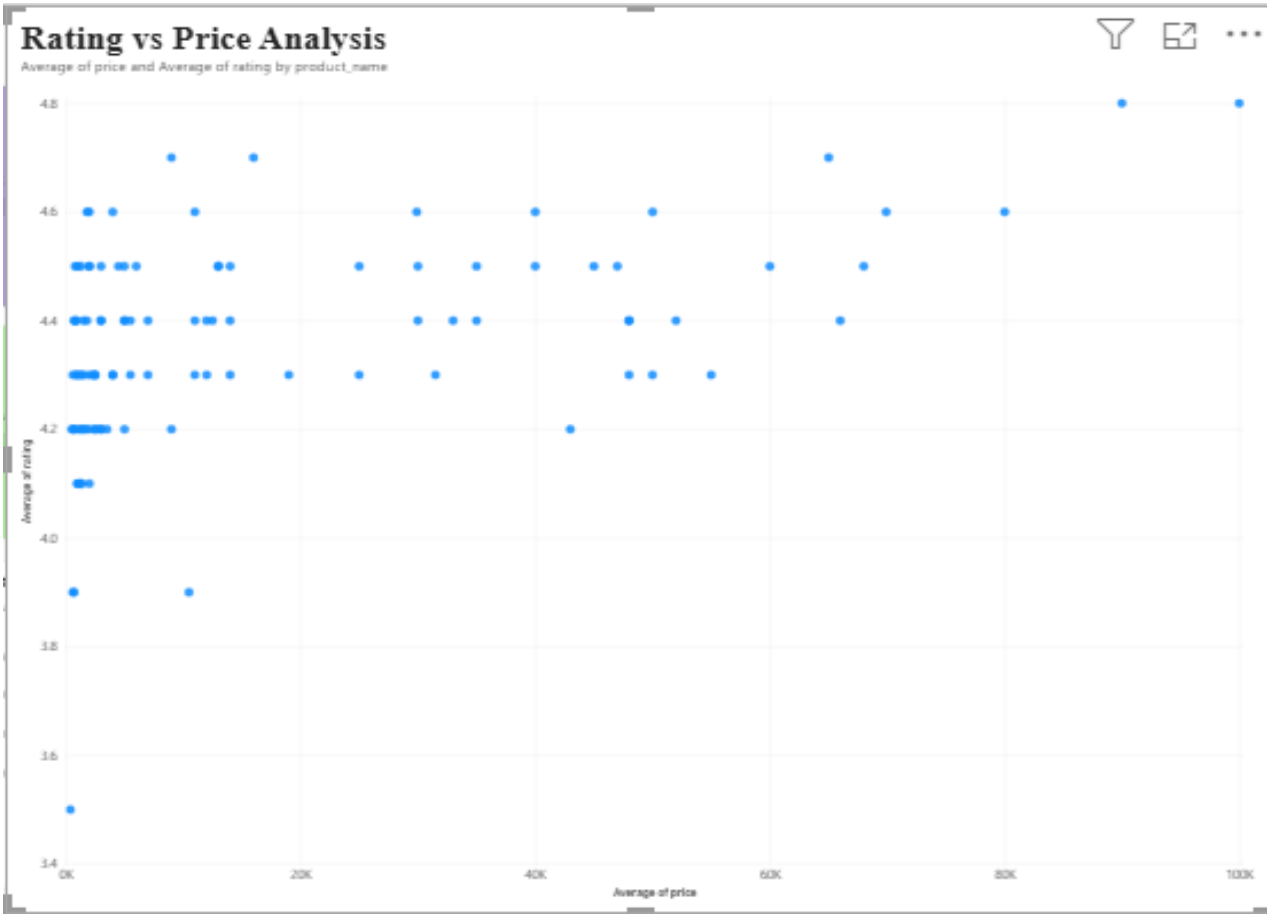
Question 3: Does product rating vary with price?

Answer:

Based on the Rating vs Price Analysis chart, product ratings vary across different price ranges. Most products have ratings between 4.2 and 4.6, while a few products have ratings around 4.8. Some products have lower ratings despite having lower prices, and certain high-priced products also receive good ratings.

Business Insight:

The chart suggests that higher prices do not always guarantee higher product ratings. Businesses should consider customer satisfaction, product quality, and pricing together when developing their pricing strategies.



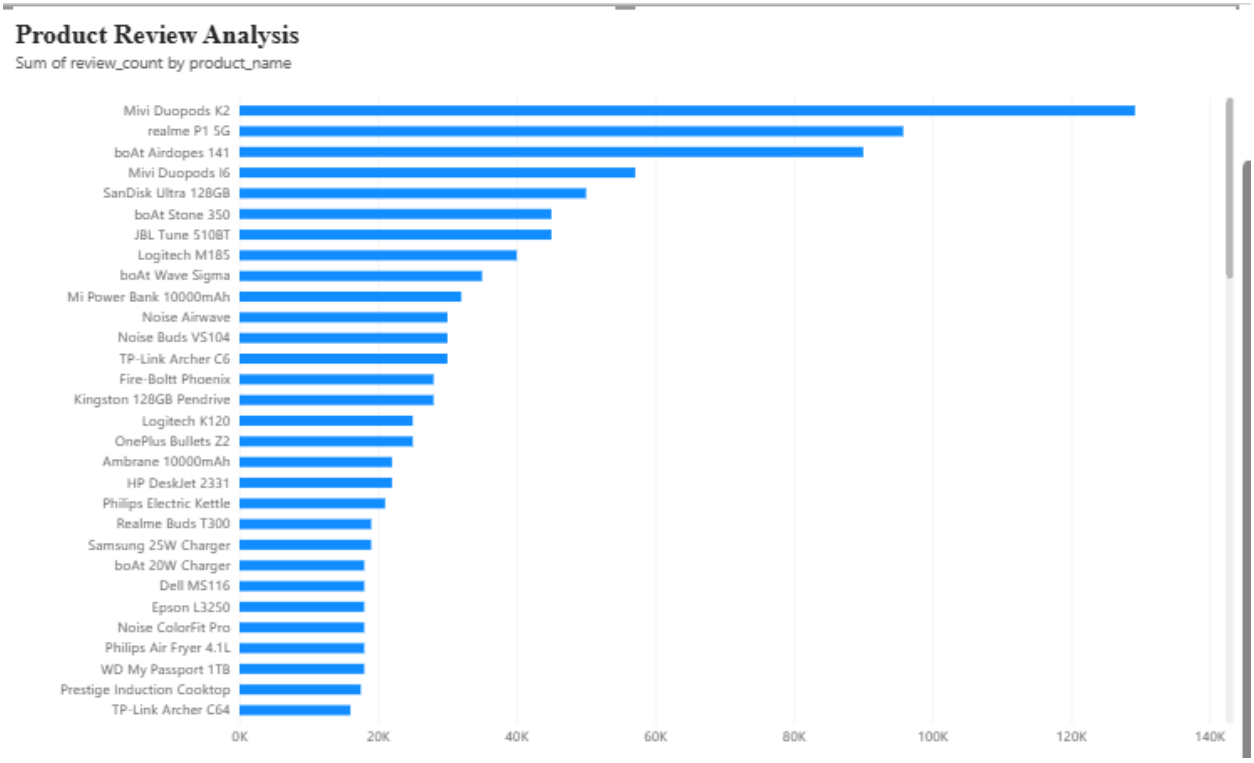
Question 4: Which products have many reviews?

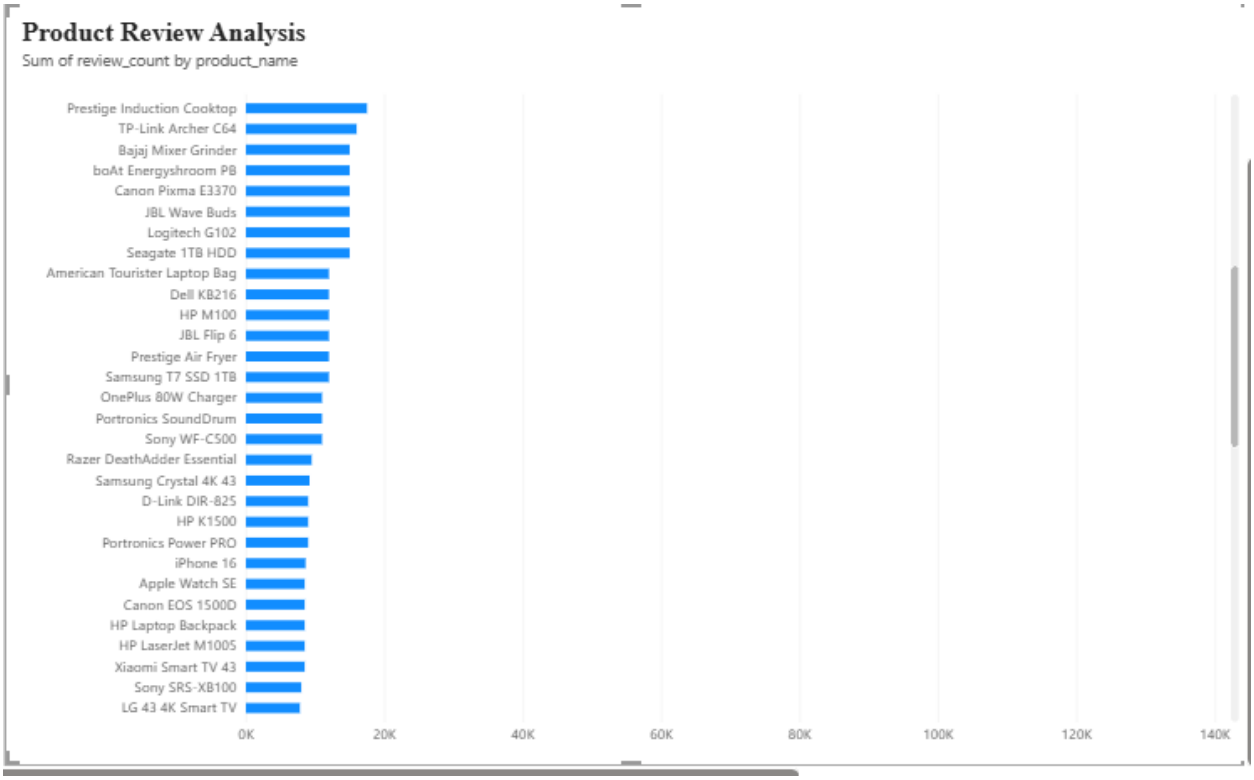
Answer:

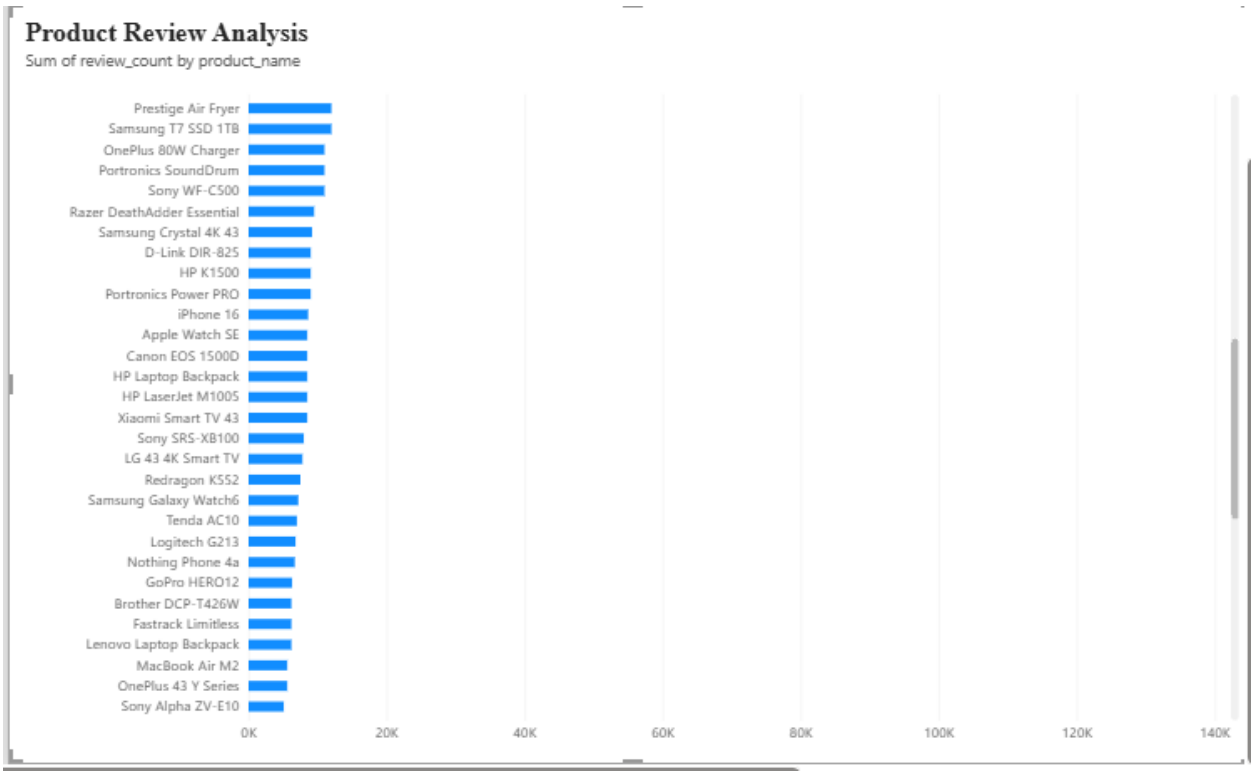
Based on the Product Review Analysis chart, the Miwi Duopods K2 has the highest review count, followed by realme P1 5G and boAt Airdopes 141. Other products with high review counts include Miwi Duopods I6 and SanDisk Ultra 128GB.

Business Insight:

These products receive comparatively more customer reviews, indicating higher customer engagement. Businesses can use this information to identify popular products, understand market demand, and make better product marketing and inventory decisions.

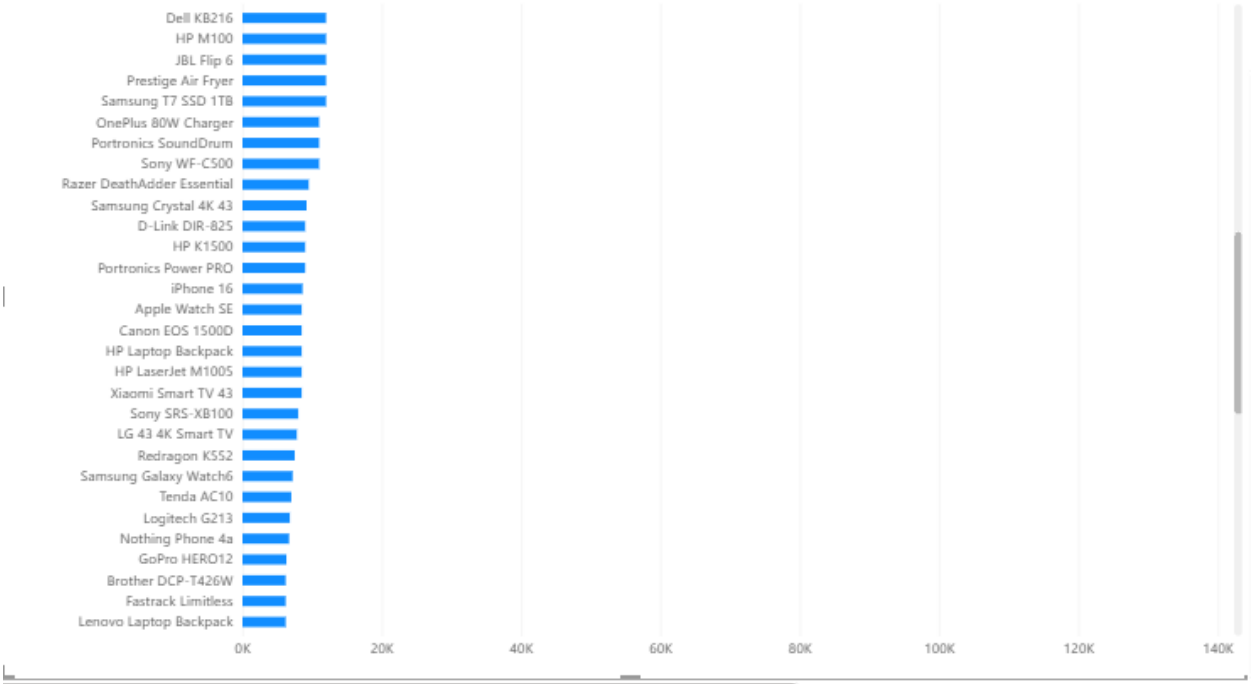



Product Review Analysis
Sum of review_count by product_name
Prestige Induction Cooktop
TP-Link Archer C64
Bajaj Mixer Grinder
boAt Energyshroom PB
Canon Pixma E3370
JBL Wave Buds
Logitech G102
Seagate 1TB HDD
American Tourister Laptop Bag
Dell KB216
HP M100
JBL Flip 6
Prestige Air Fryer
Samsung T7 SSD 1TB
OnePlus 80W Charger
Portronics SoundDrum
Sony WF-C500
Razer DeathAdder Essential
Samsung Crystal 4K 43
D-Link DIR-825
HP K1500
Portronics Power PRO
iPhone 16
Apple Watch SE
Canon EOS 1500D
HP Laptop Backpack
HP LaserJet M1005
Xiaomi Smart TV 43
Sony SRS-XB100
LG 43 4K Smart TV
0K
20K
40K
60K
80K
100K
120K
140K


Product Review Analysis
Sum of review_count by product_name
Prestige Air Fryer
Samsung T7 SSD 1TB
OnePlus 80W Charger
Portronics SoundDrum
Sony WF-C500
Razer DeathAdder Essential
Samsung Crystal 4K 43
D-Link DIR-825
HP K1500
Portronics Power PRO
iPhone 16
Apple Watch SE
Canon EOS 1500D
HP Laptop Backpack
HP LaserJet M1005
Xiaomi Smart TV 43
Sony SRS-XB100
LG 43 4K Smart TV
Redragon K552
Samsung Galaxy Watch6
Tenda AC10
Logitech G213
Nothing Phone 4a
GoPro HERO12
Brother DCP-T426W
Fastrack Limitless
Lenovo Laptop Backpack
MacBook Air M2
OnePlus 43 Y Series
Sony Alpha ZV-E10
0K
20K
40K
60K
80K
100K
120K
140K

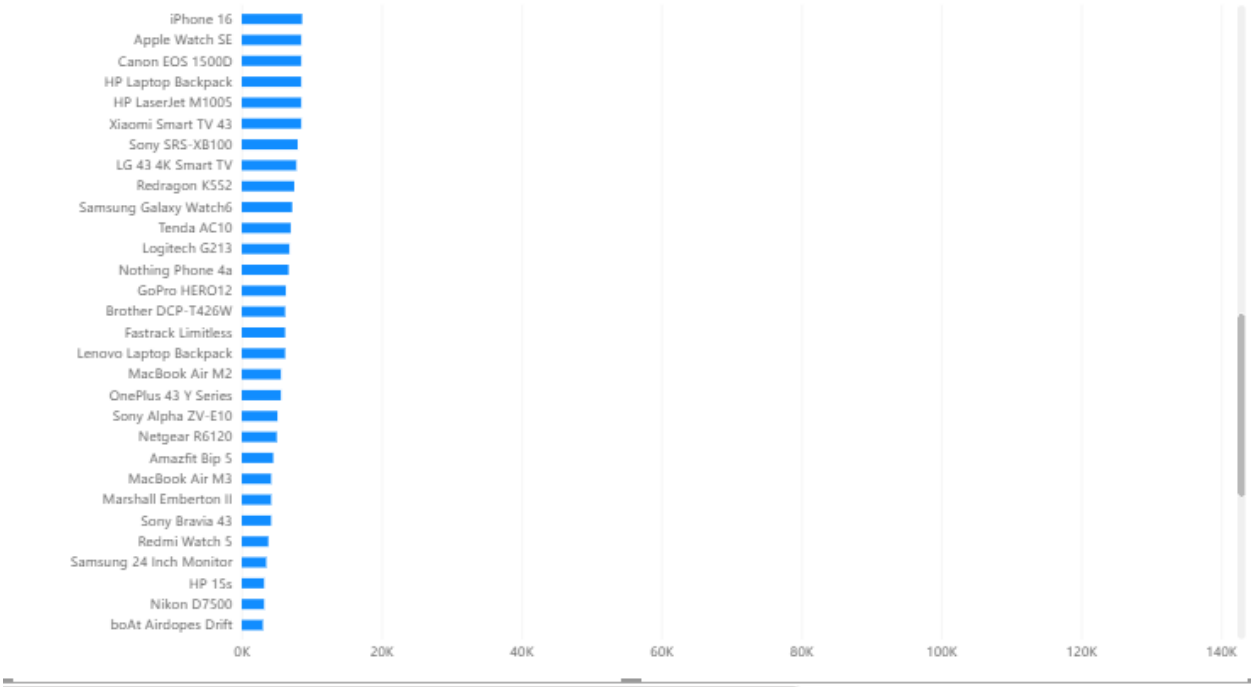
Product Review Analysis

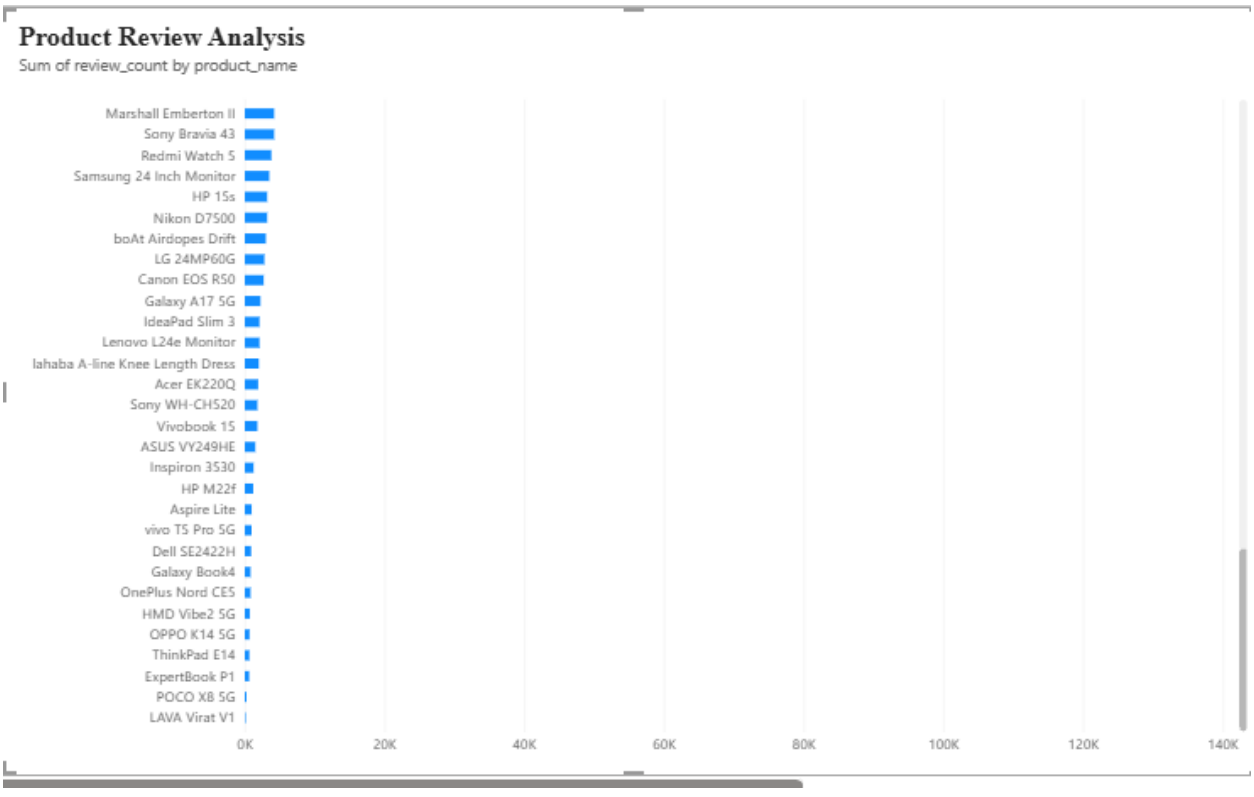
Sum of review_count by product_name



Product Review Analysis

Sum of review_count by product_name





Question 5: Which product categories offer the highest average discounts?

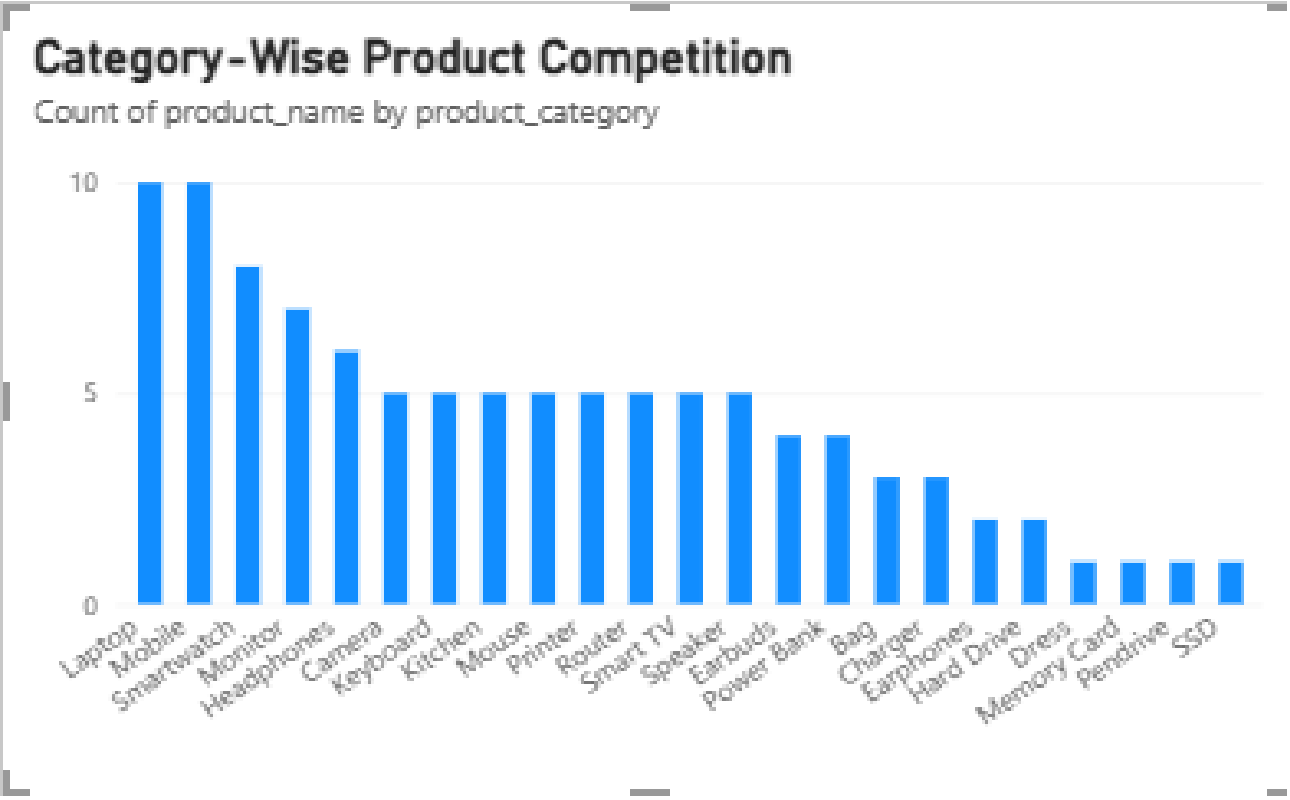
Answer:

Based on the Category-Wise Product Competition chart, **Laptops** show the highest competition, with approximately 10 products. Smartphones follow with around 9 products, while Monitors have approximately 8 products. Headphones and Cameras also show considerable competition compared with several other categories.

Business Insight:

Categories with a larger number of products may have stronger competition among brands. Businesses can use this information to compare product offerings, improve pricing strategies, and

identify opportunities to differentiate their products from competitors.



14. RECOMMENDATIONS

Based on the analysis, the following recommendations can be considered:

- Compare product prices across categories before making pricing decisions.
- Consider customer ratings and review counts when evaluating product performance.
- Use discount analysis to understand promotional strategies across categories.
- Monitor products with high review counts to identify customer interest.
- Use dashboard insights to support product comparison and business decision-making.

Final Outcome

The project demonstrates how Excel, PostgreSQL, Python, and Power BI can be used together to collect, clean, analyze, and visualize e-commerce product data. The resulting analysis and

dashboard help present product price comparisons, customer ratings, review patterns, and other available product information in an understandable format.

---

15. CONCLUSION

This project demonstrates how e-commerce product data can be collected, organized, stored, analysed, and visualized using different data analysis tools.

Excel was used for data organization, PostgreSQL for database analysis, Python for exploratory data analysis, and Power BI for dashboard creation.

The project provides an overview of the collected product data and demonstrates how data analysis can support product price monitoring, review analysis, discount comparison, and competition analysis.

16.FUTURE SCOPE

The E-Commerce Product Price & Review Analysis project can be enhanced in the future by adding more features and improving the analysis process.

1. Automated Data Collection: Product information can be collected automatically to reduce manual work.
2. Price Tracking: Product prices and discounts can be monitored over time to identify price changes.
3. Larger Dataset: More products, brands, and categories can be included to improve the analysis.
4. Advanced Data Analysis: Machine learning techniques can be explored to predict future product prices and identify customer preferences.
5. Improved Power BI Dashboard: Additional charts, filters, and interactive features can be added to provide better business insights.
6. Customer Review Analysis: Customer reviews can be analysed to understand customer opinions and satisfaction.

These improvements can make the project more useful for understanding product pricing, customer preferences, and market competition.

---

17. REFERENCES

1. Flipkart – Public product listings used for data collection.

2. Microsoft Excel – https://www.microsoft.com/microsoft-365/excel

3. PostgreSQL – https://www.postgresql.org/

4. Python Documentation – https://docs.python.org/3/

5. Microsoft Power BI – https://powerbi.microsoft.com/

****